

Changelog for GPW WATS 2.01 Native Order Gateway. Specification

Date: 25.03.2024 | Version: GPW0.62

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1. DISCLAIMER

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In case of sections of documentation at a **High** level work progress according to the current version of *GPW WATS Advancement of Documentation*, Warsaw Stock Exchange will endeavor to limit changes to these sections of documents to those related to:

1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. PREFACE

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform.

2.1. TARGET AUDIENCE

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. DOCUMENT'S PURPOSE

This document presents all adjustments that has been made in the newest release. The document is always compared with its previous version. Comparison of format changes is not included for readability purposes.

2.3. ASSOCIATED DOCUMENTS

GPW WATS 2.01 Native Order Gateway Specification is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification (this document)
- GPW WATS 2.02 FIX Order Gateway Specification .

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol

Please check the additional documentation which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway
- GPW WATS 4.02 Post Trade Gateway
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes
- GPW WATS 2.04 BenDec Message Definition Format
- GPW WATS 6.01 Connectivity.

It is recommended that you read the GPW WATS 1.01 Trading System document first.

Gateway messages are available in the Repository shared with ISV.

The contract is based on the BenDec library which is publicly available, see [Bendec library](#).

3. DOCUMENT HISTORY

Version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.
0.52	26.07.2023	Messages that have been added or updated in version 0.52: Added new messages: • MassQuote • MassQuoteResponse • RiskLimitDefinition • RiskLimitDefinitionResponse • TradeBust Modified • OrderAddResponse - field source has new Value 7 - IcebergRefill • OrderCancelResponse - field source has new Value 7 - IcebergRefill • RiskLimitBreach - field limitType has new values (301-326) • RiskLimitBreach - new fields: • riskWarningLevelAction • riskLimitUtilizationAmount • riskLimitUtilizationPercent. Cancelled: • OrderAddResponse field memo was deleted.
0.53	16.08.2023	Messages that have been updated accordingly to changes in messages v0.53: MassQuoteResponse: OrderBookRebuild value added in responses.source field. OrderAddResponse: OrderBookRebuild value added in source field. OrderCancelResponse: OrderBookRebuild value added in source field.
0.54	06.09.2023	Messages that have been updated accordingly to changes in messages v0.54: Changes in MassQuote message fields: • mifidFields.client.qualifier • mifidFields.executingTrader.qualifier MassQuoteResponse message field: • responses. OrderAdd • mifidFields.client.qualifier (values) • mifidFields.executingTrader.qualifier (values) • mifidFields.investmentDecisionMaker.qualifier (values) OrderAddResponse • reason (values) • source (values Activated added) OrderCancelResponse • reason (values) • source (values Activated added) OrderModifyResponse • reason (values) RiskLimitDefinition deleted: • elementType, side field added: • marketSegmentId, capacity field RiskLimitDefinitionResponse • reason (value MissingMicCode) TradeCaptureReportDual • algorithmicTradeIndicator (values) • tcrPartyBuy.mifidFields.client.qualifier (values) • tcrPartyBuy.mifidFields.executingTrader.qualifier (values) • tcrPartyBuy.mifidFields.investmentDecisionMaker.qualifier (values)

Version	Date	Description
		• tcrPartyBuy.mifidFields.investmentDecisionMaker.qualifier (values) • tcrPartySell.mifidFields.client.qualifier • tcrPartySell.mifidFields.executingTrader.qualifier • tcrPartySell.mifidFields.investmentDecisionMaker.qualifier TradeCaptureReportResponse • reason (values) TradeCaptureReportSingle • algorithmicTradeIndicator • tcrParty.mifidFields.client.qualifier • tcrParty.mifidFields.executingTrader.qualifier • tcrParty.mifidFields.investmentDecisionMaker.qualifier
0.55	16.10.2023	Messages that have been updated accordingly to changes in messages v0.55: Changes in OrderAdd message Listed values added in reason field: • InvalidPartyRoleForCLOB (1073) • InvalidPartyRoleQualifierForPartyId (1075) • RiskLimitNotDefined (7000) • RiskMaximumOrderVolumeExceeded (7001) • RiskMaximumOrderValueExceeded (7002) • RiskOrderPriceCollarExceeded (7003). Reject message Listed values added in rejectReason field: • NA (0) • InvalidSettlementDate (4). RiskLimitDefinition message riskLimitId field added. Modified values in clientRoleQualifier field: • NA (1) • Algorithm (2) • FirmOrLegalEntity (3) • NaturalPerson (4). TradeCaptureReportDual message Deleted fields: • tradeRequestId • lastParPx • tcrPartyBuy.side • tcrPartyBuy.counterpartyId • tcrPartySell.side • tcrPartySell.counterpartyId. TradeCaptureReportResponse message Added values in reason field: • SettlementDateCannotBeEarlierThanMinimumSettlementDate (2015) • SettlementDateCannotBeLaterThanMaximumSettlementDate (2016). TradeCaptureReportSingle message Deleted fields: • tradeRequestId • lastParPx. tcrParty.counterpartyId changed into counterpartyId
0.56	09.11.2023	Messages that have been updated accordingly to changes in messages v0.56: New message TestEvent has been added. Changes in Header message: msgType length value changed from 1 to 2 bytes.
0.57	30.11.2023	Messages that have been updated accordingly to changes in messages v0.57: • Unused messages removed: RiskLimitDefinition, RiskLimitDefinitionResponse, Test and TestEvent. • Rejection codes updated.

Version	Date	Description
		Added the value 'Stp' to the enum 'OrderSource'. Cross transactions added in sections 4.1.2 and 4.5. Order source filed values updated – section 4.4.5. New chapter 4.6 describing quotes handling.
0.58	15.12.2023	Publication of v0.58.
0.59	25.01.2024	RegularTrade value has been removed from TradeType filed from TradeCaptureReport messages. New values have been added for responses (OrderRejectReason) field: - InvalidBidAskSpread – 1208, - RequestNotAllowedForBlockInstrument – 2026, - RequestNotAllowedForCrossInstrument – 2028. New value has been added for source (OrderSource) filed: - CorporateAction – 11. New values have been added for rejectReason field: - SettlementDateRequired – 5, - TradeReportIdRequired – 6, - MissingReportIdSecondaryTradeReportIdOrTradeReportRefId – 7. MassQuoteResponse message Fields status (MassQuoteStatus) and reason (MassQuoteRejectionReason) have been added. TradeCaptureReportDual message Values for matchStatus filed have been changed to: - NA – 0, - Matched – 1, - Unmatched – 2. Values for reason (TcrRejectionReason) field have changed to: Deleted: - CannotFindMatchingFirstLeg – 2025. Added: - InvalidMatchStatus – 2029, - CrossNotAllowedOutsideOfClobInstrumentSpread – 2030, - CrossPriceNotEqualToTheReferencePrice – 2031, - CrossNotAllowedDuringClobInstrumentAuctionOrSuspension – 2032, - ForbiddenSecondaryTradereportId – 2033, - UnknownSecondaryTradereportId – 2034. TradeCaptureReportSingle message Added fields: - secondaryTradeReportId Values for matchStatus filed have been changed to: - NA – 0, - Matched – 1, - Unmatched – 2. Correction of the throttling mechanism description. Division into business and technical limits.
0.62	25.03.2024	The document has been updated with: - modifications of orders in terms of GTD validity (4.4.3.3), - enhancement of the description in section 4.5 Trade Capture for Cross and Block trades, - new section 4.8 Mass Cancellation, The following messages in section 7 have been updated: The list of messages is now described not by type, but by Id. New messages: - InitiateState - InitiateStateResponse - OrderMassCancel - OrderMassCancelResponse - RequestForExecution MassQuote:

Version	Date	Description
		• stpld length has changed to 1 • count field name has changed to quotes.count • quotes field name has changed to quotes.item; its type has changed to QuotesArray • count field name has changed to responses.count • responses field name has changed to responses.item; its type has changed to QuoteOrderResponseArray; new or changed values for reason field: ○ 1028 – Description amended to "lower", ○ 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, ○ 1039 – Name changed to InvalidTimeInForceForOrderType, ○ 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, ○ 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, ○ 1043 – New value, ○ 1044 – New value, ○ 1046 – New value, ○ 1069 – New value, ○ 1201 – New value, ○ 1203 – New value, ○ 1301 – New value, ○ 1401 – New value. • reason field – 1210 – New value added. OrderAdd: • stpld length has changed to 1 OrderAddResponse • New or changed values for reason field: ○ 1028 – Description amended to "lower", ○ 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, ○ 1039 – Name changed to InvalidTimeInForceForOrderType, ○ 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, ○ 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, ○ 1043 – New value, ○ 1044 – New value, ○ 1046 – New value, ○ 1201 – New value, ○ 1203 – New value, ○ 1301 – New value, ○ 1401 – New value. • source field – new value added: 12 MassCancel OrderModify • New field added: expire. OrderModifyResponse • New or changed values for reason field: ○ 1028 – Description amended to "lower", ○ 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, ○ 1039 – Name changed to InvalidTimeInForceForOrderType, ○ 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, ○ 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, ○ 1043 – New value, ○ 1044 – New value, ○ 1046 – New value, ○ 1069 – New value, ○ 1201 – New value, ○ 1203 – New value, ○ 1301 – New value, ○ 1401 – New value. TradeCaptureReportResponse: • secondaryTradeReportId field added, • New or changed values for reason field: ○ 1028 – Description amended to "lower",

Version	Date	Description
		○ 2024- New Value, • counterpartyCode field and its type amended.

4. SERVICE DESCRIPTION

4.1. NATIVE SERVICE DESCRIPTION

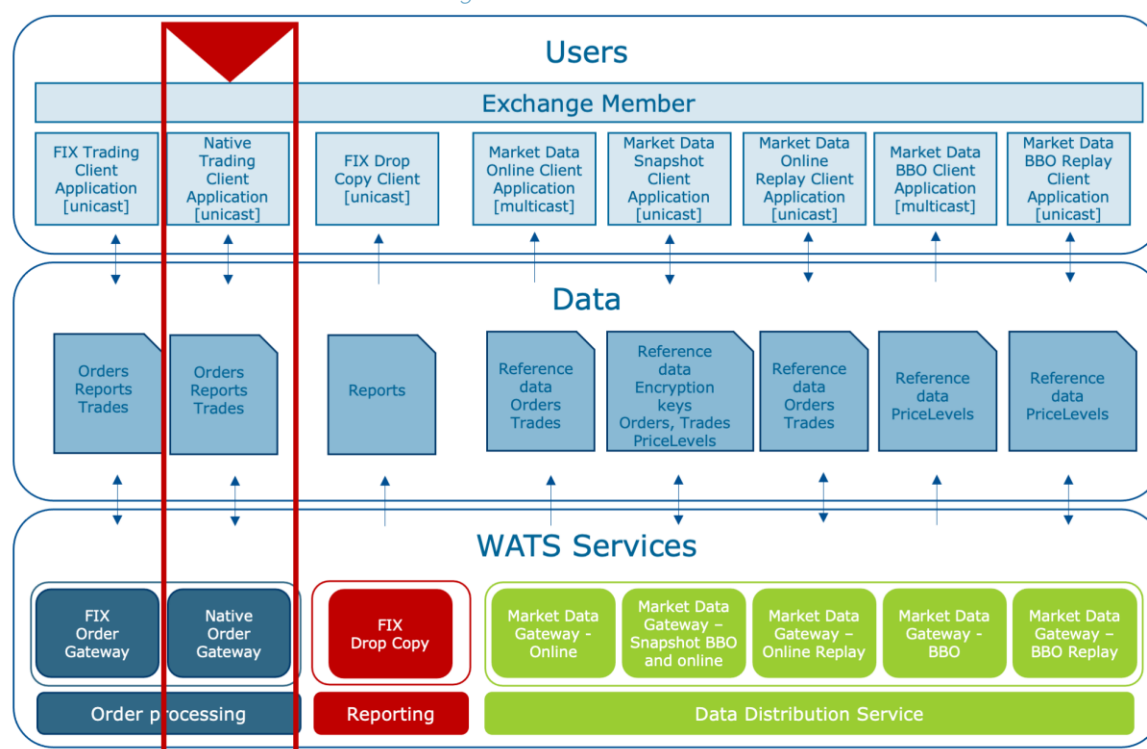
Native Order Gateway service provides a low-latency trading interface based on a proprietary binary protocol.

The main service functionalities:

- order submission
- order modification
- order cancellation
- transaction submission (block market).

Native Order Gateway is a point-to-point service based on the TCP-IP protocol.

Figure 1. GPW WATS Services



The following description is a brief introduction to the operation of the service, divided into the session and business layer. Detailed information can be found in the subsequent chapters of the document.

4.1.1. SESSION LAYER

To use the service, log in by sending a Login message. The message contains access parameters such as a token and a connection identifier. System confirms successful login with a LoginResult message which contains the result field set to "OK".

After logging in, a potential synchronization of missing messages takes place based on the sequential numbers exchanged during the login process (also called gap filling).

If no messages are sent within a specified time period, each party should send a Heartbeat message.

Participant can terminate the session at any time by sending a Logout message and closing the network connection. The Service responds to the Logout message with a LogoutResponse message.

Gateway can terminate the session by sending a ConnectionClose message and closes the network connection. The ConnectionClose message can be sent in a positive scenario similar to closing trading day, as well as in the case of significant session-level errors, e.g. communication protocol errors.

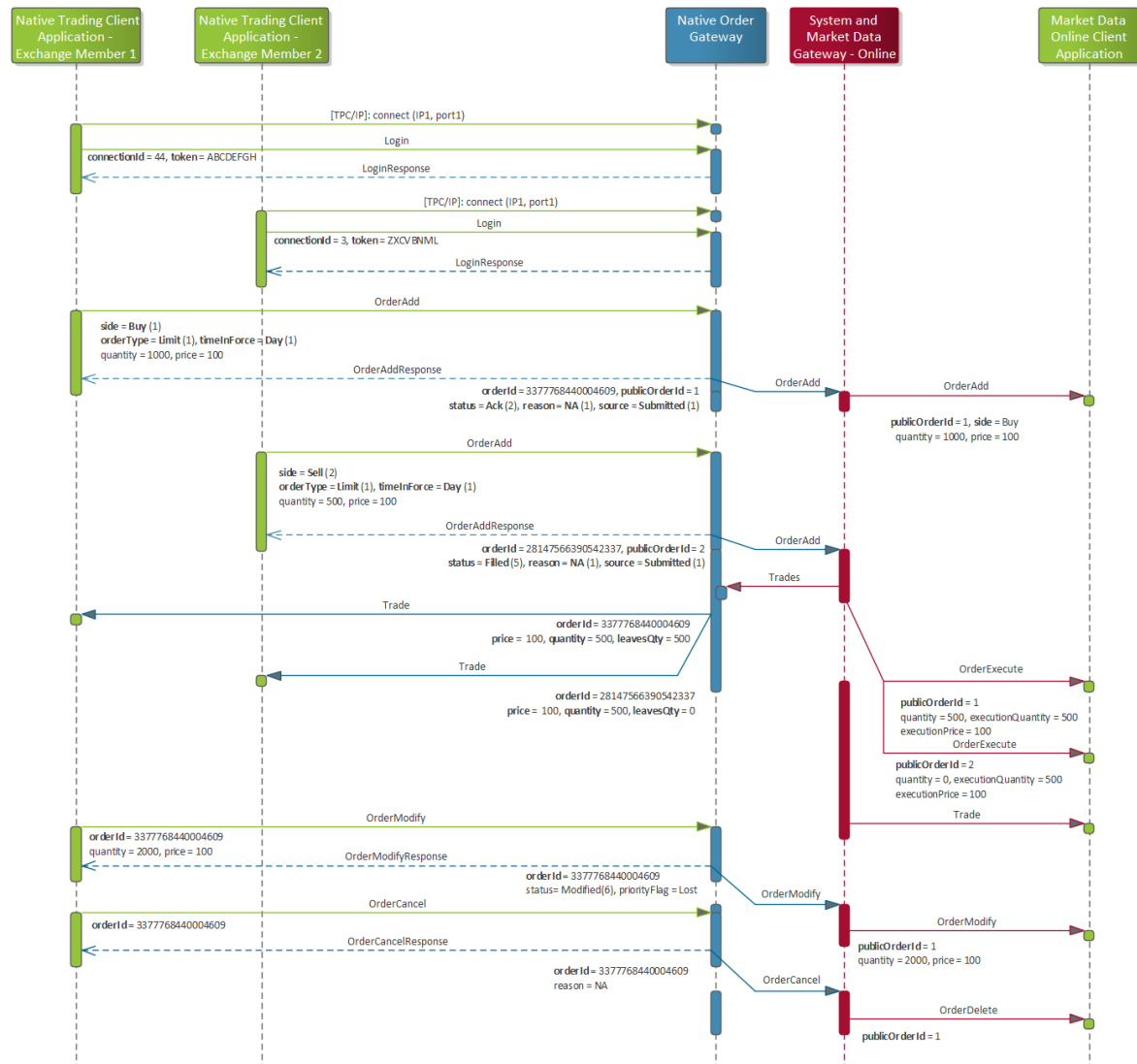
4.1.2. BUSINESS LAYER

Business communication is primarily based on the request - response pattern. We have the following main communication scenarios:

- An exchange member places an order by sending an OrderAdd message. Gateway responds with an OrderAddResponse message, which confirms the successful submission of the order or otherwise reject it.
- An exchange member modifies an order by sending an OrderModify message. Gateway responds with an OrderModifyResponse message to confirm the accuracy of the modification or otherwise reject it.
- An exchange member cancels an order by sending an OrderCancel message. Gateway responds with an OrderCancelResponse message to confirm the accuracy of the cancellation or subsequently reject it.
- An exchange member reports a block transaction conducted outside the market by sending a TradeCaptureReportSingle or TradeCaptureReportDual message. Gateway responds with a TradeCaptureReportAck message and continues to handle the blocked transaction.
- An exchange member reports a cross transaction using trade capture reports - similar to block transactions.

In addition to the above request - response pattern, where the exchange member initiates communication, it is also possible to receive ad-hoc messages from Gateway concerning the status of orders, transactions and order cancellations performed by System. Changes in the order status are sent to the exchange member in the OrderAddResponse message. Information about transactions based on the submitted orders is provided in the Trade message. With regards to order cancellations performed by System, the OrderCancelResponse message is used. Below is a diagram illustrating the usage of key business messages.

Figure 2. Scenario of messages exchange via Native Order Gateway



The diagram depicts the actions of two exchange members. Both log into the system and then submit crossed orders (an OrderAdd message). Gateway notifies about the transaction being executed by sending a Trade message. Next, Exchange Member 1 modifies the order by increasing the quantity to 2000 and cancels it. The above actions of exchange members are shown in the context of provided market data. It demonstrates the orders submission, execution, modification and cancellation.

4.2. CONDITIONALLY REQUIRED FIELDS

Conditional fields are ignored unless they are necessary in the message data (e.g. triggerPrice is ignored in orders other than Stop Loss and Stop Limit).

One sets unused conditional and optional numeric fields should to zero. One sets unused conditional and optional enumerable fields to a value which indicates their unrequired use (e.g. not applicable).

4.3. SECURITY IDENTIFICATION

System assigns the instrument identifier (e.i. instrumentId). On the basis of reference data, one obtains other identifiers, e.g. ISIN number.

4.4. ORDER MANAGEMENT

The chapter describes important information about orders, including their identifiers, types, and processing rules.

4.4.1. ORDER IDENTIFIERS

The following order identification fields are used in messages:

- orderId
- publicOrderId.

4.4.1.1. orderId

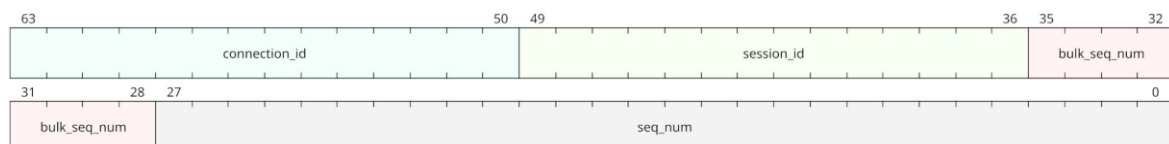
Trading System assigns a unique identifier for an order. The orderId field is unique across all trading days, all order books and all Participants. It is persistent for the entire life of an order (i.e. it does not change after an order modification). orderId is known only for the exchange member who submits the order, Exchange and possibly for supervisory entities (e.g. Polish Financial Supervision Authority - KNF). Other exchange members do not know that orderId field.

An orderId field is generated for each order as a concatenation of connection id, session id, bulk sequence number and sequence number of the OrderAdd message sent by an exchange member:

- connection Id – a connection number - connectivity parameter provided by GPW,
- session Id – a session number received during the login process,
- bulk sequence number – zero value for single orders (e.g. Market, Limit) and the value of subsequent orders for messages with multiple orders or quotes (e.g. MassQuote),
- sequence number - a sequence number of OrderAdd message.

All the above parts of the orderId field are known at the time of sending the OrderAdd message.

Figure3. orderId field construction.



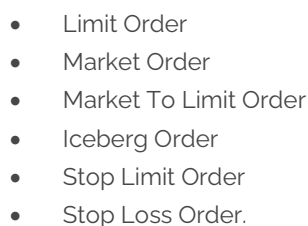
Example

- OrderAddResponse message with orderId (highlighted in bold blue):

1b 00 05 01 00 00 00 **01 00 00 00 10 00 38 04** 01 00 00 00 00 00 00 02 01 00 01,

- Raw bytes of order id: 01 00 00 00 10 00 38 04,

- Figure 4. `orderId` = 0x0438001000000001



4.4.2.1. Limit Order

Limit Order is an order to buy or sell at a stipulated limit price or a better one. Limit Order can be executed at a price equal to or better than its limit price, which means for buy order at its limit or lower price and for sell order at its limit or higher price. Unexecuted remainder of a Limit Order is added to the order book, unless order's validity attribute `timeInForce` specifies otherwise.

Relevant message fields:

- `orderType` = 1 (Limit)
- `price` = Limit Price.

4.4.2.2. Market Order

Market Order is an order to buy or sell without specified price. Market Order is executed at the best opposite prices with one or more counterpart resting orders.

During Continuous Trading market phase, Market Orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. `timeInForce` = 3, 4, 5, 7.

During Auction market phase, Market Orders can only be submitted with VFA and VFC validity attributes, i.e. `timeInForce` = 5, 7.

Relevant message fields:

- `ordType` = 2 (Market),
- `timeInForce` = 3 (Immediate Or Cancel), 4 (Fill Or Kill), 5 (Valid For Auction), 7 (Valid For Closing).

4.4.2.3. Market To Limit

Market To Limit Order is an order to buy or sell without a specified price. Market To Limit Order is executed at the best opposite price level with one or more counterpart resting orders.

During Continuous Trading market phase, Market To Limit Orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. `timeInForce` = 3, 4, 5, 7.

During Auction market phase, Market To Limit Orders can only be submitted with VFA and VFC validity attributes i.e. `timeInForce` = 5, 7.

Relevant message fields:

- `ordType` = 3 (Market To Limit)
- `timeInForce` = 3 (Immediate Or Cancel), 4 (Fill Or Kill), 5 (Valid For Auction), 7 (Valid For Closing).

4.4.2.4. Iceberg

Iceberg Order is an order, for which only a part of the total quantity is disclosed to the market via Market Data. The rest of the order is hidden (invisible) to the general market. An exchange member submitting an Iceberg Order must specify additionally the quantity that is going to be displayed initially, i.e. `displayQty`, which:

- must be lower than the total order quantity,
- must be greater or equal to the Minimum Display Quantity set in Trading System configuration,
- may be randomized by the Trading System within a specified range.

Once displayed quantity is completely filled, a new portion of the order is disclosed to the market according to the displayQty requested at the order entry (with possible randomization).

The value of each Iceberg Order (i.e. price x orderQty) at the moment of entry or during any modification must be equal or greater than the Minimum Iceberg Size set in the Trading System configuration.

Iceberg Orders can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.

Relevant message fields:

- ordType = 4 (Iceberg),
- price = Limit Price,
- displayQty = Initially Displayed Quantity,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.4.2.5. Stop Limit

Stop Limit is an order to buy or sell at a stipulated limit price or a better one, which is activated and inserted into the order book upon reaching or exceeding pre-defined price level (called Trigger Price) by the Last Traded Price on the market. Until activation, Stop Limit is hidden and cannot interact with the order book. Once activated, Stop Limit is processed the same way as a Limit Order with the same validity attribute that the original Stop Order had before its activation (nonetheless maintains its original Order Type = 5).

When submitting or modifying a Stop Limit order, the following conditions must be met:

- for buy order: Price $\geq$ Trigger Price $>$ Last Traded Price (LTP),
- for sell order: Price $\leq$ Trigger Price $<$ Last Traded Price (LTP).

Stop limit can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.

The Trigger Price may be modified for inactive Stop Order, but once the order is activated, Trigger Price cannot be modified anymore.

Relevant message fields:

- ordType = 5 (Stop Limit),
- price = Limit Price,
- triggerPrice = Activation Price,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.4.2.6. Stop Loss

Stop Loss is an order to buy or sell without specified price, which is activated and inserted into the order book upon reaching or exceeding pre-defined price level (called Trigger Price) by the Last Traded Price on the market. Until activation, Stop Loss is hidden and cannot interact with the order book. Once activated, Stop Loss is processed the same way as a Market Order with Immediate Or Cancel (IOC) validity attribute (nonetheless maintains its original Order Type = 6).

When submitting or modifying a Stop Loss Order, the following conditions must be met:

- for buy order: Trigger Price $>$ Last Traded Price (LTP),
- for sell order: Trigger Price $<$ Last Traded Price (LTP).

Stop Loss can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.

The Trigger price may be modified for inactive Stop order, but once the order is activated, Trigger Price cannot be modified anymore.

Relevant message fields:

- ordType = 6 (Stop Loss),
- triggerPrice = Activation Price,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.4.3. ORDER VALIDITY TYPES

Exchange members may submit orders with the following validity attributes (timeInForce field):

- Day (DAY),
- Good Till Cancel (GTC),
- Good Till Date (GTD),
- Good Till Time (GTT),
- Immediate Or Cancel (IOC),
- Fill Or Kill (FOK),
- Valid For Auction (VFA).
- Valid For Closing (VFC).

Not all validity attributes may be used with every order type throughout various market phases.

4.4.3.1. Day (DAY)

Day order is valid until the end of the current trading day only. Unexecuted Day orders are expired by Trading System at the end of the current trading day.

Day validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 1 (Day)

4.4.3.2. Good Till Cancel (GTC)

Good Till Cancel order is valid until it gets fully executed or canceled by the submitter or market operations (whichever comes first).

There is a maximum number of days threshold set up in Trading System configuration, that defines for how long a Good Till Cancel order may rest in the order book (ranging from 1 to 365 days). After that time elapses, GTC order gets expired as well.

Good Till Cancel validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 2 (Good Till Cancel)

4.4.3.3. Good Till Date (GTD)

Good Till Date order is valid until the end of the specified day.

If the expiry date specified for an order is not a trading day, the order gets expired at the end of the previous day.

The specified date cannot exceed the maximum number of days threshold defined by Market Operator (ranging from 1 to 365 days).

Good Till Date validity attribute may be used with all order types, except for Market Order and Market To Limit.

It is possible to modify orders with GTD validity type. Modification of the date is possible from the current date to the limit for GTD (otherwise the order is rejected). Modified orders do not change priority.

Relevant message fields:

- timeInForce = 6 (Good Till Date)
- expire = Expiry Date

4.4.3.4. Good Till Time (GTT)

Good Till Time order is valid until the specified time of the current trading day.

If the expiry time specified for an order is later than the end of the last market phase, the order gets expired at the end of the current trading day.

Good Till Time validity attribute must be used in conjunction with the expiry field set to the current date and time in the future.

Good Till Time validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 8 (Good Till Time)
- expire = Expiry Time (UTC timestamp with current date)

4.4.3.5. Immediate Or Cancel (IOC)

Immediate Or Cancel order must be executed immediately, in full or partially, upon order entry. IOC orders may be matched against one or more opposite orders (with the price limits within Trade Price Collars). Any unexecuted part of an IOC order is automatically canceled by Trading System.

Immediate Or Cancel orders may be submitted during Continuous Trading only.

Immediate Or Cancel validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 3 (Immediate Or Cancel)

4.4.3.6. Fill Or Kill (FOK)

Fill Or Kill order must be executed in full, immediately upon order entry. FOK orders may be matched against one or more opposite orders (with the price limits within Trade Price Collars). An unexecuted FOK order is automatically canceled by Trading System.

Fill Or Kill orders may be submitted during Continuous Trading market phase only.

Fill Or Kill validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 4 (Fill Or Kill)

4.4.3.7. Valid For Auction (VFA)

Valid For Auction order is activated and inserted into the order book at the beginning of the next Auction market phase (whether scheduled or unscheduled). VFA order submitted in Continuous Trading is hidden until its activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFA order is valid until the end of Auction and gets expired by Trading System immediately after uncrossing (if left unexecuted). VFA orders are valid for the current trading day only. Activated VFA orders retain their original priority timestamp from the moment of submission.

Valid For Auction validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 5 (Valid For Auction)

4.4.3.8. Valid For Closing (VFC)

Valid For Closing order is activated and inserted into the order book at the beginning of the Closing Auction market phase only. VFC order submitted in Continuous Trading or Auction other than Closing, is hidden until its activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFC order is valid until the end of the Closing Auction and gets expired by Trading System immediately after the uncrossing (if left unexecuted). VFC orders are valid for the current trading day only. Activated VFC orders retain their original priority timestamp from the moment of submission.

Valid For Closing validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 7 (Valid For Closing)

4.4.4. ORDER CAPACITY

Order capacity defines the capacity of the company, which submits an order. The capacity field values:

- 1 - Agency mapped as AOTC (Any other capacity)
- 2 - Principal mapped as DEAL (Dealing on own account)
- 3 - Riskless Principal mapped as MTCH (Matched principal)

It is required to set the correct capacity field value by the exchange member.

4.4.5. ORDER STATUS

The order status informs on the state in which it is in GPW WATS. With regards to orders submitted by an exchange member, the status may take the values acknowledged, cancelled, rejected or filled. Successful modification of the order by the exchange member results in the modified status. Similarly, a successful cancellation changes the status of the order to be cancelled.

An Exchange member may receive the following status for orders from Gateway:

- Acknowledged (1) – when an incoming unexecuted order is accepted into System or restated the next day in the morning (in the case of GTC/GTD orders),
- Cancelled (2) – when an order is cancelled by the exchange member, System itself or by Market Operations.
- Rejected (3) – when an order is rejected by System upon entry due to various business validations.
- Filled (4) – when an order is fully or partially executed.
- Modified (5) – when an order is modified by Participant as a confirmation of modification.

Information about the status of orders is sent in OrderAddResponse and OrderModifyResponse messages.

The reason for the order status change may be an order modification or cancelation by the exchange member, orders matching, System actions (e.g. cancelation via Cancel on Disconnect mechanism) and session supervision team operations.

More information concerning the reason for the order state change can be found in the source field description (see source field description in OrderAddResponse message). This field specifies the origin of the order message (e.i. OrderAddResponse), consequently, makes it possible to determine the possible cause of the status change. It takes the following values:

- Submitted – order message coming from Gateway,
- CoD– cancelled order by Cancel on Disconnect mechanism,
- Expired – cancelled order by System in the case of expiry,
- StopOrder – System (i.e. the order management service) notifies on the activation of a stop order,
- Suspended – cancelled order by System (i.e. canceller service) in the case of instrument suspension,
- Reinstated – reinstated order,
- IcebergRefill – iceberg order refill,
- OrderBookRebuild – order book rebuild after auction,
- Activated – VFA/C order got activated, from now on it will take part in matching,
- Stp – cancelled order due to self-trade prevention,
- CorporateAction – cancelled order due to submitted Corporate Action.

4.4.6. CANCELLATION

Cancellation of a previously submitted order by the exchange member is only possible on the same connection id on which the order was submitted. For example, an order submitted on the connection id = 123 cannot be removed from the connection id = 242.

The order is cancelled by sending the OrderCancel message with the appropriate orderId field (i.e. the orderId received in the OrderAddResponse message)

Connection allocation management is the responsibility of the exchange member.

Deleting orders is subject to business validation, in particular, it is not possible to delete an order which has already been fully executed.

More information about connections is presented in the Connectivity chapter.

4.4.7. MODIFICATION

Modification of the order, similarly to its cancellation, is only possible on the connection on which the order was previously sent (i.e. the same connection id).

The order is modified by sending the OrderModify message with the appropriate orderId field (i.e. the orderId received in the OrderResponse message).

4.4.8. ACCOUNT STRUCTURE

There are the following types of accounts (i.e. accountType field):

- Customer - account is carried on the customer side of the books.
- House – house trader.

If there is no account placed in the order, the accountType field should be set to "Missing" and the account field should be filled with zeros.

4.4.9. TEXT FIELDS

A character in the text field is treated as an unsigned byte type. In the trading_port.json contract, the above type is called AnsiChar. The string is created as an AnsiChar array. Examples of text fields are account, token, micCode etc.

If the length of the text is less than the length of the field, the empty space should be filled with zeros.

ASCII characters, in the range 0 – 127, should be used. Messages containing text fields with characters outside the above range is rejected.

4.4.10. ORDER BOOK RESTATEMENT

At the beginning of each trading day, exchange members receive incoming orders from Gateway which were placed during the previous session and are still valid (e.g., orders with a Time In Force equal to GTD or GTC), known as restated orders.

More information can be found in the document GPW WATS 1.01 Trading System, chapter Trading Schedules.

Restated orders are sent using an OrderAddResponse message with fields set to the values from the end of the previous day. An order status is also set to the last status from the previous day (e.g. modified, filled). The field source is set to a Reinstated value.

Reinstated orders are not sent publicly in Market Data, only to the exchange members which submitted them.

Expired orders are not transferred – reinstated (e.g. order with TIF = Day, or Good Till Date orders with a past expiry date).

UNSOLICITED ORDER CANCELLATIONS GENERATED BY TRADING SYSTEM

Unsolicited order cancellations generated by System is carried out using OrderCancelResponse messages with the following characteristics:

1. order source (i.e. source field) takes a different value than submitted. This value explains the source of the order cancellation e.g. Cancel on Disconnect mechanism,

2. reason – a reason for cancelation if available.

4.5. TRADE CAPTURE

There are several situations where transaction capture reports are exchanged:

1. Block transactions (field tradeType = 38 - BlockTrade)
 - o An exchange member may send a dual transaction capture report in a situation where a transaction involving clients of a given exchange member was made on the block market. The TradeCaptureReportDual message is used by the exchange member as a dual transaction capture report. Gateway response should be the TradeCaptureReportResponse message.
 - o An exchange member may send a single transaction capture report which requires confirmation by the other party. System ensures correct message exchange between the transaction parties. The TradeCaptureReportSingle message is used by the exchange member as a single transaction capture report. Gateway response should be the TradeCaptureReportResponse message.
2. Cross transactions (field tradeType = 22 - PrivatelyNegotiatedTrade)
 - o An exchange member may send a dual transaction capture report (a TradeCaptureReportDual message) in a situation where a cross transaction is submitted.

FIX Order Gateway equivalent of the message about transaction capture reports are TradeCaptureReport (AE)/ TradeCaptureReportAck (AR) messages.

4.5.1. BLOCK MARKET MODEL

Block trading is a market model in which trades are concluded bilaterally in the process of the negotiations between the counterparties (beneficiary clients). The beneficiary clients may belong to one or a few Exchange Members. The negotiations usually take place beyond Trading System infrastructure. Block trading is concluded when the respective trade entry is recorded in Trading System based on the Trade Capture Report message/messages submitted by Exchange Members.

Block Trade facility enables the Exchange Members to report pre-arranged large trades through the Market Operator Trading System.

The value of Block Trades must not be lower than the value e.g. Large In Scale value. The price of the Block Trade must be within thresholds set by the exchange. Price thresholds can be set on different levels depending on the session phase of the same instrument in the Central Limit Order Book. Only instruments traded in the Central Limit Order Book are subject to Block Trades.

Block Trade is reported to the exchange by entering "one-party" or "two-party" Trade Capture Report. In the "one-party for pass-through" Block Trade the beneficiary clients belong to different Exchange Members. The initiator who enters one-party Block Trade requires confirmation by the counterparty to achieve the Block Trade. In "two-party" Block Trade both beneficiary clients belong to the same Exchange Member. "Two-party" Block Trade is used to report internal trade where the trade can be concluded and reported between two clients of the exchange member (Order Capacity; Agency vs Agency) or between the firm and the client (order Capacity: Principal vs Agency).

Block Trade features:

- Despite the fact that MiFID 2 tick size regime does not apply to Block Trades, simplified tick size tables for block instruments have been adopted.
- Both parties to the Block Trade must be the Exchange Member.
- Block Trades are reported by using the Trade Capture Reports messages.
- Block Trades are not pre-transparent thus not matched Trade Capture Reports are not disseminated in the market data stream.
- Block Trades are post-trade transparent thus matched Block Trades are immediately disseminated in the Market Data stream.
- Block Trades do not update the Last Trade Price.
- Block Trades are not taken into account when calculating the opening price or the closing price.

4.5.1.1. Trade Capture Report Dual for Block Trades

TradeCaptureReportDual is applied when both beneficiary parties to the trade belong to the reporting firm (exchange member). The TradeCaptureReportDual is accepted by Gateway by sending back TradeCaptureReportResponse with the field status = 2 (Accepted). If TradeCaptureReportDual is rejected, then the exchange member receives the TradeCaptureReportResponse with status = 3 (Rejected) with the respective information about the reason for rejection provided in the **reason** field.

4.5.1.2. Trade Capture Report Single for Block Trades

TradeCaptureReportSingle is applied when the beneficiary parties to the trade belong to different exchange members. The applied one-sided model is the "one-party report for pass-through". The initiator submits a report (first leg to the trade), which is accepted by the counterparty to the trade.

This is the basic TradeCaptureReportSingle flow enabling exchange members to conclude the block trade:

- The initiator submits the first leg to the trade by sending TradeCaptureReportSingle with the values:
 - a) tradeReportTransType = 1 (New).
 - b) tradeReportType = 1 (Submit).
- The first leg is accepted by System by sending TradeCaptureReportResponse with the status = 2 (Accepted).
- System forwards the TradeCaptureReportSingle to the counterparty by sending notification with the values:
 - a) tradeReportTransType = 1 (New).
 - b) tradeReportType = 2 (Alleged).
- The counterparty accepts the first leg by sending TradeCaptureReportSingle with the values:
 - a) tradeReportTransType = 3 (Replace).
 - b) tradeReportType = 3 (Accept).
- System sends the TradeCaptureReportSingle informing the initiator and the counterparty about the concluded trade with the values:
 - a) execType = 16 (Trade Correct).
 - b) matchStatus = 1 (Matched).

4.5.2. CROSS TRADES

For a given instrumentId, Cross trades can only be submitted if the corresponding instrumentId with the same ISIN code in the Order Book is in the Continuous Trading phase (variable price and fixed price), i.e. in:

- Continuous price and time (ContinuousPriceTime).
- Continuous time at reference price (ContinuousPriceTime).
- Trade at last (ContinuousLastAuctionTime).

Cross trades do not change Last Trade Price for an instrument with the same ISIN code in the central order book.

Cross trades are transmitted to System via Trade Capture Report message. This is a dual-sided TradeCaptureReportDual.

The content of the TradeCaptureReportDual message for Cross trades is analogous to that for bilateral Block trades, if the following tags have the following fixed values:

- tradeType = 22 (PrivatelyNegotiatedTrade)
- settlementDate = T+2

The values in tradeReportType have been reduced to the requirements of a Cross transaction to:

- 1 = Submit
- 7= Trade Report Cancel

Kinematics of Cross transaction:

1. Transmission of a TCR Cross to System with a tradeReportType field with the value 1 = Submit.
2. For a valid TCR Cross, System sends a TradeCaptureReportResponse with an Accepted status (status = 2 (Accepted))
3. For an invalid TCR Cross, System sends a TradeCaptureReportResponse with status Rejected (status = 3 (Rejected))

4.6. QUOTE HANDLING

Market makers send quotes into a market using a MassQuote message. For a given instrument, it is possible to send one quotation. The MassQuote message enables the sending of up to 30 quotations.

The trading platform sends a confirmation of the accepted and rejected quotations by transmitting a MassQuoteResponse message.

Quotations rules:

- Two-sided quotes are allowed,
- Only one market maker quotation is allowed per instrument,
- MassQuoteResponse message includes all submitted quotations along with their acceptance status (e.g. Accepted, Rejected),

- Modification of submitted quotations involves placing another MassQuote message with changed values for price or quantity,
- Cancellation of submitted quotations involves placing another MassQuote message with the quantity set to 0.

4.7. CANCEL ON DISCONNECT

Cancel on Disconnect (CoD) mechanism prevents orders from remaining outside the exchange member's control after an incorrect disconnection from System, i.e. without logging out.

Disconnection from Gateway is understood as breaking the TCP/IP connection or lack of communication from the Client Application, including the lack of Heartbeat messages for a period of 12 seconds (so-called grace period).

The CoD is a sub-service of Gateway service and can be enabled for any exchange member connection. Elimination of orders follows as per the CoD mechanism rules on a given trading connection. If the user has more than one connection, then the mechanism can be activated on each of these connections or only on chosen connections. The GPW is responsible for the configuration of the service and performs it at the exchange member request.

In the case of an unexpected party disconnection, the service would cancel orders with the following TIFs: DAY, GTT, VFA and VFC. Orders with the following TIFs: IOC, FOK, GTC, GTD are excluded from cancellation.

4.8. MASS CANCELLATION

Mass Cancel functionality allows an Exchange Member to cancel all active orders according to the selection criteria they have specified. In the case of a previously partially executed order, only the remaining unexecuted part of the order is cancelled. This means that Mass Cancel doesn't affect the already executed volume of the order. Mass Cancel functionality works only for CLOB market model (i.e. the Mass Cancel functionality does not include the cancellation of unmatched Trade Capture Reports).

This functionality can only be used by an exchange member in respect of orders that they have sent themselves. Sending a Mass Cancel message is not possible via the DEA (Direct Electronic Access). An Exchange Member must indicate the Executing Trader performing the operation (PartyRole= Executing Trader) and provide shortcode in PartyID field.

The Exchange Member performs Mass Cancel by sending the OrderMassCancel message. The OrderMassCancel message allows the cancellation of all active orders placed on all connections (ConnectionID) of a Participant. OrderMassCancel message contains fields that define the selection criteria for cancelled orders. These fields can be divided into two groups:

- General criteria specified in the massCancelRequestType field.
- Narrowing criteria contained in the body of the message (e.g. targetPartyId).

Selection Criteria

Three main selection criteria are available for the mass cancellation of orders. These main criteria are disjoint and are contained in the required massCancelRequestType field:

- 1 = CancelForSecurity

- 7 = CancelAllOrders
- 9 = CancelOrdersForMarketSegment

In addition, a number of narrowing criteria is available that only work in combination with one of the above three main criteria for Mass Cancel. It is allowed to select only one narrowig criterion from the targetPartyRole, which automatically means that the value of the targetPartyId field should correspond to the selected role:

- None and LiquidityProvider – targetPartyId equals 0.
- SenderLocation - targetPartyId is set to connection_id.
- ClientID, ExecutingTrader and InvestmentDecisionMaker - targetPartyId is set to the short code of a party.

~~4.8:~~ 4.9. REJECT MESSAGES

Messages sent by the exchange member can be rejected at the session level and at the application level.

The Reject message informs on the message rejection at the session level i.e. in a situation where the message cannot be considered valid and can no longer be processed. Reason for rejection are given in the rejectReason field (e.g. Invalid message type).

Session-level valid messages related to submitting, cancelling and modifying orders may be rejected at the application level. Relevant information about the rejection is provided in the status and reason fields of the message which are a Response to the exchange member's message.

~~4.9:~~ 4.10. TIMESTAMPS AND DATES

The time and date are stored on the Timestamp type, which is based on a 64-bit unsigned integer. On the Timestamp type, the time is given within nano second accuracy.

Below are the fields in which the time is given and the rules for filling them in:

- Expire – in the case of GTT orders, only the time is filled in, in the case of GTD orders, only the date is filled in.

5. CONNECTIVITY

This chapter provides information regarding: logon authentication, network parameters, failover and recovery.

5.1. SESSION IDENTIFICATION

5.1.1. LOGIN PARAMETERS

Establishing a session with GPW WATS requires logging into System. To do this, it is necessary to have a token associated with a unique connection ID. The token and connection ID are fields of the login message. Both parameters are obtained from GPW. You cannot log in with the same connection ID multiple times.

5.1.2. NETWORK CONNECTION

IP addresses and ports for Gateway are published in a separate configuration document. There are 2 IP addresses available for each connection. The primary address for typical use and secondary address used in the case of failure.

5.1.3. TECHNICAL SESSION IDENTIFICATION

The technical session begins when System starts up and ends when it is turned off. Thus, it typically covers a trading day. In special cases, such as an emergency system restart, another technical session is started. The technical session ID is published in the StartOfTechnicalSession market data message as a field sessionId. It is also possible to obtain the current session id when logging in. Then the session id is given in response to the login message.

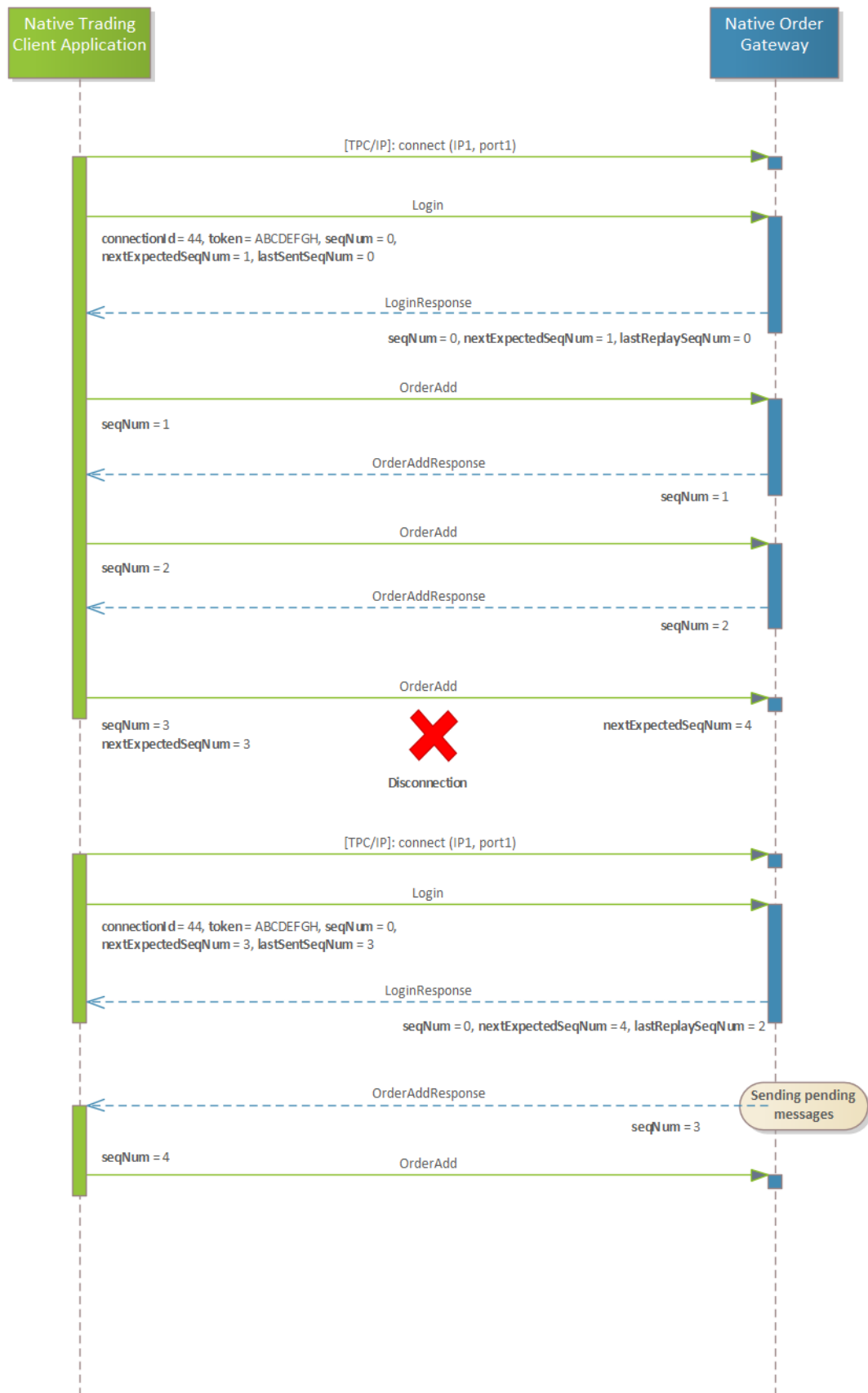
The session ID is used to create an order ID (see section "Order Identifiers").

5.2. FAILOVER AND RECOVERY

Due to a reduction to operational risk within GPW WATS, the failover and recovery mechanism has been designed.

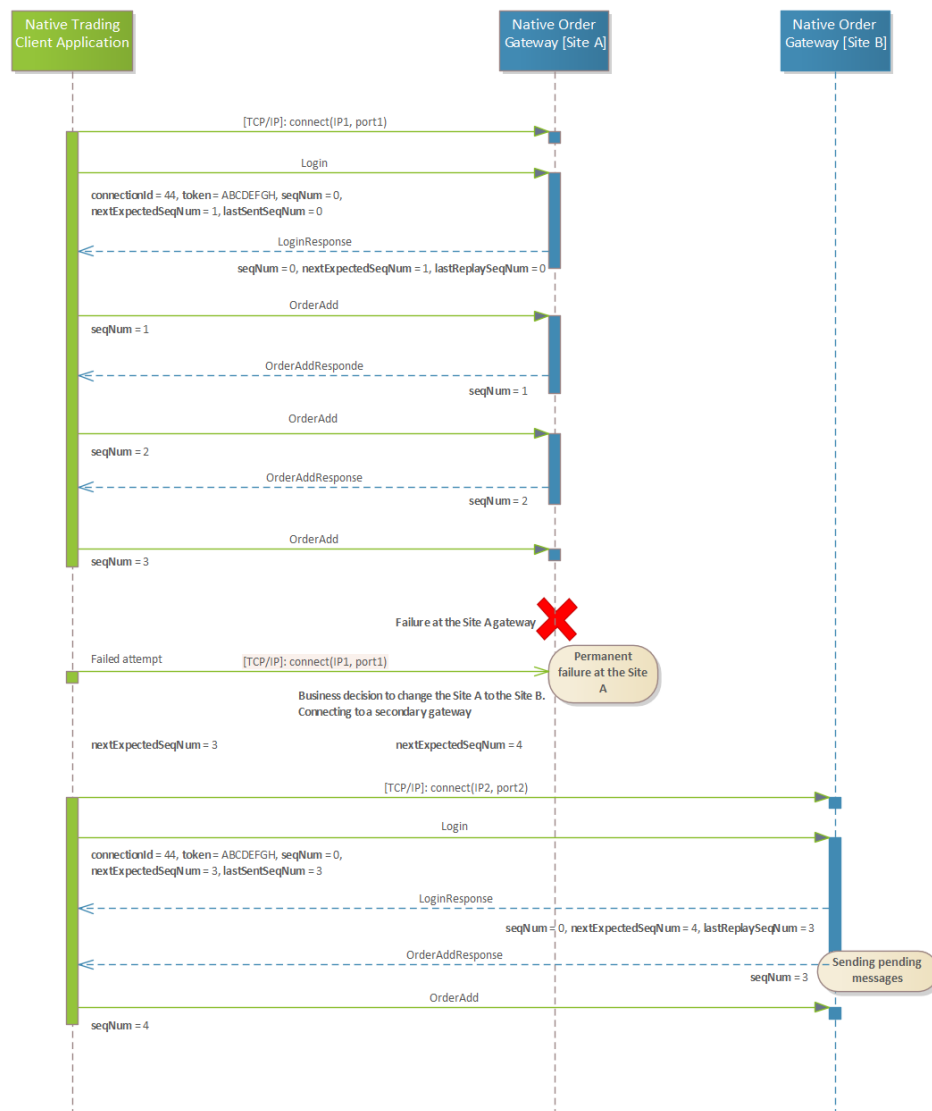
In the case of temporary and short unavailability of Gateway, the exchange member should attempt to re-establish the connection and login to Gateway. Subsequently the messages are resynchronized between the exchange member and Gateway based on the sequence numbers included in the Login and LoginResponse messages. There are nextExpectedSeqNum and lastSentSeqNum in the Login message. The nextExpectedSeqNum field means the next expected message sequence number to be received and the lastSentSeqNum field means the last sent sequence number. The LoginResponse message also contains the nextExpectedSeqNum field and, additionally, the lastReplaySeqNum field indicating the number of the last sent message or the last message currently waiting to be sent, if any. Any gap in the messages is filled by sending missing messages by each party. The diagram below shows such a situation.

Figure 5. The synchronization after disconnection scenario.



When the primary Gateway is unavailable for a long period, the procedure to switch to a secondary is launched. In such a situation, the exchange member receives information from the GPW concerning the need to switch to the secondary Gateway. The connection is established with the same login parameters but on a new IP address. After logging in, business messages are synchronized in the same manner as in the first case. The diagram below shows such a situation.

Figure 6. Switching gateway location in the case of disconnection.



Please note that when using the CoD mechanism and disconnecting without a logout message, which is natural for most failures, orders are cancelled (see section Cancel On Disconnect).

5.3. MESSAGE THROTTLING

The throttling mechanism protects Gateway against overloading with an excessive number of input messages.

There are 2 limits set in Gateway:

1. business throttling - after exceeding the business limit level, consecutive messages are rejected using Reject messages with the reason set to MaxThroughputExceeded,

2. technical throttling - after exceeding the technical limit level, the connection is closed by sending a ConnectionClose message with the reason set to AntiFloodingThresholdExceeded and then the socket is closed.

Limit calculations take into account all messages. However, Logout and OrderCancel messages will not be rejected.

The MassQuote messages are treated differently. When calculating limits for MassQuote, each side of each quotation is counted as a separate message. For example, a MassQuote message containing 8 quotations is counted as 16 messages.

The above limits are calculated over a period of time configurable in Gateway (i.e., measurement period).

The throttling parameters are configurable at Gateway service as follows:

- The business limit level is configured for each connection and depends on the level of service provision agreed upon between the System Operator and the exchange member. The measurement period - last 5 seconds.
- The technical limit level is configured as an internal gateway parameter and is set to 5000 messages per second. The measurement period - last 100 milliseconds.

In addition, after exceeding the technical limit level, it is not possible to log into System for 10 seconds.

Note: the measurement period means the period of time during which messages are counted and the limits which have possibly been exceeded are determined.

5.4. CONNECTIVITY POLICY

Connectivity policy is described in the document **GPW WATS 6.01 CONNECTIVITY**.

6. SESSION LAYER

6.1. ESTABLISHING A SESSION

Starting a session between the Native Trading Client Application and Gateway begins with establishing a TCP/IP connection and sending the correct Login message. The message should contain the connection parameters presented in the Connectivity chapter. In particular, the Login message should contain a token and a connection identifier. Both of the above parameters are provided by System Operator.

When successfully logging in, the client receives a LoginResponse message with the result field equals "OK". When there is a business level error, a reason is given in the result field, e.g. invalid token.

When there are protocol errors or it has not been possible to establish a session, Gateway sends the ConnectionClose message which contains the reason and then the TCP/IP connection would be closed.

Each user can only be logged in once with the same credentials. When the already logged user tries login again during another established session, Gateway sends the LoginResponse message with the result field set to "AlreadyLoggedIn".

If Participant exceeds the number of login attempts, Gateway blocks the account and the LoginResponse message is set to "AccountLocked". The limit of failed login attempts is set to 5.

The maximum limit of the login attempts with Gateway is set up as once every 3 seconds. In the situation of a failed login attempt, the exchange member is advised to contact the GPW service desk for further assistance.

6.2. MAINTAINING A SESSION

6.2.1. SEQUENCE NUMBER

Sequence numbers are only assigned to business messages within a trading day. Each trading day is associated with a session id available after logging in (see. a LoginResponse message and the sessionId field) or in market data (see. StartOfTechnicalSession and EndOfTechnicalSession messages). The sequence numbers start with 1 and are incremented by 1 in each sent business message. Session messages have a sequence number set to 0.

The following is a list of session message types (i.e. messages with a sequence number of zero):

- Login
- LoginResponse
- Logout
- LogoutResponse
- ConnectionClose
- Heartbeat
- Reject.

Message types other than those listed above belong to the business message set and require setting a sequence number.

6.2.2. HEARTBEATS

The heartbeat message used by Gateway and Client Application to provide information about the communication line during time gaps of inactivity and to verify that interfaces at each end are accessible. The length of heartbeat interval is 4 seconds. The heartbeat send time (i.e. interval) is counted from the time of the last sent message.

The Logout procedure and closing the TCP/IP connection is done once the server noticed an inactivity for 3 heartbeat intervals.

6.2.3. REJECTIONS

Client Application message is rejected at the session level by sending a "Reject" message. The reason for rejecting is presented in the "reason" field and includes:

- exceeding throughput limits (MaxThroughputExceeded),
- sending the wrong type of a message (InvalidMsgType),
- wrong expire field format (InvalidExpireTimePrecision).
- invalid settlement date in trade capture report (InvalidSettlementDate).

In the case of communication protocol errors, instead of rejecting the message, the connection is terminated (see next section).

6.3. TERMINATING A CONNECTION

Client Application should terminate all used connections at the end of trading day by sending a Logout message. Then Gateway service closes TCP/IP connection.

If Client Application does not terminate a connection, Gateway does it automatically once it shuts down (a ConnectionClose message is sent).

During the trading day, in extraordinary cases, Gateway may initiate the termination of connection by sending the ConnectionClose message.

7. MESSAGES

Name	Type ID	Length	Description
ConnectionClose	ConnectionClose12	17	The ConnectionClose message confirms the termination of a session through the trading port service.
Heartbeat	Heartbeat13	16	Heartbeat message
InitiateState (an initial message version)	26	21	Initiate state change to Regular or BuyOnly.
InitiateStateResponse (an initial message version)	27	17	The response to the attempt at state change.
Login	Login2	36	The login message authenticates a user establishing a connection to the trading port service. The login message must be the first message sent by the client application to request the initiation of a trading port session.
LoginResponse	LoginResponse3	27	The login response message. The result field describes the login status, indicating whether the login was successful or not (i.e. successful login).
Logout	Logout11	16	The logout message is sent from the client application to terminate the communication session with the trading port.
LogoutResponse	LogoutResponse14	16	The logout response message confirms the client logout message.
MassQuote	MassQuote24	Message1173	Mass Quote
MassQuoteResponse	MassQuoteResponse25	718	The response to a MassQuote message.
OrderAdd	OrderAdd4	Message139	Message used to add new orders to the system.
OrderAddResponse	OrderAddResponse5	36	The message is a response to an OrderAdd message and includes the order execution status.
OrderCancel	OrderCancel6	24	Message used to cancel the previously submitted order.
OrderCancelResponse	OrderCancel	27	The message is a response to an order cancel request and contains information about its execution, in particular whether the order to cancel was found or not.

	OrderMassCancel	29	30	Message used to cancel multiple existing orders.
	OrderMassCancelResponse	30	44	Response to message used to cancel multiple existing orders.
	OrderModify	OrderModifyResponse	Message 64	Message used to modify the submitted order.
	OrderModifyResponse	OrderModifyResponse	28	The response message for an OrderModify.
	Reject	Reject	21	The reject message is sent by the trading port service when receiving an erroneous message that cannot be further processed.
	RequestForExecution (an initial message version)	28	20	The information for the MM that one of the quotes has been crossed.
	RiskLimitBreach	RiskLimitBreach	53	Message to inform about violation of risk limit.
	Trade	Trade	52	The message used to report trades between counterparties (i.e. generated when two or more orders are matched).
	TradeBust	TradeBust	20	Message used to cancel previously accepted Trade Capture Report.
	TradeCaptureReportDual	TradeCaptureReportDual	210	Trade Capture Report - dual sided.
	TradeCaptureReportResponse	TradeCaptureReportResponse	55	The message is a response to an Trade Capture Report message, containing the state of TCR execution.
	TradeCaptureReportSingle	TradeCaptureReportSingle	175	Trade Capture Report - single side.

7.1.1. HEADER:

Name	Type	Kind	Length	Description
length	MsgLength	Alias (u16)	2	Total length of the message.
msgType	MsgType	Enum	2	Type of the message (e.g. Login).
seqNum	SeqNum	Alias (u32)	4	Sequence number of the message added by the sender.
timestamp	Timestamp	Alias (u64)	8	Sending time.

7.1.2. CONNECTIONCLOSE:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
reason	ConnectionCloseReason	Enum	1	Connection close reason.		
				Name	Value	Description
				ProtocolError	1	Invalid message or frame length.
				InvalidSeqNum	2	Message came with an incorrect sequence number.
				EndOfDay	3	The session day has come to an end.
				SyncFail	4	The synchronization of messages has failed.
				AntiFloodingThresholdExceeded	5	The second level of the throttling limit has been exceeded.

7.1.3. HEARTBEAT:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.

7.1.4. INITIATESTATE

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Message header.		
instrumentId	ElementId	Alias (u32)	4	Instrument ID		
state	HybridMarketState	Enum	1	The state the mm wants the hybrid market to switch to.		
				Name	Value	Description
				Regular	1	Regular trading.
				BuyOnly	2	Buy only trading.

7.1.5. INITIALSTATERESPONSE

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Message header.		
result	InitiateStateResult	Enum	1	The result of an attempt at changing phase.		
				Name	Value	Description
				Success	1	

Name	Type	Kind	Length	Description	
				Failure	2

7.1.4.7.1.6. LOGIN:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
version	MsgVersion	Alias (u16)	2	Indicates the version of the protocol in which the message is defined.
token	Token	Array (AnsiChar)	8	The security data required for authentication, which is a token received by the exchange member during the registration process.
connectionId	ConnectionId	Alias (u16)	2	ID of the connection.
nextExpectedSeqNum	SeqNum	Alias (u32)	4	Next expected message sequence number value to be received.
lastSentSeqNum	SeqNum	Alias (u32)	4	Last sent sequence number.

7.1.5.7.1.7. LOGINRESPONSE:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
result	LoginResult	Enum	1	Login response status code.		
				Name	Value	Description
				Ok	1	Successful login.
				NotFound	2	User not found.
				InvalidToken	3	Authorization failure.
				AlreadyLoggedIn	4	Already logged in.
				AccountLocked	5	Account locked.
				LoginNotAllowed	6	Login is currently unavailable due to reasons such as service unavailability.
				InvalidLoginParameters	7	The login parameters, such as the connection ID, are invalid.
				ThrottlingTemporaryLock	8	The login attempt failed due to exceeding the anti-flooding threshold.
Other	9	Other errors.				
nextExpectedSeqNum	SeqNum	Alias (u32)	4	Next expected message sequence number value to be received.		
lastReplaySeqNum	SeqNum	Alias (u32)	4	Last replay sequence number.		
sessionId	SessionId	Alias (u16)	2	ID of the session.		

7.1.6.7.1.8. LOGOUT:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.

7.1.7.7.1.9. LOGOUTRESPONSE:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.

7.1.8.7.1.10. MASSQUOTE

Name	Type	Kind	Length	Description																		
header	Header	Struct	16	Message header.																		
onBehalfOf	OnBehalfOf	Alias (u16)	2																			
stpId	STPID	Alias (u16)	2	An ID assigned by the client used in Self Match Prevention mechanism.																		
capacity	Capacity	Enum	1	Capacity of the party making the order (either principal or agency). <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Agency</td><td>1</td><td>Agency (mapped to AOTC).</td></tr><tr><td>Principal</td><td>2</td><td>Principal (mapped to DEAL).</td></tr><tr><td>RisklessPrincipal</td><td>3</td><td>Riskless Principal (mapped to MTCH)</td></tr></table>	Name	Value	Description	Agency	1	Agency (mapped to AOTC).	Principal	2	Principal (mapped to DEAL).	RisklessPrincipal	3	Riskless Principal (mapped to MTCH)						
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account	Account	Array (AnsiChar)	16	Account mnemonic as agreed between buy and sell sides.																		
accountType	AccountType	Enum	1	Type of account associated with the order. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Missing</td><td>1</td><td>Account is missing. Account is expected to be filled with 0x00.</td></tr><tr><td>Customer</td><td>2</td><td>Account is carried on customer side of the books.</td></tr><tr><td>House</td><td>3</td><td>House trader.</td></tr></table>	Name	Value	Description	Missing	1	Account is missing. Account is expected to be filled with 0x00.	Customer	2	Account is carried on customer side of the books.	House	3	House trader.						
Name	Value	Description																				
Missing	1	Account is missing. Account is expected to be filled with 0x00.																				
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House	3	House trader.																				
mifidFields.flags	MifidFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>None</td><td>0b0000</td><td></td></tr><tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td></td></tr><tr><td>DirectOrSponsoredAccess</td><td>0b0010</td><td></td></tr><tr><td>AlgorithmicTrade</td><td>0b0100</td><td></td></tr><tr><td>MarketMakerOrSpecialist</td><td>0b1000</td><td></td></tr></table>	Name	Value	Description	None	0b0000		LiquidityProvisionActivity	0b0001		DirectOrSponsoredAccess	0b0010		AlgorithmicTrade	0b0100		MarketMakerOrSpecialist	0b1000	
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None	0b0000																					
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mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr><tr><td>FirmOrLegalEntity</td><td>3</td><td></td></tr><tr><td>NaturalPerson</td><td>4</td><td></td></tr></table>	Name	Value	Description	NA	1		Algorithm	2		FirmOrLegalEntity	3		NaturalPerson	4				
Name	Value	Description																				
NA	1																					
Algorithm	2																					
FirmOrLegalEntity	3																					
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mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr></table>	Name	Value	Description	NA	1		Algorithm	2										
Name	Value	Description																				
NA	1																					
Algorithm	2																					

Name	Type	Kind	Length	Description				
				FirmOrLegalEntity	3			
				NaturalPerson	4			
mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.				
mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.				
				Name	Value	Description		
				NA	1			
				Algorithm	2			
				FirmOrLegalEntity	3			
				NaturalPerson	4			
memo	Memo	Array (AnsiChar)	18	Free text.				
clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code.				
clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier.				
				Name	Value	Description		
				NotApplicable	33	Not Applicable.		
				Lei	78	Legal Entity Identifier.		
				Bic	66	Business Identifier Code.		
				Custom	68	Custom clearing identifier.		
quotes.count	u8	Primitive	1	The number of quotes this message contains.				
quotes.items	QuotesArray	Array (Quote)	1080	The array of quotes. Quote:				
				Name	Type	Kind	Length	Description
				instrumentId	ElementId	Alias (u32)	4	ID of the instrument being traded.
				bid.price	Price	Alias (Number)	8	Indicates the price of the given order.
				bid.quantity	Quantity	Alias (u64)	8	Indicates the quantity of the instrument included in the order.
				ask.price	Price	Alias (Number)	8	Indicates the price of the given order.
				ask.quantity	Quantity	Alias (u64)	8	Indicates the quantity of the instrument included in the order.

7.1.9-7.1.11. MASSQUOTERESPONSE

Name	Type	Kind	Length	Description						
header	Header	Struct	16	Message header.						
massQuoteId	OrderId	Alias (u64)	8	Quote id						
responses.count	u8	Primitive	1	How many responses this message contains.						
responses.items	QuoteOrderResponsesArray	Array (QuoteOrderResponse)	690	The array of responses.						
				QuoteOrderResponse:						
				Name	Data type	Kind	Length	Description		
				instrumentId	ElementId	Alias (u32)	4	ID of the instrument being traded.		
				bidOrderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.		
				askOrderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.		
				status	OrderStatus	Enum	1	Status of the given order.		
								Name	Value	Description
								Ack	1	Order acknowledged by system.
								Cancelled	2	Order canceled.
Rejected	3	Order rejected.								
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order.						
				Name	Value	Description				
				NA	1	Not applicable.				
				ExchangeClosed	2	Exchange closed.				
				InvalidPriceIncrement	18	Invalid price increment.				
Other	99	Other.								

Name	Type	Kind	Length	Description						
								UnknownInstrumentId	100	Unknown instrument.
								InstrumentPhaseNoTrading	106	Trading is not available for the instrument in its current phase.
								UnknownOrder	1001	The order id is unrecognized.
								InvalidExecutionTrader	1005	Invalid execution trader.
								InvalidDecisionMaker	1006	Invalid decision maker.
								InvalidClientId	1007	Invalid client.
								InvalidPartyRoleQualifierForClientId	1008	Invalid Party Role Qualifier for Client Id Party group
								InvalidPartyRoleQualifierForExecutingTrader	1009	Invalid Party Role Qualifier for Executing Trader Party group
								InvalidPartyRoleQualifierForInvestmentDecisionMaker	1010	Invalid Party Role Qualifier for Investment Decision Maker Party group
								WrongDisplayQtyValue	1013	The display quantity (displayQty) cannot

Name	Type	Kind	Length	Description						
									exceed the order quantity.	
								InvalidDisplayQty	1014	The Display quantity (displayQty) not allowed for specified order type - only for Iceberg.
								WrongIcebergOrderValue	1015	The value of the iceberg order is less than the required.
								OrderQuantityMustBeGreaterThanMinimumQuantity	1025	The order quantity must be greater than the minimum quantity.
								OrderQuantityMustBeLowerThanMaximumQuantity	1026	The order quantity must be lower than the maximum quantity.
								OrderPriceMustBeGreaterThanMinimumPrice	1027	The order price must be greater than minimum price.
								OrderPriceMustBeLowerThanMaximumPrice	1028	The order price must be <u>greater</u> <u>lower</u> than maximum price.

Name	Type	Kind	Length	Description						
								OrderPriceMustBeNonzero	1029	The order price must be non-zero.
								OrderValueMustBeGreaterThanMinimumValue	1030	The order value must be greater than minimum value.
								OrderValueMustBeLowerThanMaximumValue	1031	The order value must be lower than maximum value.
								MarketOrderNotAllowedInAuctionInvalidOrdTypeForSelectedMarketModel	1032	Market orders are not permitted during the auction. Invalid OrdType for selected market model
								LeavesQuantityMustBeGreaterThanZeroAfterModification	1034	The remaining quantity (LeavesQty) must be greater than 0 after the modification.
								PriceNotAllowed	1035	The price is not allowed for a market order. In the binary protocol, the price field should have a value of 0.

Name	Type	Kind	Length	Description				
							InvalidMarketOrderTimeInForce <u>InvalidTimeInForceForOrderType</u>	1039A market order can only have a TimeInForce value of 3 (IOC – Immediate or Cancel) or 4 (FOK – Fill or Kill). <u>InvalidTimeInForce for specified OrderType</u>
							InvalidAuctionTimeInForce <u>InvalidTimeInForceForCurrentMarketPhase</u>	1040The TimeInForce value of 3 (IOC – Immediate or Cancel) or 4 (FOK – Fill or Kill) is not allowed during the auction. <u>InvalidTimeInForce for current market phase</u>
							InvalidIcebergAndStopOrderTimeInForce <u>InvalidTimeInForceForSelectedMarketModel</u>	1041Iceberg and Stop orders can only have a TimeInForce value of DAY, GTC (Good Till Cancel), GTD (Good Till Date), or GTT (Good Till Time). <u>Invalid</u>

Name	Type	Kind	Length	Description				
								TimeInForce for selected market model
								<u>ExpireTimeCannotBeModified</u> 1043 ExpireTime (126) cannot be modified.
								<u>ObsoleteExpireDate</u> 1044 ExpireDate (432) not allowed for selected TimeInForce (59).
								ExpireDateInPast 1045 The expiration date is earlier than the current date.
								<u>ObsoleteExpireTime</u> 1046 ExpireTime (126) not allowed for selected TimeInForce (59).
								ExpireTimeInPast 1047 The expiration time is earlier than the current time.
								AmbiguousExpire 1048 Please provide either a time or a date, but not both.
								ExpireDateExceedsLimit 1049 The expiration date exceeds the allowed limit.

Name	Type	Kind	Length	Description						
								PriceBelowLowCollar	1037	The validation for collars has failed. The price is too low.
								PriceAboveHighCollar	1038	The validation for collars has failed. The price is too high.
								TriggerPriceNotAllowed	1063	The trigger price not allowed for a specified order type.
								TriggerPriceNotHigherThanLTP	1064	The trigger price is not higher than the last trade price.
								TriggerPriceNotLowerThanLTP	1065	The trigger price is not lower than the last trade price.
								TriggerPriceLowerThanPrice	1066	The trigger price is lower than the limit price.
								TriggerPriceHigherThanPrice	1067	The trigger price is higher than the limit price.
								TriggerPriceModifiedForActivatedOrder	1068	The trigger price has been modified for the

Name	Type	Kind	Length	Description				
								activated order.
							<u>TriggerPriceMustBeGreaterThanZero</u>	1069 TriggerPrice (1102) must be greater than 0.
							InvalidPartyIdForClientId	1070 Invalid PartyID (448) for Client ID
							InvalidPartyIdForExecutingTrader	1071 Invalid PartyID (448) for Executing Trader
							InvalidPartyIdForInvestmentDecisionMaker	1072 Invalid PartyID (448) for Investment Decision Maker
							InvalidPartyRoleQualifierForPartyId	1075 Invalid PartyRoleQualifier (2376) for PartyID (448)
							<u>UnknownInstrument</u>	1201 SecurityID (48) not recognized by the system.
							<u>InstrumentClosed</u>	1203 Instrument closed for trading.
							InvalidBidAskSpread	1208 OfferPx (133) must be greater than BidPx (132).
							<u>BuyOrderNotAllowed</u>	1301 Buy orders from participants are not allowed in

Name	Type	Kind	Length	Description				
								BuyOnly phase.
							OnlyOneSellOrderIsAllowedForIpo	1401 Only one sell order is allowed for IPO instrument
							RequestNotAllowedForBlockInstrument	2026 Request not allowed for BLOCK instrument
							RequestNotAllowedForCrossInstrument	2028 Request not allowed for CROSS instrument
							RiskLimitNotDefined	7000 The risk limit has not been defined.
							RiskMaximumOrderVolumeExceeded	7001 The maximum order volume for the risk limit has been exceeded.
							RiskMaximumOrderValueExceeded	7002 The maximum order value for the risk limit has been exceeded.
							RiskOrderPriceCollarExceeded	7003 The order price has exceeded the risk limit.

Name	Type	Kind	Length	Description																																																			
status	MassQuoteStatus	Enum	1	Status of the given mass quote order.																																																			
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Accepted</td><td>1</td><td>Mass quote acknowledged by system.</td></tr><tr><td>Rejected</td><td>2</td><td>Mass quote rejected.</td></tr></table>	Name	Value	Description	Accepted	1	Mass quote acknowledged by system.	Rejected	2	Mass quote rejected.																																										
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				Accepted	1	Mass quote acknowledged by system.																																																	
Rejected	2	Mass quote rejected.																																																					
reason	MassQuoteRejectionReason	Enum	2	Reason for rejecting the given mass quote order.																																																			
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td>Not applicable.</td></tr><tr><td>ExchangeClosed</td><td>2</td><td>Exchange closed.</td></tr><tr><td>Other</td><td>99</td><td>Other.</td></tr><tr><td>InvalidExecutionTrader</td><td>1005</td><td>Invalid execution trader.</td></tr><tr><td>InvalidDecisionMaker</td><td>1006</td><td>Invalid decision maker.</td></tr><tr><td>InvalidClientId</td><td>1007</td><td>Invalid client.</td></tr><tr><td>InvalidPartyRoleQualifierForClientId</td><td>1008</td><td>Invalid Party Role Qualifier for Client Id Party group</td></tr><tr><td>InvalidPartyRoleQualifierForExecutingTrader</td><td>1009</td><td>Invalid Party Role Qualifier for Executing Trader Party group</td></tr><tr><td>InvalidPartyRoleQualifierForInvestmentDecisionMaker</td><td>1010</td><td>Invalid Party Role Qualifier for Investment Decision Maker Party group</td></tr><tr><td>InvalidPartyIdForClientId</td><td>1070</td><td>Invalid PartyID (448) for Client ID</td></tr><tr><td>InvalidPartyIdForExecutingTrader</td><td>1071</td><td>Invalid PartyID (448) for Executing Trader</td></tr><tr><td>InvalidPartyIdForInvestmentDecisionMaker</td><td>1072</td><td>Invalid PartyID (448) for Investment Decision Maker</td></tr><tr><td>InvalidPartyRoleQualifierForPartyId</td><td>1075</td><td>Invalid PartyRoleQualifier (2376) for PartyID (448)</td></tr><tr><td>DuplicateInstrument</td><td>1202</td><td>Multiple quotes for the same instrument within Mass Quote message.</td></tr><tr><td>InvalidQuotesCount</td><td>1204</td><td>Invalid quotes count value.</td></tr><tr><td>ForbiddenOrderCapacityValue</td><td>1210</td><td>Forbidden OrderCapacity (528) value for Mass Quote.</td></tr></table>	Name	Value	Description	NA	1	Not applicable.	ExchangeClosed	2	Exchange closed.	Other	99	Other.	InvalidExecutionTrader	1005	Invalid execution trader.	InvalidDecisionMaker	1006	Invalid decision maker.	InvalidClientId	1007	Invalid client.	InvalidPartyRoleQualifierForClientId	1008	Invalid Party Role Qualifier for Client Id Party group	InvalidPartyRoleQualifierForExecutingTrader	1009	Invalid Party Role Qualifier for Executing Trader Party group	InvalidPartyRoleQualifierForInvestmentDecisionMaker	1010	Invalid Party Role Qualifier for Investment Decision Maker Party group	InvalidPartyIdForClientId	1070	Invalid PartyID (448) for Client ID	InvalidPartyIdForExecutingTrader	1071	Invalid PartyID (448) for Executing Trader	InvalidPartyIdForInvestmentDecisionMaker	1072	Invalid PartyID (448) for Investment Decision Maker	InvalidPartyRoleQualifierForPartyId	1075	Invalid PartyRoleQualifier (2376) for PartyID (448)	DuplicateInstrument	1202	Multiple quotes for the same instrument within Mass Quote message.	InvalidQuotesCount	1204	Invalid quotes count value.	ForbiddenOrderCapacityValue	1210	Forbidden OrderCapacity (528) value for Mass Quote.
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7.1.10-7.1.12. ORDERADD:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
onBehalfOf	OnBehalfOf	Alias (u16)	2	ID of the client party on behalf which order is submitted. The default value is 0.		
stpld	STPLd	Alias (u16)	2	ID assigned by the client used in the Self Trade Prevention mechanism.		
instrumentId	ElementId	Alias (u32)	4	ID of the instrument being traded.		
orderType	OrderType	Enum	1	Indicates the order type.		
				Name	Value	Description
				Limit	1	Limit order type.
				Market	2	Market order type.
				MarketToLimit	3	Market to limit order type.
				Iceberg	4	Iceberg order type.

Name	Type	Kind	Length	Description																											
				<table><tr><td>StopLimit</td><td>5</td><td>Stop limit order type.</td></tr><tr><td>StopLoss</td><td>6</td><td>Stop loss order type.</td></tr></table>	StopLimit	5	Stop limit order type.	StopLoss	6	Stop loss order type.																					
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StopLoss	6	Stop loss order type.																													
timeInForce	TimeInForce	Enum	1	Indicates the order's time in force (e.g. GTC). <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Day</td><td>1</td><td>A day order is valid until the end of the trading day.</td></tr><tr><td>GTC</td><td>2</td><td>A GTC order is good till canceled.</td></tr><tr><td>IOC</td><td>3</td><td>An Immediate or Cancel order must be filled immediately or canceled.</td></tr><tr><td>FOK</td><td>4</td><td>A Fill or Kill order must be filled or canceled.</td></tr><tr><td>VFA</td><td>5</td><td>A market or limit-price order to be executed at the opening of the stock or not at all; all or part of any order not executed at the opening is treated as canceled.</td></tr><tr><td>GTD</td><td>6</td><td>A Good Till Date order must be filled before timestamp provided in 'Expire' field or canceled.</td></tr><tr><td>VFC</td><td>7</td><td>Indicated price is to be around the closing price, however, not held to the closing price.</td></tr><tr><td>GTT</td><td>8</td><td>A Good Till Time order must be filled before timestamp provided in 'Expire' field or canceled within the day of submission.</td></tr></table>	Name	Value	Description	Day	1	A day order is valid until the end of the trading day.	GTC	2	A GTC order is good till canceled.	IOC	3	An Immediate or Cancel order must be filled immediately or canceled.	FOK	4	A Fill or Kill order must be filled or canceled.	VFA	5	A market or limit-price order to be executed at the opening of the stock or not at all; all or part of any order not executed at the opening is treated as canceled.	GTD	6	A Good Till Date order must be filled before timestamp provided in 'Expire' field or canceled.	VFC	7	Indicated price is to be around the closing price, however, not held to the closing price.	GTT	8	A Good Till Time order must be filled before timestamp provided in 'Expire' field or canceled within the day of submission.
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side	OrderSide	Enum	1	Indicates the order's side (buy or sell). <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Buy</td><td>1</td><td>Indicates a buy-side order.</td></tr><tr><td>Sell</td><td>2</td><td>Indicates a sell-side order.</td></tr></table>	Name	Value	Description	Buy	1	Indicates a buy-side order.	Sell	2	Indicates a sell-side order.																		
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price	Price	Alias (Number)	8	Indicates the price of the order.																											
triggerPrice	Price	Alias (Number)	8	Indicates the trigger price (Last Trade Price - LTP) after which the order should be added to the order book.																											
quantity	Quantity	Alias (u64)	8	Indicates the quantity of the instrument included in the order.																											
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mifidFields.flags	MifidFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive.																											

Name	Type	Kind	Length	Description																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>None</td><td>0b0000</td><td></td></tr><tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td></td></tr><tr><td>DirectOrSponsoredAccess</td><td>0b0010</td><td></td></tr><tr><td>AlgorithmicTrade</td><td>0b0100</td><td></td></tr><tr><td>MarketMakerOrSpecialist</td><td>0b1000</td><td></td></tr></table>	Name	Value	Description	None	0b0000		LiquidityProvisionActivity	0b0001		DirectOrSponsoredAccess	0b0010		AlgorithmicTrade	0b0100		MarketMakerOrSpecialist	0b1000	
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mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr><tr><td>FirmOrLegalEntity</td><td>3</td><td></td></tr><tr><td>NaturalPerson</td><td>4</td><td></td></tr></table>	Name	Value	Description	NA	1		Algorithm	2		FirmOrLegalEntity	3		NaturalPerson	4				
				Name	Value	Description																
				NA	1																	
				Algorithm	2																	
FirmOrLegalEntity	3																					
NaturalPerson	4																					
mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr><tr><td>FirmOrLegalEntity</td><td>3</td><td></td></tr><tr><td>NaturalPerson</td><td>4</td><td></td></tr></table>	Name	Value	Description	NA	1		Algorithm	2		FirmOrLegalEntity	3		NaturalPerson	4				
				Name	Value	Description																
				NA	1																	
				Algorithm	2																	
FirmOrLegalEntity	3																					
NaturalPerson	4																					
mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr><tr><td>FirmOrLegalEntity</td><td>3</td><td></td></tr><tr><td>NaturalPerson</td><td>4</td><td></td></tr></table>	Name	Value	Description	NA	1		Algorithm	2		FirmOrLegalEntity	3		NaturalPerson	4				
				Name	Value	Description																
				NA	1																	
				Algorithm	2																	
FirmOrLegalEntity	3																					
NaturalPerson	4																					
expire	Timestamp	Alias (u64)	8	Expiration time indicating the validity of the order - relevant only when TimeInForce is set to GTD (Good Till Date) or GTT (Good Till Time).																		
memo	Memo	Array (AnsiChar)	18	Free text.																		
clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code.																		
clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>33</td><td>Not Applicable.</td></tr><tr><td>Lei</td><td>78</td><td>Legal Entity Identifier.</td></tr><tr><td>Bic</td><td>66</td><td>Business Identifier Code.</td></tr><tr><td>Custom</td><td>68</td><td>Custom clearing identifier.</td></tr></table>	Name	Value	Description	NotApplicable	33	Not Applicable.	Lei	78	Legal Entity Identifier.	Bic	66	Business Identifier Code.	Custom	68	Custom clearing identifier.			
				Name	Value	Description																
				NotApplicable	33	Not Applicable.																
				Lei	78	Legal Entity Identifier.																
Bic	66	Business Identifier Code.																				
Custom	68	Custom clearing identifier.																				

~~7.1.11.~~7.1.13. ORDERADDRESSRESPONSE:

Name	Type	Kind	Length	Description			
header	Header	Struct	16	Header.			
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.			
publicOrderId	PublicOrderId	Alias (u64)	8	Unique for single trading day order identifier assigned by the trading system and shared publicly in market data (public information).			
status	OrderStatus	Enum	1	Status of the given order.			
				Name	Value	Description	
				Ack	1	Order acknowledged by system.	
				Cancelled	2	Order canceled.	
				Rejected	3	Order rejected.	
				Filled	4	Order filled.	
Modified	5	Order modified.					
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order.			
				Name		Value	Description
				NA		1	Not applicable.
				ExchangeClosed		2	Exchange closed.
				InvalidPriceIncrement		18	Invalid price increment.
				Other		99	Other.
				UnknownInstrumentId		100	Unknown instrument.
				InstrumentPhaseNoTrading		106	Trading is not available for the instrument in its current phase.
				UnknownOrder		1001	The order id is unrecognized.
				InvalidExecutionTrader		1005	Invalid execution trader.
				InvalidDecisionMaker		1006	Invalid decision maker.
				InvalidClientId		1007	Invalid client.
				InvalidPartyRoleQualifierForClientId		1008	Invalid Party Role Qualifier for Client Id Party group
				InvalidPartyRoleQualifierForExecutingTrader		1009	Invalid Party Role Qualifier for Executing Trader Party group
InvalidPartyRoleQualifierForInvestmentDecisionMaker		1010	Invalid Party Role Qualifier for Investment				

Name	Type	Kind	Length	Description
				Decision Maker Party group
				WrongDisplayQtyValue 1013 The display quantity (displayQty) cannot exceed the order quantity.
				InvalidDisplayQty 1014 The Display quantity (displayQty) not allowed for specified order type - only for Iceberg.
				WrongIcebergOrderValue 1015 The value of the iceberg order is less than the required.
				OrderQuantityMustBeGreaterThanMinimumQuantity 1025 The order quantity must be greater than the minimum quantity.
				OrderQuantityMustBeLowerThanMaximumQuantity 1026 The order quantity must be lower than the maximum quantity.
				OrderPriceMustBeGreaterThanMinimumPrice 1027 The order price must be greater than minimum price.
				OrderPriceMustBeLowerThanMaximumPrice 1028 The order price must be greater lower than maximum price.
				OrderPriceMustBeNonzero 1029 The order price must be non-zero.
				OrderValueMustBeGreaterThanMinimumValue 1030 The order value must be greater than minimum value.
				OrderValueMustBeLowerThanMaximumValue 1031 The order value must be lower

Name	Type	Kind	Length	Description		
						than maximum value.
				MarketOrderNotAllowedInAuction <u>InvalidOrdTypeForSelectedMarketModel</u>	1032	Market orders are not permitted during the auction. <u>Invalid OrdType for selected market model</u>
				LeavesQuantityMustBeGreaterThanZeroAfterModification	1034	The remaining quantity (LeavesQty) must be greater than 0 after the modification.
				PriceNotAllowed	1035	The price is not allowed for a market order. In the binary protocol, the price field should have a value of 0.
				InvalidMarketOrderTimeInForce <u>InvalidTimeInForceForOrderType</u>	1039	A market order can only have a TimeInForce value of 3 (IOC – Immediate or Cancel) or 4 (FOK – Fill or Kill). <u>Invalid TimeInForce for specified OrderType</u>
				InvalidAuctionTimeInForce <u>InvalidTimeInForceForCurrentMarketPhase</u>	1040	The TimeInForce value of 3 (IOC – Immediate or Cancel) or 4 (FOK – Fill or Kill) is not allowed during the auction. <u>Invalid TimeInForce for current market phase</u>
				InvalidIcebergAndStopOrderTimeInForce <u>InvalidTimeInForceForSelectedMarketModel</u>	1041	Iceberg and Stop orders can only

Name	Type	Kind	Length	Description
				have a TimeInForce value of DAY, GTC (Good Till Cancel), GTD (Good Till Date), or GTT (Good Till Time). Invalid TimeInForce for selected market model
				<u>ExpireTimeCannotBeModified</u> 1043 ExpireTime (126) cannot be modified.
				<u>ObsoleteExpireDate</u> 1044 ExpireDate (432) not allowed for selected TimeInForce (59).
				ExpireDateInPast 1045 The expiration date is earlier than the current date.
				<u>ObsoleteExpireTime</u> 1046 ExpireTime (126) not allowed for selected TimeInForce (59).
				ExpireTimeInPast 1047 The expiration time is earlier than the current time.
				AmbiguousExpire 1048 Please provide either a time or a date, but not both.
				ExpireDateExceedsLimit 1049 The expiration date exceeds the allowed limit.
				PriceBelowLowCollar 1037 The validation for collars has failed. The price is too low.
				PriceAboveHighCollar 1038 The validation for collars has failed. The price is too high.

Name	Type	Kind	Length	Description		
				TriggerPriceNotAllowed	1063	The trigger price not allowed for a specified order type.
				TriggerPriceNotHigherThanLTP	1064	The trigger price is not higher than the last trade price.
				TriggerPriceNotLowerThanLTP	1065	The trigger price is not lower than the last trade price.
				TriggerPriceLowerThanPrice	1066	The trigger price is lower than the limit price.
				TriggerPriceHigherThanPrice	1067	The trigger price is higher than the limit price.
				TriggerPriceModifiedForActivatedOrder	1068	The trigger price has been modified for the activated order.
				InvalidPartyIdForClientId	1070	Invalid PartyID (448) for Client ID
				InvalidPartyIdForExecutingTrader	1071	Invalid PartyID (448) for Executing Trader
				InvalidPartyIdForInvestmentDecisionMaker	1072	Invalid PartyID (448) for Investment Decision Maker
				InvalidPartyRoleQualifierForPartyId	1075	Invalid PartyRoleQualifier (2376) for PartyID (448)
				<u>UnknownInstrument</u>	<u>1201</u>	<u>SecurityID (48) not recognized by the system.</u>
				<u>InstrumentClosed</u>	<u>1203</u>	<u>Instrument closed for trading.</u>
				InvalidBidAskSpread	1208	OfferPx (133) must be greater than BidPx (132).

Name	Type	Kind	Length	Description		
				BuyOrderNotAllowed	1301	Buy orders from participants are not allowed in BuyOnly phase.
				OnlyOneSellOrderIsAllowedForIpo	1401	Only one sell order is allowed for IPO instrument
				RequestNotAllowedForBlockInstrument	2026	Request not allowed for BLOCK instrument
				RequestNotAllowedForCrossInstrument	2028	Request not allowed for CROSS instrument
				RiskLimitNotDefined	7000	The risk limit has not been defined.
				RiskMaximumOrderVolumeExceeded	7001	The maximum order volume for the risk limit has been exceeded.
				RiskMaximumOrderValueExceeded	7002	The maximum order value for the risk limit has been exceeded.
				RiskOrderPriceCollarExceeded	7003	The order price has exceeded the risk limit.
source	OrderSource	Enum	1	Response message sources can include trading ports, order cancellation mechanisms, and more.		
				Name	Value	Description
				Submitted	1	Order message coming from the trading port.
				Cod	2	Order cancelled by the canceller on client disconnect.
				Expired	3	Order cancelled by the canceller due to expiration.
				StopOrder	4	Order message coming from the stop order subsystem, after the stop order got triggered.
				Suspended	5	Order cancelled by canceller due to instrument suspension.
				Reinstated	6	Order reinstated
				IcebergRefill	7	Iceberg order refill
				OrderBookRebuild	8	Order book rebuild after auction
				Activated	9	VFA/C order got activated, from now on it will take part in matching
				Stp	10	Order cancelled due to self-trade prevention
				CorporateAction	11	Order cancelled due to submitted Corporate Action.
MassCancel	12	Order cancelled due to mass cancel				

~~7.1.12~~7.1.14. ORDERCANCEL:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.

~~7.1.13.~~7.1.15. ORDERCANCELRESPONSE:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.		
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order.		
				Name	Value	Description
				NA	1	Not applicable.
				ExchangeClosed	2	Exchange closed.
				InvalidPriceIncrement	18	Invalid price increment.
				Other	99	Other.
				UnknownInstrumentId	100	Unknown instrument.
				InstrumentPhaseNoTrading	106	Trading is not available for the instrument in its current phase.
				UnknownOrder	1001	The order id is unrecognized.
				InvalidExecutionTrader	1005	Invalid execution trader.
				InvalidDecisionMaker	1006	Invalid decision maker.
				InvalidClientId	1007	Invalid client.
				InvalidPartyRoleQualifierForClientId	1008	Invalid Party Role Qualifier for Client Id Party group
				InvalidPartyRoleQualifierForExecutingTrader	1009	Invalid Party Role Qualifier for Executing Trader Party group
				InvalidPartyRoleQualifierForInvestmentDecisionMaker	1010	Invalid Party Role Qualifier for Investment Decision Maker Party group
				WrongDisplayQtyValue	1013	The display quantity (displayQty) cannot exceed the order quantity.
				InvalidDisplayQty	1014	The Display quantity (displayQty) not allowed

Name	Type	Kind	Length	Description
				for specified order type - only for Iceberg.
WrongIcebergOrderValue				1015 The value of the iceberg order is less than the required.
OrderQuantityMustBeGreaterThanMinimumQuantity				1025 The order quantity must be greater than the minimum quantity.
OrderQuantityMustBeLowerThanMaximumQuantity				1026 The order quantity must be lower than the maximum quantity.
OrderPriceMustBeGreaterThanMinimumPrice				1027 The order price must be greater than minimum price.
OrderPriceMustBeLowerThanMaximumPrice				1028 The order price must be greater lower than maximum price.
OrderPriceMustBeNonzero				1029 The order price must be non-zero.
OrderValueMustBeGreaterThanMinimumValue				1030 The order value must be greater than minimum value.
OrderValueMustBeLowerThanMaximumValue				1031 The order value must be lower than maximum value.
MarketOrderNotAllowedInAuctionInvalidOrdTypeForSelectedMarketModel				1032 Market orders are not permitted during the auction. Invalid OrdType for selected market model
LeavesQuantityMustBeGreaterThanZeroAfterModification				1034 The remaining quantity (LeavesQty) must be greater than 0 after the modification.
PriceNotAllowed				1035 The price is not allowed for a market order. In the binary protocol, the price field should have a value of 0.
InvalidMarketOrderTimeInForceInvalidTimeInForceForOrderType				1039 A market order can only have a TimeInForce value of 3 (IOC - Immediate or Cancel) or 4 (FOK - Fill or

Name	Type	Kind	Length	Description
				Kill Invalid TimeInForce for specified OrderType
				InvalidAuctionTimeInForce InvalidTimeInForceForCurrentMarketPhase
				1040 The TimeInForce value of 3 (IOC – Immediate or Cancel) or 4 (FOK – Fill or Kill) is not allowed during the auction.Invalid TimeInForce for current market phase
				InvalidIcebergAndStopOrderTimeInForce InvalidTimeInForceForSelectedMarketModel
				1041 Iceberg and Stop orders can only have a TimeInForce value of DAY, GTC (Good Till Cancel), GTD (Good Till Date), or GTT (Good Till Time).Invalid TimeInForce for selected market model
				<u>ExpireTimeCannotBeModified</u>
				1043 ExpireTime (126) cannot be modified.
				<u>ObsoleteExpireDate</u>
				1044 ExpireDate (432) not allowed for selected TimeInForce (59).
				ExpireDateInPast
				1045 The expiration date is earlier than the current date.
				<u>ObsoleteExpireTime</u>
				1046 ExpireTime (126) not allowed for selected TimeInForce (59).
				ExpireTimeInPast
				1047 The expiration time is earlier than the current time.
				AmbiguousExpire
				1048 Please provide either a time or a date, but not both.
				ExpireDateExceedsLimit
				1049 The expiration date exceeds the allowed limit.
				PriceBelowLowCollar
				1037 The validation for collars has failed. The price is too low.
				PriceAboveHighCollar
				1038 The validation for collars has failed. The price is too high.

Name	Type	Kind	Length	Description		
				TriggerPriceNotAllowed	1063	The trigger price not allowed for a specified order type.
				TriggerPriceNotHigherThanLTP	1064	The trigger price is not higher than the last trade price.
				TriggerPriceNotLowerThanLTP	1065	The trigger price is not lower than the last trade price.
				TriggerPriceLowerThanPrice	1066	The trigger price is lower than the limit price.
				TriggerPriceHigherThanPrice	1067	The trigger price is higher than the limit price.
				TriggerPriceMustBeGreaterThanZero	1069	TriggerPrice (1102) must be greater than 0.
				TriggerPriceModifiedForActivatedOrder	1068	The trigger price has been modified for the activated order.
				InvalidPartyIdForClientId	1070	Invalid PartyID (448) for Client ID
				InvalidPartyIdForExecutingTrader	1071	Invalid PartyID (448) for Executing Trader
				InvalidPartyIdForInvestmentDecisionMaker	1072	Invalid PartyID (448) for Investment Decision Maker
				InvalidPartyRoleQualifierForPartyId	1075	Invalid PartyRoleQualifier (2376) for PartyID (448)
				UnknownInstrument	1201	SecurityID (48) not recognized by the system.
				InstrumentClosed	1203	Instrument closed for trading.
				InvalidBidAskSpread	1208	OfferPx (133) must be greater than BidPx (132).
				BuyOrderNotAllowed	1301	Buy orders from participants are not allowed in BuyOnly phase.
				OnlyOneSellOrderIsAllowedForIpo	1401	Only one sell order is allowed for IPO instrument

Name	Type	Kind	Length	Description			
				RequestNotAllowedForBlockInstrument		2026	Request not allowed for BLOCK instrument
				RequestNotAllowedForCrossInstrument		2028	Request not allowed for CROSS instrument
				RiskLimitNotDefined		7000	The risk limit has not been defined.
				RiskMaximumOrderVolumeExceeded		7001	The maximum order volume for the risk limit has been exceeded.
				RiskMaximumOrderValueExceeded		7002	The maximum order value for the risk limit has been exceeded.
				RiskOrderPriceCollarExceeded		7003	The order price has exceeded the risk limit.
source	OrderSource	Enum	1	Response message sources can include trading ports, order cancellation mechanisms, and more.			
				Name	Value	Description	
				Submitted	1	Order message coming from the trading port.	
				Cod	2	Order cancelled by the canceller on client disconnect.	
				Expired	3	Order cancelled by the canceller due to expiration.	
				StopOrder	4	Order message coming from the stop order subsystem, after the stop order got triggered.	
				Suspended	5	Order cancelled by canceller due to instrument suspension.	
				Reinstated	6	Order reinstated	
				IcebergRefill	7	Iceberg order refill	
				OrderBookRebuild	8	Order book rebuild after auction	
				Activated	9	VFA/C order got activated, from now on it will take part in matching	
				Stp	10	Order cancelled due to self-trade prevention	
				CorporateAction	11	Order cancelled due to submitted Corporate Action.	
MassCancel	12	Order cancelled due to mass cancel					

7.1.16. ORDERMASSCANCEL

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
massCancelRequestType	MassCancelRequestType	Enum	1	Mass cancel request type.		
				Name	Value	Description
				CancelForSecurity	1	Cancel orders for a security.
				CancelAllOrders	7	Cancel all orders.
				CancelOrdersForMarketSegment	9	Cancel orders for a market segment.
targetPartyRole	TargetPartyRole	Enum	1	Target party role filter selection field.		
				Name	Value	Description
				None	0	No filtering.
				LiquidityProvider	35	Liquidity provider.
				SenderLocation	54	Sender location.

Name	Type	Kind	Length	Description
				MarketMaker 66 Market maker. ClientID 3 Client ID. ExecutingTrader 12 Executing trader. InvestmentDecisionMaker 122 Investment decision maker.
targetPartyId	u32	Primitive	4	Used to identify the party targeted for the action specified in the message.
marketSegmentId	ElementId	Alias (u32)	4	Identifies the market segment for request type CancelOrdersForMarketSegment.
instrumentId	ElementId	Alias (u32)	4	

7.1.17. ORDERMASSCANCELRESPONSE

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
totalAffectedOrders	u64	Primitive	8	Total number of orders affected by the mass cancel request.
massCancelRequestType	MassCancelRequestType	Enum	1	Mass cancel request type.
				NameValueDescription
				CancelForSecurity1Cancel orders for a security.
				CancelAllOrders7Cancel all orders.
CancelOrdersForMarketSegment9Cancel orders for a market segment.				
massCancelId	OrderId	Alias (u64)	8	Mass cancel request ID.
targetPartyRole	TargetPartyRole	Enum	1	Target party role filter selection field.
				NameValueDescription
				None0No filtering.
				LiquidityProvider35Liquidity provider.
				SenderLocation54Sender location.
				MarketMaker66Market maker.
				ClientID3Client ID.
ExecutingTrader12Executing trader.				
InvestmentDecisionMaker122Investment decision maker.				
targetPartyId	u32	Primitive	4	Used to identify the party targeted for the action specified in the message.
marketSegmentId	ElementId	Alias (u32)	4	Identifies the market segment for request type CancelOrdersForMarketSegment.
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given mass order cancel request.
				NameValueDescription
				NA101Not applicable.
				UnknownInstrumentId100Unknown instrument.
UnknownMarketSegmentId1016Unknown market segment ID.				

7.1.14: 7.1.18. ORDERMODIFY:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
price	Price	Alias (Number)	8	Indicates the price of the given order.
triggerPrice	Price	Alias (Number)	8	Indicates the trigger price (Last Trade Price - LTP) after which the order should be added to the order book.
quantity	Quantity	Alias (u64)	8	Indicates the quantity of the instrument included in the order.
displayQty	Quantity	Alias (u64)	8	Used only for iceberg order. The quantity to be displayed.
expire	Timestamp	Alias (u64)	8	Expiration time indicating the validity of the order - relevant only when TimeInForce is set to GTD (Good Till Date).

7.1.15: 7.1.19. ORDERMODIFYRESPONSE:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.		
status	OrderStatus	Enum	1	Status of the given order.		
				Name	Value	Description
				Ack	1	Order acknowledged by system.
				Cancelled	2	Order canceled.
				Rejected	3	Order rejected.
				Filled	4	Order filled.
				Modified	5	Order modified.
priorityFlag	PriorityFlag	Enum	1	Indicates whether the priority flag is lost or retained after modification.		
				Name	Value	Description
				Lost	1	The priority flag was lost.
				Retained	2	The priority flag was retained.
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order.		
				Name	Value	Description
				NA	1	Not applicable.
				ExchangeClosed	2	Exchange closed.
				InvalidPriceIncrement	18	Invalid price increment.
				Other	99	Other.
				UnknownInstrumentId	100	Unknown instrument.
				InstrumentPhaseNoTrading	106	Trading is not available for the instrument in its current phase.

Name	Type	Kind	Length	Description		
				UnknownOrder	1001	The order id is unrecognized.
				InvalidExecutionTrader	1005	Invalid execution trader.
				InvalidDecisionMaker	1006	Invalid decision maker.
				InvalidClientId	1007	Invalid client.
				InvalidPartyRoleQualifierForClientId	1008	Invalid Party Role Qualifier for Client Id Party group
				InvalidPartyRoleQualifierForExecutingTrader	1009	Invalid Party Role Qualifier for Executing Trader Party group
				InvalidPartyRoleQualifierForInvestmentDecisionMaker	1010	Invalid Party Role Qualifier for Investment Decision Maker Party group
				WrongDisplayQtyValue	1013	The display quantity (displayQty) cannot exceed the order quantity.
				InvalidDisplayQty	1014	The Display quantity (displayQty) not allowed for specified order type - only for Iceberg.
				WrongIcebergOrderValue	1015	The value of the iceberg order is less than the required.
				OrderQuantityMustBeGreaterThanOrEqualToMinimumQuantity	1025	The order quantity must be greater than the minimum quantity.
				OrderQuantityMustBeLowerThanMaximumQuantity	1026	The order quantity must be lower than the maximum quantity.
				OrderPriceMustBeGreaterThanOrEqualToMinimumPrice	1027	The order price must be greater than minimum price.

Name	Type	Kind	Length	Description		
				OrderPriceMustBeLowerThanMaximumPrice	1028	The order price must be greater lower than maximum price.
				OrderPriceMustBeNonzero	1029	The order price must be non-zero.
				OrderValueMustBeGreaterThanMinimumValue	1030	The order value must be greater than minimum value.
				OrderValueMustBeLowerThanMaximumValue	1031	The order value must be lower than maximum value.
				MarketOrderNotAllowedInAuction InvalidOrdTypeForSelectedMarketModel	1032	Market orders are not permitted during the auction. Invalid OrdType for selected market model
				LeavesQuantityMustBeGreaterThanZeroAfterModification	1034	The remaining quantity (LeavesQty) must be greater than 0 after the modification.
				PriceNotAllowed	1035	The price is not allowed for a market order. In the binary protocol, the price field should have a value of 0.
				InvalidMarketOrderTimeInForce InvalidTimeInForceForOrderType	1039	A market order can only have a TimeInForce value of 3 (IOC-Immediate or Cancel) or 4 (FOK-Fill or Kill). Invalid TimeInForce for specified OrderType
				InvalidAuctionTimeInForce InvalidTimeInForceForCurrentMarketPhase	1040	The TimeInForce value of 3 (IOC-Immediate or

Name	Type	Kind	Length	Description
				Cancel) or 4 (FOK-Fill or Kill) is not allowed during the auction. Invalid TimeInForce for current market phase
				InvalidIcebergAndStopOrderTimeInForce InvalidTimeInForceForSelectedMarketModel 1041 Iceberg and Stop orders can only have a TimeInForce value of DAY, GTC (Good Till Cancel), GTD (Good Till Date), or GTT (Good Till Time). Invalid TimeInForce for selected market model
				<u>ExpireTimeCannotBeModified</u> 1043 ExpireTime (126) cannot be modified.
				<u>ObsoleteExpireDate</u> 1044 ExpireDate (432) not allowed for selected TimeInForce (59).
				ExpireDateInPast 1045 The expiration date is earlier than the current date.
				<u>ObsoleteExpireTime</u> 1046 ExpireTime (126) not allowed for selected TimeInForce (59).
				ExpireTimeInPast 1047 The expiration time is earlier than the current time.
				AmbiguousExpire 1048 Please provide either a time or a date, but not both.
				ExpireDateExceedsLimit 1049 The expiration date exceeds the allowed limit.
				PriceBelowLowCollar 1037 The validation for collars has failed. The price is too low.
				PriceAboveHighCollar 1038 The validation for collars has failed.

Name	Type	Kind	Length	Description
				The price is too high.
TriggerPriceNotAllowed				1063 The trigger price not allowed for a specified order type.
TriggerPriceNotHigherThanLTP				1064 The trigger price is not higher than the last trade price.
TriggerPriceNotLowerThanLTP				1065 The trigger price is not lower than the last trade price.
TriggerPriceLowerThanPrice				1066 The trigger price is lower than the limit price.
TriggerPriceHigherThanPrice				1067 The trigger price is higher than the limit price.
TriggerPriceModifiedForActivatedOrder				1068 The trigger price has been modified for the activated order.
<u>TriggerPriceMustBeGreaterThanZero</u>				<u>1069</u> <u>TriggerPrice (1102) must be greater than 0.</u>
InvalidPartyIdForClientId				1070 Invalid PartyID (448) for Client ID
InvalidPartyIdForExecutingTrader				1071 Invalid PartyID (448) for Executing Trader
InvalidPartyIdForInvestmentDecisionMaker				1072 Invalid PartyID (448) for Investment Decision Maker
InvalidPartyRoleQualifierForPartyId				1075 Invalid PartyRoleQualifier (2376) for PartyID (448)
<u>UnknownInstrument</u>				<u>1201</u> <u>SecurityID (48) not recognized by the system.</u>
<u>InstrumentClosed</u>				<u>1203</u> <u>Instrument closed for trading.</u>

Name	Type	Kind	Length	Description		
				InvalidBidAskSpread	1208	OfferPx (133) must be greater than BidPx (132).
				<u>BuyOrderNotAllowed</u>	<u>1301</u>	<u>Buy orders from participants are not allowed in BuyOnly phase.</u>
				<u>OnlyOneSellOrderIsAllowedForIpo</u>	<u>1401</u>	<u>Only one sell order is allowed for IPO instrument</u>
				RequestNotAllowedForBlockInstrument	2026	Request not allowed for BLOCK instrument
				RequestNotAllowedForCrossInstrument	2028	Request not allowed for CROSS instrument
				RiskLimitNotDefined	7000	The risk limit has not been defined.
				RiskMaximumOrderVolumeExceeded	7001	The maximum order volume for the risk limit has been exceeded.
				RiskMaximumOrderValueExceeded	7002	The maximum order value for the risk limit has been exceeded.
				RiskOrderPriceCollarExceeded	7003	The order price has exceeded the risk limit.

7.1.16:7.1.20. REJECT:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
refSeqNum	SeqNum	Alias (u32)	4	Sequence number of the rejected message.		
rejectReason	RejectReason	Enum	1	Reject reason.		
				Name	Value	Description
				NA	0	Not applicable
				MaxThroughputExceeded	1	Max throughput exceeded.
				InvalidMsgType	2	Invalid message type.
				InvalidExpireTimePrecision	3	Invalid expire time precision.
				InvalidSettlementDate	4	Invalid settlement date.
				SettlementDateRequired	5	Settlement date required.

Name	Type	Kind	Length	Description		
				TradeReportIdRequired	6	Trade report ID required.
				MissingReportIdSecondaryTradeReportIdOrTradeReportRefId	7	Missing report id (SecondaryTradeReportId or TradeReportRefId).

7.1.21. REQUESTFOREXECUTION

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	16	Message header.
instrumentId	ElementId	Alias (u32)	4	Instrument ID

7.1.17.7.1.22. RISKLIMITBREACH:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
id	RiskDefinitionId	Alias (u64)	8	Risk limit definition id.		
amount	RiskLimitAmount	Alias (u64)	8	Amount of the overrun.		
riskLimitRequestType	RiskLimitRequestType	Enum	1	Type of risk limit information.		
				Name	Value	Description
				Definitions	1	Definitions.
				Utilization	2	Utilization.
unsolicitedIndicator	bool	Primitive	1	DefinitionsAndUtilizations	3	Definitions and utilization.
				Indicates whether or not message is being sent as a result of a subscription request or not.		

Name	Type	Kind	Length	Description		
limitType	RiskLimitType	Enum	1	Risk limit definition limit type.		
				Name	Value	Description
				PerBuyOrderVolume	301	Per buy order volume.
				PerBuyOrderNotionalValue	302	Per buy order notional value.
				PerSellOrderVolume	311	Per sell order volume.
				PerSellOrderNotionalValue	312	Per sell order notional value.
				PerBuyPriceLimit	313	Per buy price limit.
				PerSellPriceLimit	314	Per Sell Price Limit.
				PerTotalBuyTradedValue	315	Per total buy traded value.
				PerTotalSellTradedValue	316	Per total sell traded value.
				PerTotalTradedValue	317	Per total traded value.
				PerTotalBuyOpenOrdersValue	318	Per total buy open orders value.
				PerTotalSellOpenOrdersValue	319	Per total sell open orders value.
				PerTotalOpenOrdersValue	320	Per total open orders value.
				PerTotalBuyRiskValue	321	Per total buy risk value.
				PerTotalSellRiskValue	322	Per total sell risk value.
				PerTotalRiskValue	323	Per total risk value.
				PerTotalNetRiskValue	324	Per total net risk value.
PerTotalDailyNumberOfOrders	325	Per total daily number Of orders.				
RiskLimitNotDefined	326	Required risk limits are missing.				
riskWarningLevelAction	RiskWarningLevelAction	Enum	1	Action that was take because warning level was exceeded.		
				Name	Value	Description
				Reject	3	Reject.
riskLimitUtilizationAmount	RiskLimitAmount	Alias (u64)	8	Absolute amount of utilization of a party's set risk limit.		
riskLimitUtilizationPercent	Percentage	Alias (i64)	8	Percentage of utilization of a party's set risk limit.		

7.1.18:7.1.23. TRADE:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
id	TradeId	Alias (u32)	4	ID of the trade.
price	Price	Alias (Number)	8	Price of the given trade.
quantity	Quantity	Alias (u64)	8	Quantity of the instrument involved in the given trade.
leavesQty	Quantity	Alias (u64)	8	How much of the given security is left on the market after the trade is concluded.

~~7.1.19:~~ 7.1.24. **TRADEBUST:**

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.

~~7.1.20:~~ 7.1.25. **TRADECAPTUREREPORTDUAL:**

Name	Type	Kind	Length	Description																		
header	Header	Struct	16	Header.																		
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.																		
tradeReportId	TradeReportId	Array (AnsiChar)	20	Unique identifier of trade capture report.																		
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.																		
tradeReportTransType	TradeReportTransType	Enum	1	Identifies Trade Report message transaction type.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>New</td><td>1</td><td>New.</td></tr><tr><td>Cancel</td><td>2</td><td>Cancel.</td></tr><tr><td>Replace</td><td>3</td><td>Replace.</td></tr></table>	Name	Value	Description	New	1	New.	Cancel	2	Cancel.	Replace	3	Replace.						
				Name	Value	Description																
				New	1	New.																
Cancel	2	Cancel.																				
Replace	3	Replace.																				
tradeReportType	TradeReportType	Enum	1	Type of Trade Report.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Submit</td><td>1</td><td>Submit</td></tr><tr><td>Alleged</td><td>2</td><td>Alleged</td></tr><tr><td>Accept</td><td>3</td><td>Accept</td></tr><tr><td>Decline</td><td>4</td><td>Decline</td></tr><tr><td>TradeReportCancel</td><td>7</td><td>Trade Report Cancel</td></tr></table>	Name	Value	Description	Submit	1	Submit	Alleged	2	Alleged	Accept	3	Accept	Decline	4	Decline	TradeReportCancel	7	Trade Report Cancel
				Name	Value	Description																
				Submit	1	Submit																
				Alleged	2	Alleged																
				Accept	3	Accept																
Decline	4	Decline																				
TradeReportCancel	7	Trade Report Cancel																				
tradeType	TradeType	Enum	1	Type of trade.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>PrivatelyNegotiatedTrade</td><td>22</td><td>Privately negotiated trade.</td></tr><tr><td>BlockTrade</td><td>38</td><td>Block trade.</td></tr></table>	Name	Value	Description	PrivatelyNegotiatedTrade	22	Privately negotiated trade.	BlockTrade	38	Block trade.									
				Name	Value	Description																
				PrivatelyNegotiatedTrade	22	Privately negotiated trade.																
BlockTrade	38	Block trade.																				
algorithmicTradeIndicator	AlgorithmicTradeIndicator	Enum	1	Indicates algorithmic trade.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td>Not applicable.</td></tr><tr><td>NonAlgorithmicTrade</td><td>1</td><td>Non-algorithmic trade.</td></tr><tr><td>AlgorithmicTrade</td><td>2</td><td>Algorithmic trade.</td></tr></table>	Name	Value	Description	NA	1	Not applicable.	NonAlgorithmicTrade	1	Non-algorithmic trade.	AlgorithmicTrade	2	Algorithmic trade.						
				Name	Value	Description																
				NA	1	Not applicable.																
NonAlgorithmicTrade	1	Non-algorithmic trade.																				
AlgorithmicTrade	2	Algorithmic trade.																				
execType	ExecType	Enum	1	Type of execution being reported. Uses subset of ExecType for trade capture reports.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>New</td><td>1</td><td>New.</td></tr><tr><td>Rejected</td><td>8</td><td>Rejected.</td></tr><tr><td>Trade</td><td>15</td><td>Trade.</td></tr><tr><td>TradeCorrect</td><td>16</td><td>Trade Correct.</td></tr></table>	Name	Value	Description	New	1	New.	Rejected	8	Rejected.	Trade	15	Trade.	TradeCorrect	16	Trade Correct.			
				Name	Value	Description																
				New	1	New.																
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Name	Type	Kind	Length	Description																					
				<table><tr><td>TradeCancel</td><td>17</td><td>Trade Cancel.</td></tr></table>	TradeCancel	17	Trade Cancel.																		
TradeCancel	17	Trade Cancel.																							
tradeReportRefId	TradeReportRefID	Array (AnsiChar)	20	Reference identifier used with Cancel and Replace transaction types. The TradeReportID that is being referenced for trade correction or cancelation.																					
lastQty	Quantity	Alias (u64)	8	Quantity (e.g. shares) bought/sold on this (last) fill.																					
lastPx	Price	Alias (Number)	8	Price of this (last) fill.																					
settlementDate	Date	Alias (u32)	4	Settlement date of the trade is equal to current date plus actual settlement offset calendar days.																					
matchStatus	MatchStatus	Enum	1	<table><tr><td colspan="3">The status of this trade with respect to matching or comparison.</td></tr><tr><td>Name</td><td>Value</td><td>Description</td></tr><tr><td>NA</td><td>0</td><td>Not applicable.</td></tr><tr><td>Matched</td><td>1</td><td>Compared, matched or affirmed.</td></tr><tr><td>Unmatched</td><td>2</td><td>Uncompared, unmatched, or unaffirmed.</td></tr></table>	The status of this trade with respect to matching or comparison.			Name	Value	Description	NA	0	Not applicable.	Matched	1	Compared, matched or affirmed.	Unmatched	2	Uncompared, unmatched, or unaffirmed.						
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tcrPartyBuy.mifidFields.flags	MifidFlags	Enum	1	<table><tr><td colspan="3">Flags raised on an order in compliance with the MiFID directive.</td></tr><tr><td>Name</td><td>Value</td><td>Description</td></tr><tr><td>None</td><td>0b0000</td><td></td></tr><tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td></td></tr><tr><td>DirectOrSponsoredAccess</td><td>0b0010</td><td></td></tr><tr><td>AlgorithmicTrade</td><td>0b0100</td><td></td></tr><tr><td>MarketMakerOrSpecialist</td><td>0b1000</td><td></td></tr></table>	Flags raised on an order in compliance with the MiFID directive.			Name	Value	Description	None	0b0000		LiquidityProvisionActivity	0b0001		DirectOrSponsoredAccess	0b0010		AlgorithmicTrade	0b0100		MarketMakerOrSpecialist	0b1000	
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tcrPartyBuy.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																					
tcrPartyBuy.mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	<table><tr><td colspan="3">Qualifier of MiFID participant.</td></tr><tr><td>Name</td><td>Value</td><td>Description</td></tr><tr><td>NA</td><td>1</td><td></td></tr><tr><td>Algorithm</td><td>2</td><td></td></tr><tr><td>FirmOrLegalEntity</td><td>3</td><td></td></tr><tr><td>NaturalPerson</td><td>4</td><td></td></tr></table>	Qualifier of MiFID participant.			Name	Value	Description	NA	1		Algorithm	2		FirmOrLegalEntity	3		NaturalPerson	4				
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tcrPartyBuy.account	Account	Array (AnsiChar)	16	Account number.																		
tcrPartyBuy.accountType	AccountType	Enum	1	Type of account associated with the order. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Missing</td><td>1</td><td>Account is missing. Account is expected to be filled with 0x00.</td></tr><tr><td>Customer</td><td>2</td><td>Account is carried on customer side of the books.</td></tr><tr><td>House</td><td>3</td><td>House trader.</td></tr></table>	Name	Value	Description	Missing	1	Account is missing. Account is expected to be filled with 0x00.	Customer	2	Account is carried on customer side of the books.	House	3	House trader.						
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tcrPartyBuy.orderCapacity	Capacity	Enum	1	Designates the capacity of the firm placing the order. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Agency</td><td>1</td><td>Agency (mapped to AOTC).</td></tr><tr><td>Principal</td><td>2</td><td>Principal (mapped to DEAL).</td></tr><tr><td>RisklessPrincipal</td><td>3</td><td>Riskless Principal (mapped to MTCH)</td></tr></table>	Name	Value	Description	Agency	1	Agency (mapped to AOTC).	Principal	2	Principal (mapped to DEAL).	RisklessPrincipal	3	Riskless Principal (mapped to MTCH)						
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tcrPartyBuy.orderRestrictions	ElementId	Alias (u32)	4	Restrictions associated with an order.																		
tcrPartyBuy.orderOrigination	ElementId	Alias (u32)	4	Identifies the origin of the order.																		
tcrPartyBuy.memo	Memo	Array (AnsiChar)	18	Free text.																		
tcrPartySell.mifidFields.flags	MifidFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>None</td><td>0b0000</td><td></td></tr><tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td></td></tr><tr><td>DirectOrSponsoredAccess</td><td>0b0010</td><td></td></tr><tr><td>AlgorithmicTrade</td><td>0b0100</td><td></td></tr><tr><td>MarketMakerOrSpecialist</td><td>0b1000</td><td></td></tr></table>	Name	Value	Description	None	0b0000		LiquidityProvisionActivity	0b0001		DirectOrSponsoredAccess	0b0010		AlgorithmicTrade	0b0100		MarketMakerOrSpecialist	0b1000	
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tcrPartySell.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		
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tcrPartySell.mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.																		

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tcrPartySell.mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.															
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tcrPartySell.account	Account	Array (AnsiChar)	16	Account number.															
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tcrPartySell.orderCapacity	Capacity	Enum	1	Designates the capacity of the firm placing the order.															
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tcrPartySell.orderRestrictions	ElementId	Alias (u32)	4	Restrictions associated with an order.															
tcrPartySell.orderOrigination	ElementId	Alias (u32)	4	Identifies the origin of the order.															
tcrPartySell.memo	Memo	Array (AnsiChar)	18	Free text.															

7.1.21.7.1.26. TRADECAPTUREREPORTRESPONSE:

Name	Type	Kind	Length	Description		
header	Header	Struct	16	Header.		
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.		
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.		
tradeReportId	TradeReportId	Array (AnsiChar)	20	Unique identifier of trade capture report.		
secondaryTradeReportId	OrderId	Alias (u64)	8	ID of the trade capture report.		
status	TcrStatus	Enum	1	Status of the given order.		
				Name	Value	Description
				New	1	New.
				Accepted	2	Accepted.
				Rejected	3	Rejected.
Cancelled	4	Cancelled.				

Name	Type	Kind	Length	Description																																																																								
reason	TcrRejectionReason	Enum	2	Reason for rejecting the given TCR.																																																																								
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td>Not applicable.</td></tr><tr><td>ExchangeClosed</td><td>2</td><td>Exchange closed.</td></tr><tr><td>InvalidPriceIncrement</td><td>18</td><td>Invalid price increment.</td></tr><tr><td>Other</td><td>99</td><td>Other.</td></tr><tr><td>UnknownInstrumentId</td><td>100</td><td>Unknown instrument.</td></tr><tr><td>InstrumentPhaseNoTrading</td><td>106</td><td>Trading is not available for the instrument in its current phase.</td></tr><tr><td>InvalidExecutionTrader</td><td>1005</td><td>Invalid execution trader.</td></tr><tr><td>InvalidDecisionMaker</td><td>1006</td><td>Invalid decision maker.</td></tr><tr><td>InvalidClientId</td><td>1007</td><td>Invalid client.</td></tr><tr><td>InvalidPartyRoleQualifierForClientId</td><td>1008</td><td>Invalid Party Role Qualifier for Client Id Party group</td></tr><tr><td>InvalidPartyRoleQualifierForExecutingTrader</td><td>1009</td><td>Invalid Party Role Qualifier for Executing Trader Party group</td></tr><tr><td>InvalidPartyRoleQualifierForInvestmentDecisionMaker</td><td>1010</td><td>Invalid Party Role Qualifier for Investment Decision Maker Party group</td></tr><tr><td>OrderQuantityMustBeGreaterThanMinimumQuantity</td><td>1025</td><td>The order quantity must be greater than the minimum quantity.</td></tr><tr><td>OrderQuantityMustBeLowerThanMaximumQuantity</td><td>1026</td><td>The order quantity must be lower than the maximum quantity.</td></tr><tr><td>OrderPriceMustBeGreaterThanMinimumPrice</td><td>1027</td><td>The order price must be greater than minimum price.</td></tr><tr><td>OrderPriceMustBeLowerThanMaximumPrice</td><td>1028</td><td>The order price must be greaterlower than maximum price.</td></tr><tr><td>OrderPriceMustBeNonzero</td><td>1029</td><td>The order price must be non-zero.</td></tr><tr><td>OrderValueMustBeGreaterThanMinimumValue</td><td>1030</td><td>The order value must be greater than minimum value.</td></tr><tr><td>OrderValueMustBeLowerThanMaximumValue</td><td>1031</td><td>The order value must be lower than maximum value.</td></tr><tr><td>PriceBelowLowCollar</td><td>1037</td><td>The validation for collars has failed. The price is too low.</td></tr><tr><td>PriceAboveHighCollar</td><td>1038</td><td>The validation for collars has failed. The price is too high.</td></tr><tr><td>InvalidPartyIdForClientId</td><td>1070</td><td>Invalid PartyID (448) for Client ID</td></tr><tr><td>InvalidPartyIdForExecutingTrader</td><td>1071</td><td>Invalid PartyID (448) for Executing Trader</td></tr></table>	Name	Value	Description	NA	1	Not applicable.	ExchangeClosed	2	Exchange closed.	InvalidPriceIncrement	18	Invalid price increment.	Other	99	Other.	UnknownInstrumentId	100	Unknown instrument.	InstrumentPhaseNoTrading	106	Trading is not available for the instrument in its current phase.	InvalidExecutionTrader	1005	Invalid execution trader.	InvalidDecisionMaker	1006	Invalid decision maker.	InvalidClientId	1007	Invalid client.	InvalidPartyRoleQualifierForClientId	1008	Invalid Party Role Qualifier for Client Id Party group	InvalidPartyRoleQualifierForExecutingTrader	1009	Invalid Party Role Qualifier for Executing Trader Party group	InvalidPartyRoleQualifierForInvestmentDecisionMaker	1010	Invalid Party Role Qualifier for Investment Decision Maker Party group	OrderQuantityMustBeGreaterThanMinimumQuantity	1025	The order quantity must be greater than the minimum quantity.	OrderQuantityMustBeLowerThanMaximumQuantity	1026	The order quantity must be lower than the maximum quantity.	OrderPriceMustBeGreaterThanMinimumPrice	1027	The order price must be greater than minimum price.	OrderPriceMustBeLowerThanMaximumPrice	1028	The order price must be greater lower than maximum price.	OrderPriceMustBeNonzero	1029	The order price must be non-zero.	OrderValueMustBeGreaterThanMinimumValue	1030	The order value must be greater than minimum value.	OrderValueMustBeLowerThanMaximumValue	1031	The order value must be lower than maximum value.	PriceBelowLowCollar	1037	The validation for collars has failed. The price is too low.	PriceAboveHighCollar	1038	The validation for collars has failed. The price is too high.	InvalidPartyIdForClientId	1070	Invalid PartyID (448) for Client ID	InvalidPartyIdForExecutingTrader	1071	Invalid PartyID (448) for Executing Trader
				Name	Value	Description																																																																						
				NA	1	Not applicable.																																																																						
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				OrderPriceMustBeGreaterThanMinimumPrice	1027	The order price must be greater than minimum price.																																																																						
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InvalidPartyIdForExecutingTrader	1071	Invalid PartyID (448) for Executing Trader																																																																										

Name	Type	Kind	Length	Description
				InvalidPartyIdForInvestmentDecisionMaker 1072 Invalid PartyID (448) for Investment Decision Maker
				InvalidPartyRoleQualifierForPartyId 1075 Invalid PartyRoleQualifier (2376) for PartyID (448)
				UnknownTradeReport 2001 Unknown trade report
				DuplicateTradeReportId 2002 Duplicate TradeReportId
				TradeReportTypeNotCompatibleWithTradeReportTransType 2005 TradeReportType not compatible with TradeReportTransType
				InvalidExecType 2008 Invalid ExecType
				TradeReportRefIdNotAllowed 2009 TradeReportRefId not allowed
				SettlementDateCannotBeEarlierThanMinimumSettlementDate 2015 Settlement date cannot be earlier than minimum settlement date
				SettlementDateCannotBeLaterThanMaximumSettlementDate 2016 Settlement date cannot be later than maximum settlement date
				UnknownContraFirm 2022 Unknown contra firm.
				<u>SentAttributeDoesNotMatchOriginalValue</u> 2024 <u>Sent attribute does not match original value</u>
				RequestNotAllowedForBlockInstrument 2026 Request not allowed for BLOCK instrument
				RequestNotAllowedForClobInstrument 2027 Request not allowed for CLOB instrument
				RequestNotAllowedForCrossInstrument 2028 Request not allowed for CROSS instrument
				InvalidMatchStatus 2029 Invalid MatchStatus
				CrossNotAllowedOutsideOfClobInstrumentSpread 2030 Cross not allowed outside of CLOB instrument spread
				CrossPriceNotEqualToTheReferencePrice 2031 Cross price not equal to the reference price
				CrossNotAllowedDuringClobInstrumentAuctionOrSuspension 2032 Cross not allowed during CLOB instrument Auction or Suspension
				ForbiddenSecondaryTradereportId 2033 Forbidden SecondaryTradeReportID
				UnknownSecondaryTradereportId 2034 Unknown SecondaryTradeReportID

7.1.22-7.1.27. TRADECAPTUREREPORTSINGLE:

Name	Type	Kind	Length	Description
header	Header	Struct	16	Header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.

Name	Type	Kind	Length	Description																		
tradeReportId	TradeReportId	Array (AnsiChar)	20	Unique identifier of the trade capture report.																		
secondaryTradeReportId	OrderId	Alias (u64)	8	ID of the trade capture report.																		
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.																		
tradeReportTransType	TradeReportTransType	Enum	1	Identifies Trade Report message transaction type.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>New</td><td>1</td><td>New.</td></tr><tr><td>Cancel</td><td>2</td><td>Cancel.</td></tr><tr><td>Replace</td><td>3</td><td>Replace.</td></tr></table>	Name	Value	Description	New	1	New.	Cancel	2	Cancel.	Replace	3	Replace.						
				Name	Value	Description																
				New	1	New.																
Cancel	2	Cancel.																				
Replace	3	Replace.																				
tradeReportType	TradeReportType	Enum	1	Type of Trade Report.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Submit</td><td>1</td><td>Submit</td></tr><tr><td>Alleged</td><td>2</td><td>Alleged</td></tr><tr><td>Accept</td><td>3</td><td>Accept</td></tr><tr><td>Decline</td><td>4</td><td>Decline</td></tr><tr><td>TradeReportCancel</td><td>7</td><td>Trade Report Cancel</td></tr></table>	Name	Value	Description	Submit	1	Submit	Alleged	2	Alleged	Accept	3	Accept	Decline	4	Decline	TradeReportCancel	7	Trade Report Cancel
				Name	Value	Description																
				Submit	1	Submit																
				Alleged	2	Alleged																
				Accept	3	Accept																
Decline	4	Decline																				
TradeReportCancel	7	Trade Report Cancel																				
tradeType	TradeType	Enum	1	Type of trade.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>PrivatelyNegotiatedTrade</td><td>22</td><td>Privately negotiated trade.</td></tr><tr><td>BlockTrade</td><td>38</td><td>Block trade.</td></tr></table>	Name	Value	Description	PrivatelyNegotiatedTrade	22	Privately negotiated trade.	BlockTrade	38	Block trade.									
				Name	Value	Description																
				PrivatelyNegotiatedTrade	22	Privately negotiated trade.																
BlockTrade	38	Block trade.																				
algorithmicTradeIndicator	AlgorithmicTradeIndicator	Enum	1	Indicates algorithmic trader.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NA</td><td>1</td><td>Not applicable.</td></tr><tr><td>NonAlgorithmicTrade</td><td>2</td><td>Non-algorithmic trade.</td></tr><tr><td>AlgorithmicTrade</td><td>3</td><td>Algorithmic trade.</td></tr></table>	Name	Value	Description	NA	1	Not applicable.	NonAlgorithmicTrade	2	Non-algorithmic trade.	AlgorithmicTrade	3	Algorithmic trade.						
				Name	Value	Description																
				NA	1	Not applicable.																
NonAlgorithmicTrade	2	Non-algorithmic trade.																				
AlgorithmicTrade	3	Algorithmic trade.																				
execType	ExecType	Enum	1	Type of execution being reported. Uses subset of ExecType for trade capture reports.																		
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>New</td><td>1</td><td>New.</td></tr><tr><td>Rejected</td><td>8</td><td>Rejected.</td></tr><tr><td>Trade</td><td>15</td><td>Trade.</td></tr><tr><td>TradeCorrect</td><td>16</td><td>Trade Correct.</td></tr><tr><td>TradeCancel</td><td>17</td><td>Trade Cancel.</td></tr></table>	Name	Value	Description	New	1	New.	Rejected	8	Rejected.	Trade	15	Trade.	TradeCorrect	16	Trade Correct.	TradeCancel	17	Trade Cancel.
				Name	Value	Description																
				New	1	New.																
				Rejected	8	Rejected.																
				Trade	15	Trade.																
TradeCorrect	16	Trade Correct.																				
TradeCancel	17	Trade Cancel.																				
tradeReportRefId	TradeReportRefId	Array (AnsiChar)	20	Reference identifier used with Cancel and Replace transaction types. The TradeReportId that is being referenced for trade correction or cancellation.																		
lastQty	Quantity	Alias (u64)	8	Quantity (e.g. shares) bought/sold on this (last) fill.																		
lastPx	Price	Alias (Number)	8	Price of this (last) fill.																		

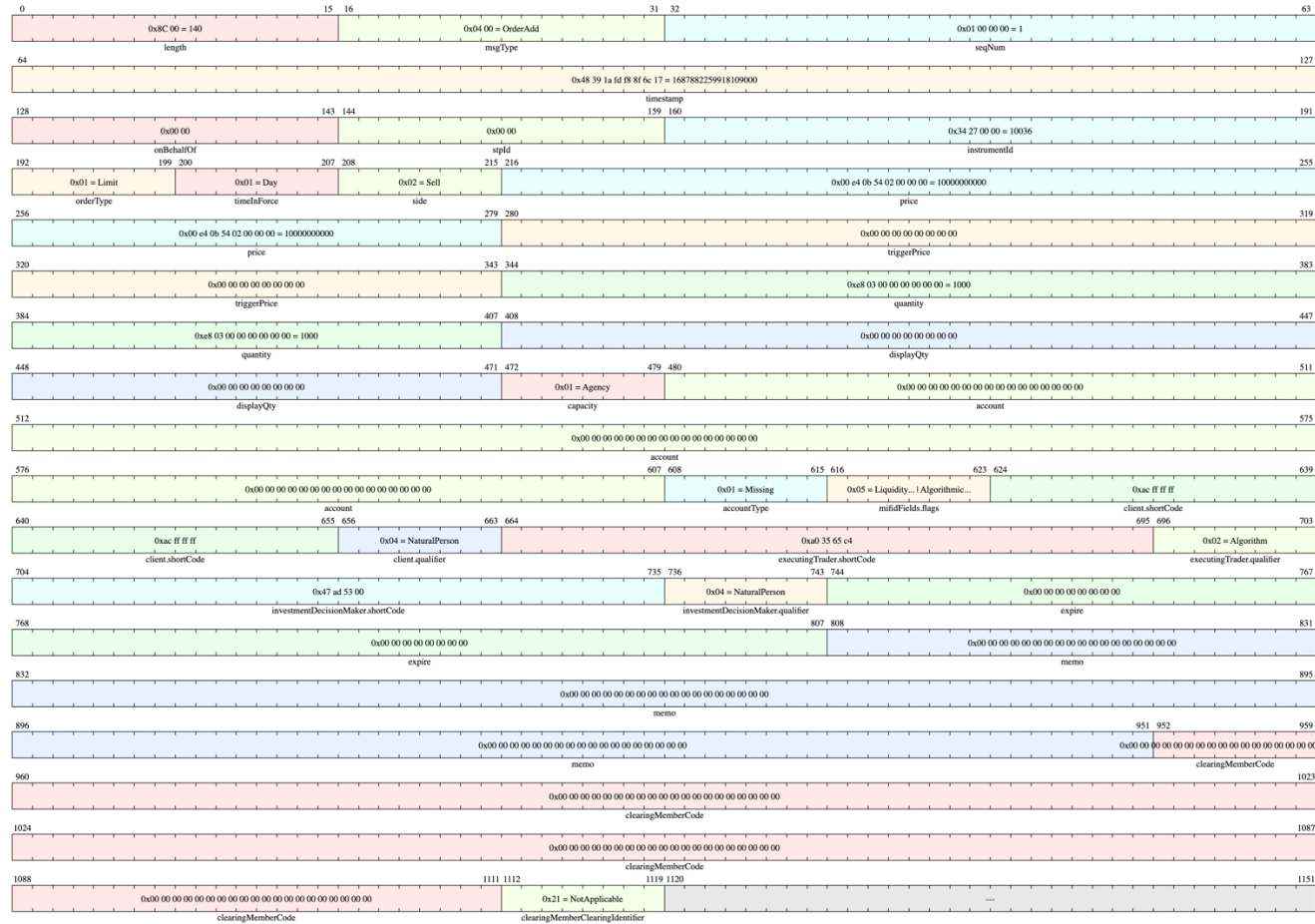
Name	Type	Kind	Length	Description		
settlementDate	Date	Alias (u32)	4	Settlement date of the trade is equal to current date plus actual settlement offset calendar days.		
matchStatus	MatchStatus	Enum	1	The status of this trade with respect to matching or comparison.		
				Name	Value	Description
				NA	0	Not applicable.
				Matched	1	Compared, matched or affirmed.
Unmatched	2	Uncompared, unmatched, or unaffirmed.				
side	OrderSide	Enum	1	Side of order.		
				Name	Value	Description
				Buy	1	Indicates a buy-side order.
Sell	2	Indicates a sell-side order.				
counterpartyCode	CompanyCCPCode	Array (AnsiChar)	16	Company IDCCP code of the counterparty.		
tcrParty.mifidFields.flags	MifidFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive.		
				Name	Value	Description
				None	0b0000	
				LiquidityProvisionActivity	0b0001	
				DirectOrSponsoredAccess	0b0010	
				AlgorithmicTrade	0b0100	
MarketMakerOrSpecialist	0b1000					
tcrParty.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.		
tcrParty.mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.		
				Name	Value	Description
				NA	1	
				Algorithm	2	
				FirmOrLegalEntity	3	
NaturalPerson	4					
tcrParty.mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.		
tcrParty.mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.		
				Name	Value	Description
				NA	1	
				Algorithm	2	
				FirmOrLegalEntity	3	
NaturalPerson	4					
tcrParty.mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.		
tcrParty.mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.		
				Name	Value	Description
				NA	1	
				Algorithm	2	
				FirmOrLegalEntity	3	
NaturalPerson	4					

Name	Type	Kind	Length	Description												
tcrParty.account	Account	Array (AnsiChar)	16	Account number.												
tcrParty.accountType	AccountType	Enum	1	Type of account associated with the order.												
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Missing</td><td>1</td><td>Account is missing. Account is expected to be filled with 0x00.</td></tr><tr><td>Customer</td><td>2</td><td>Account is carried on customer side of the books.</td></tr><tr><td>House</td><td>3</td><td>House trader.</td></tr></table>	Name	Value	Description	Missing	1	Account is missing. Account is expected to be filled with 0x00.	Customer	2	Account is carried on customer side of the books.	House	3	House trader.
				Name	Value	Description										
				Missing	1	Account is missing. Account is expected to be filled with 0x00.										
Customer	2	Account is carried on customer side of the books.														
House	3	House trader.														
tcrParty.orderCapacity	Capacity	Enum	1	Designates the capacity of the firm placing the order.												
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Agency</td><td>1</td><td>Agency (mapped to AOTC).</td></tr><tr><td>Principal</td><td>2</td><td>Principal (mapped to DEAL).</td></tr><tr><td>RisklessPrincipal</td><td>3</td><td>Riskless Principal (mapped to MTCH)</td></tr></table>	Name	Value	Description	Agency	1	Agency (mapped to AOTC).	Principal	2	Principal (mapped to DEAL).	RisklessPrincipal	3	Riskless Principal (mapped to MTCH)
				Name	Value	Description										
				Agency	1	Agency (mapped to AOTC).										
Principal	2	Principal (mapped to DEAL).														
RisklessPrincipal	3	Riskless Principal (mapped to MTCH)														
tcrParty.orderRestrictions	ElementId	Alias (u32)	4	Restrictions associated with an order.												
tcrParty.orderOrigination	ElementId	Alias (u32)	4	Identifies the origin of the order.												
tcrParty.memo	Memo	Array (AnsiChar)	18	Free text.												

7.2. ORDER ADD EXAMPLE

Below is an example of an OrderAdd message, illustrating sample values for each field along with their binary representation.

Figure 7 OrderAdd message frame



8. MESSAGE KINEMATICS

Scope:

The purpose of this appendix is to present the message flow within the Native communication protocol, as used in GPW WATS. The appendix does not include all available message flows in GPW WATS, but only those selected for presentation purposes.

Remarks:

Due to ensuring transparency of message flow diagrams, the scope of presented fields is limited to basic components and does not include the complete set of all fields that a given message may contain. The scope of the fields for the same messages may also vary depending on the specific example described in the diagram. The complete range of fields in the messages is available in the description of each individual message within the protocol.

Actors:

For the purpose of the presentations, the following actors have been utilized in the context of kinematics:

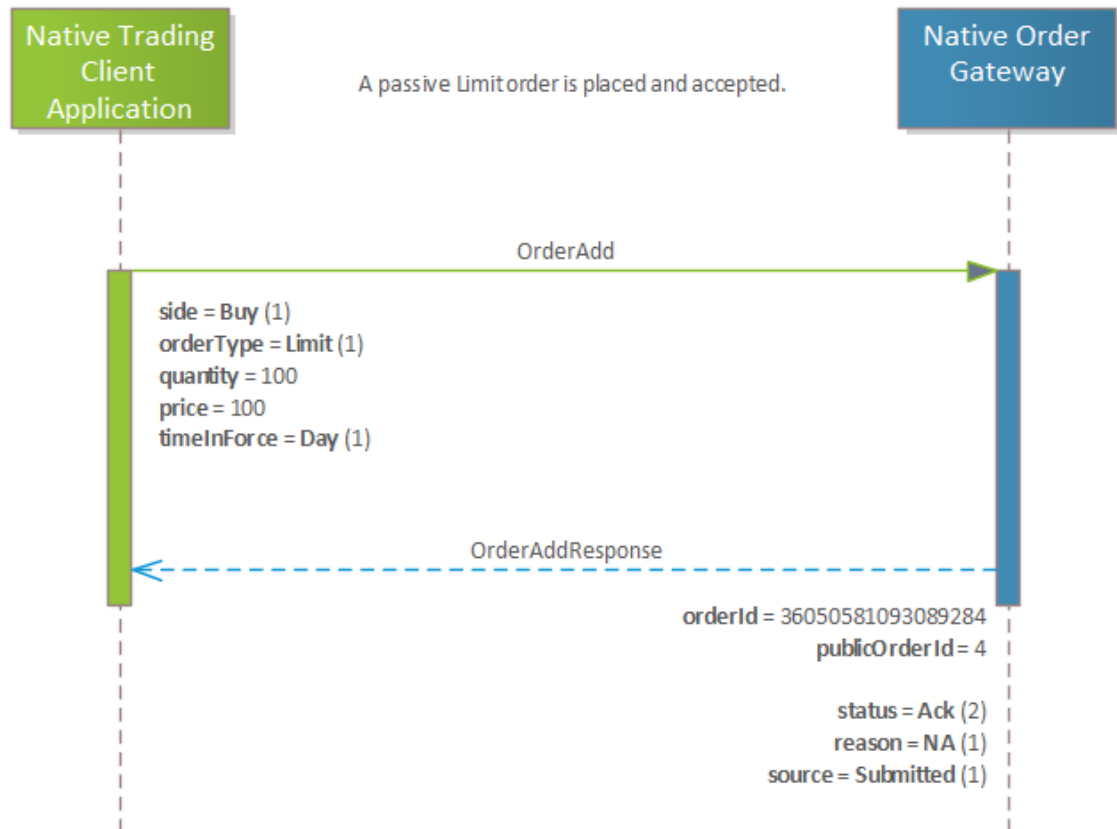
Native Trading Client Application – an exchange member authorized to submit buy and sell orders, communicating with the system represented by the Native Order Gateway.

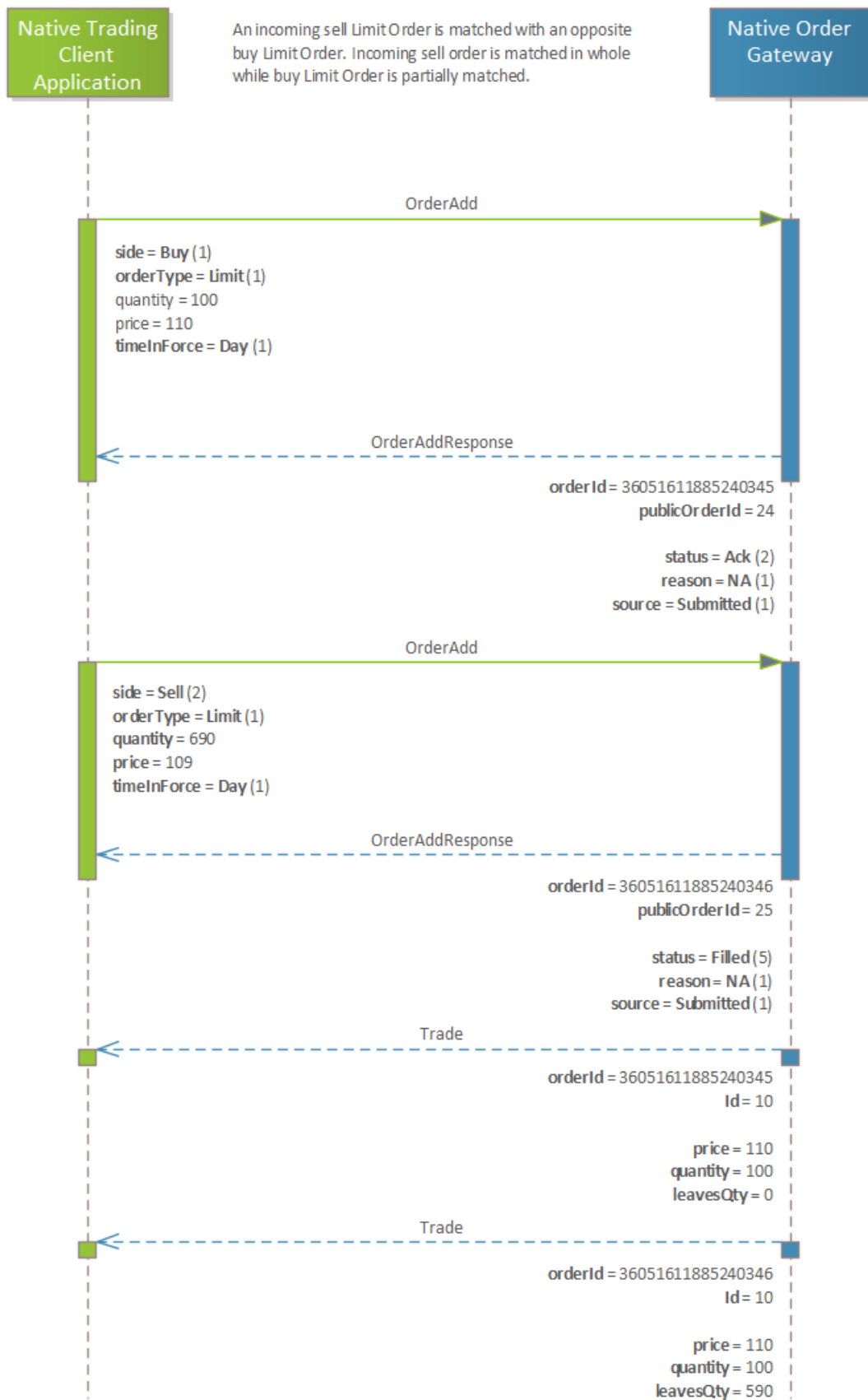
Native Order Gateway – a system that communicates with the Native Protocol Client Application.

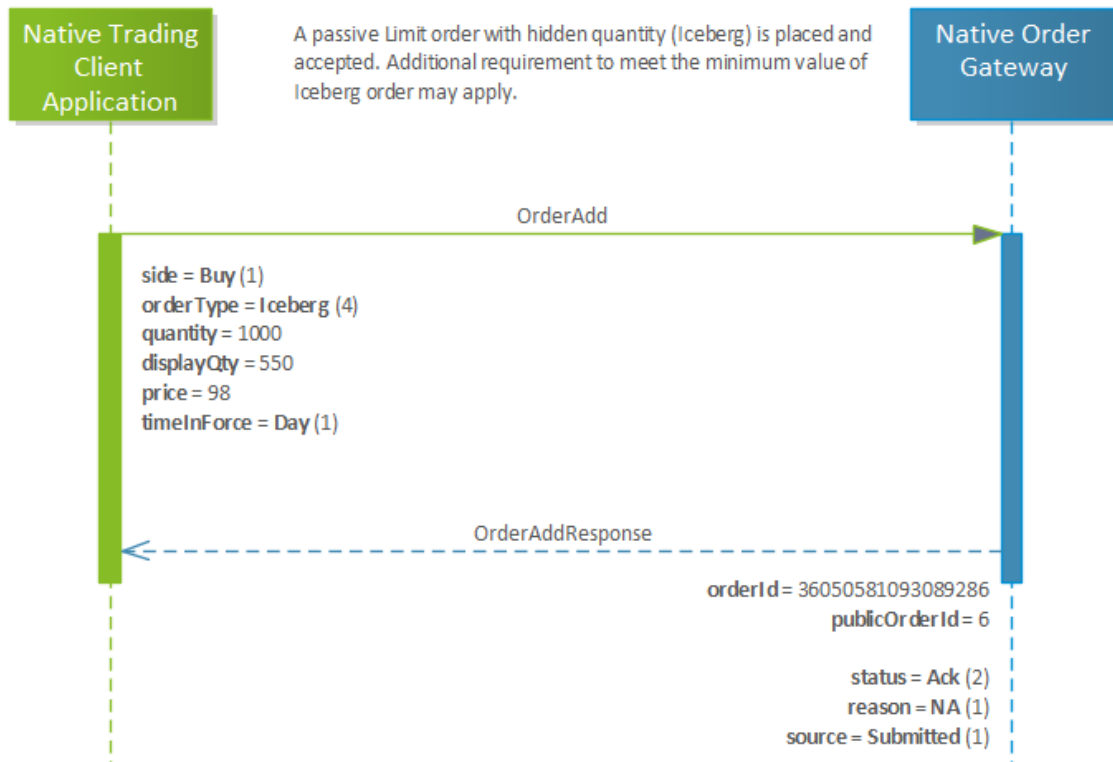
Note: presentation of simultaneous communication between multiple clients and Gateway was omitted. As a result, the Native Trading Client Application can either represent a single exchange member or simulate the exchange of messages between two different exchange members including buying, selling and conducting transactions.

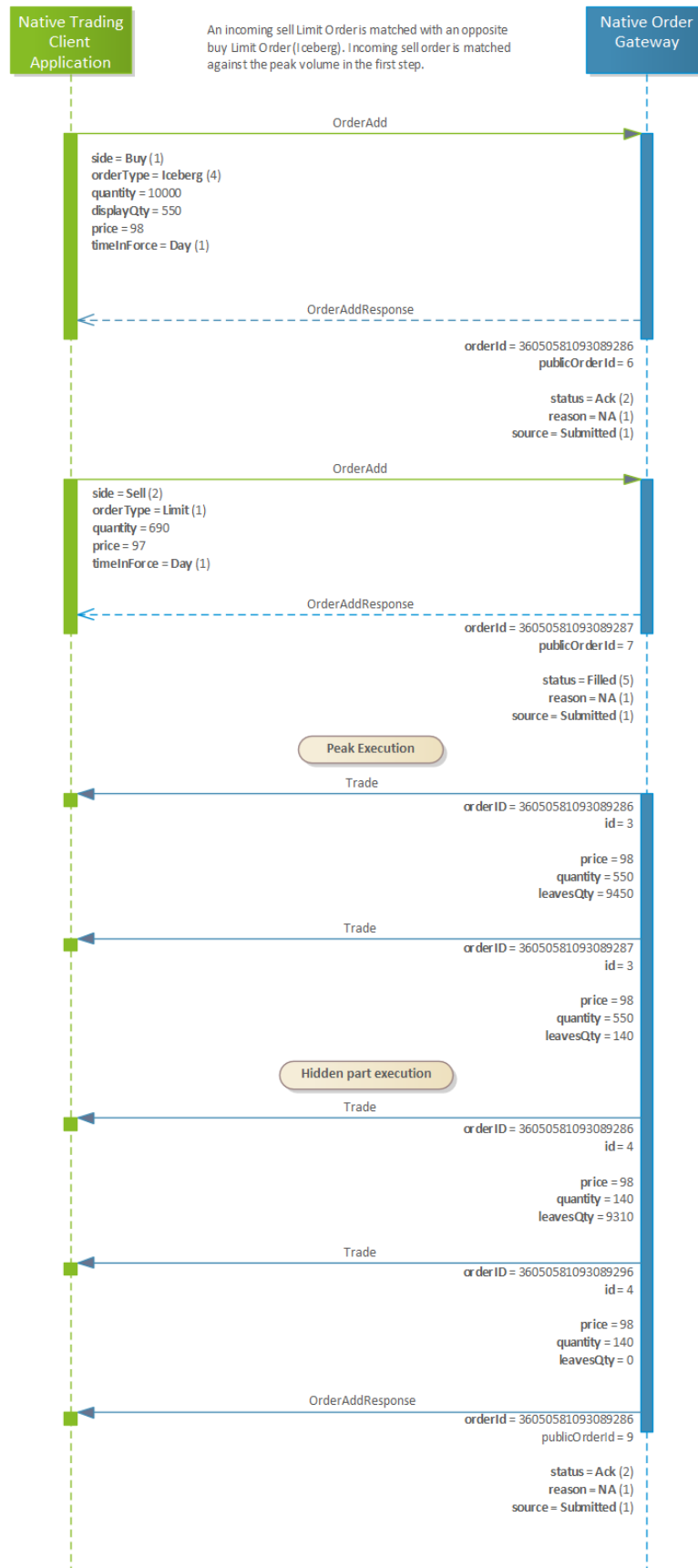
8.1. CLOB

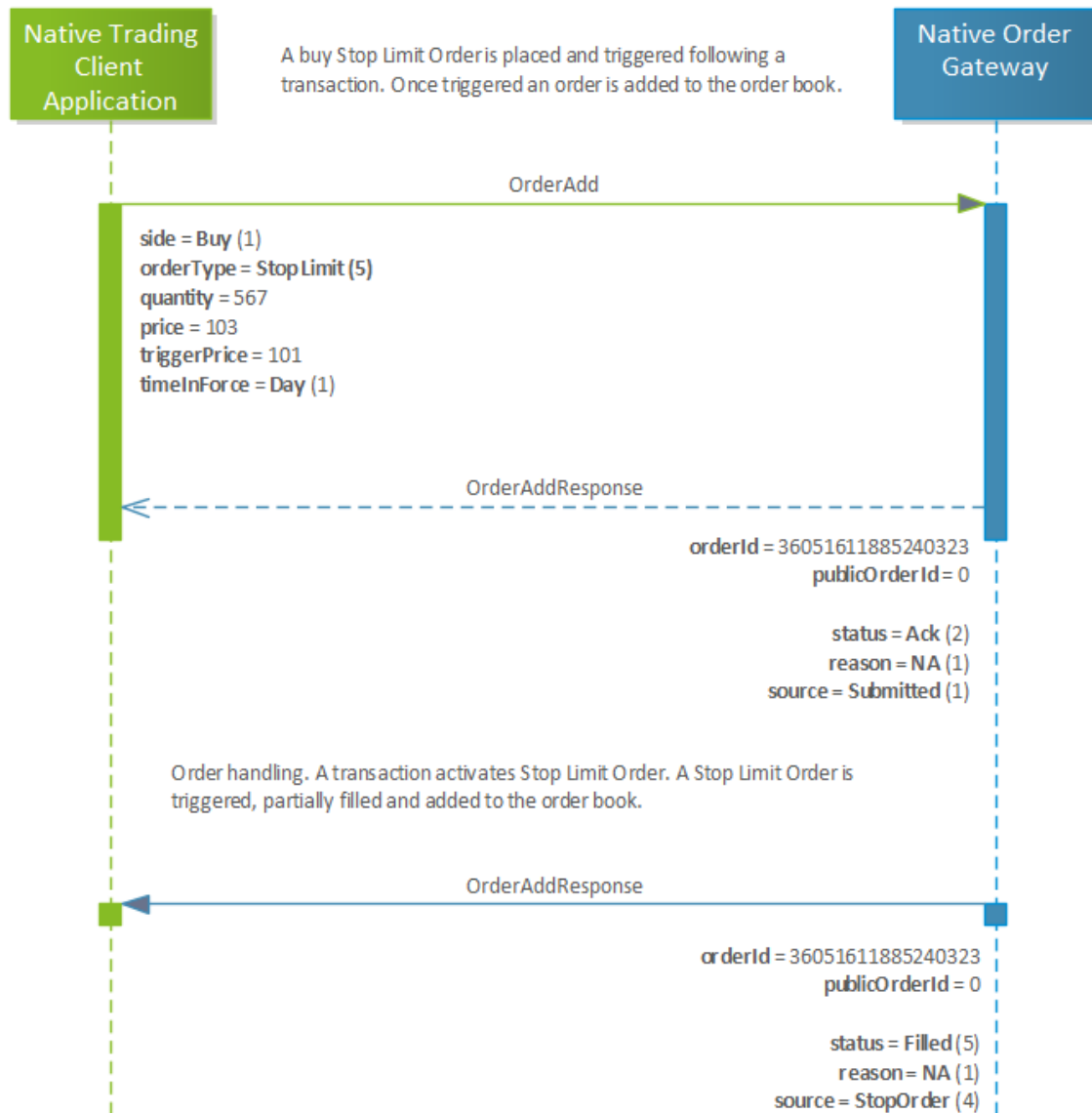
Order - Accepted

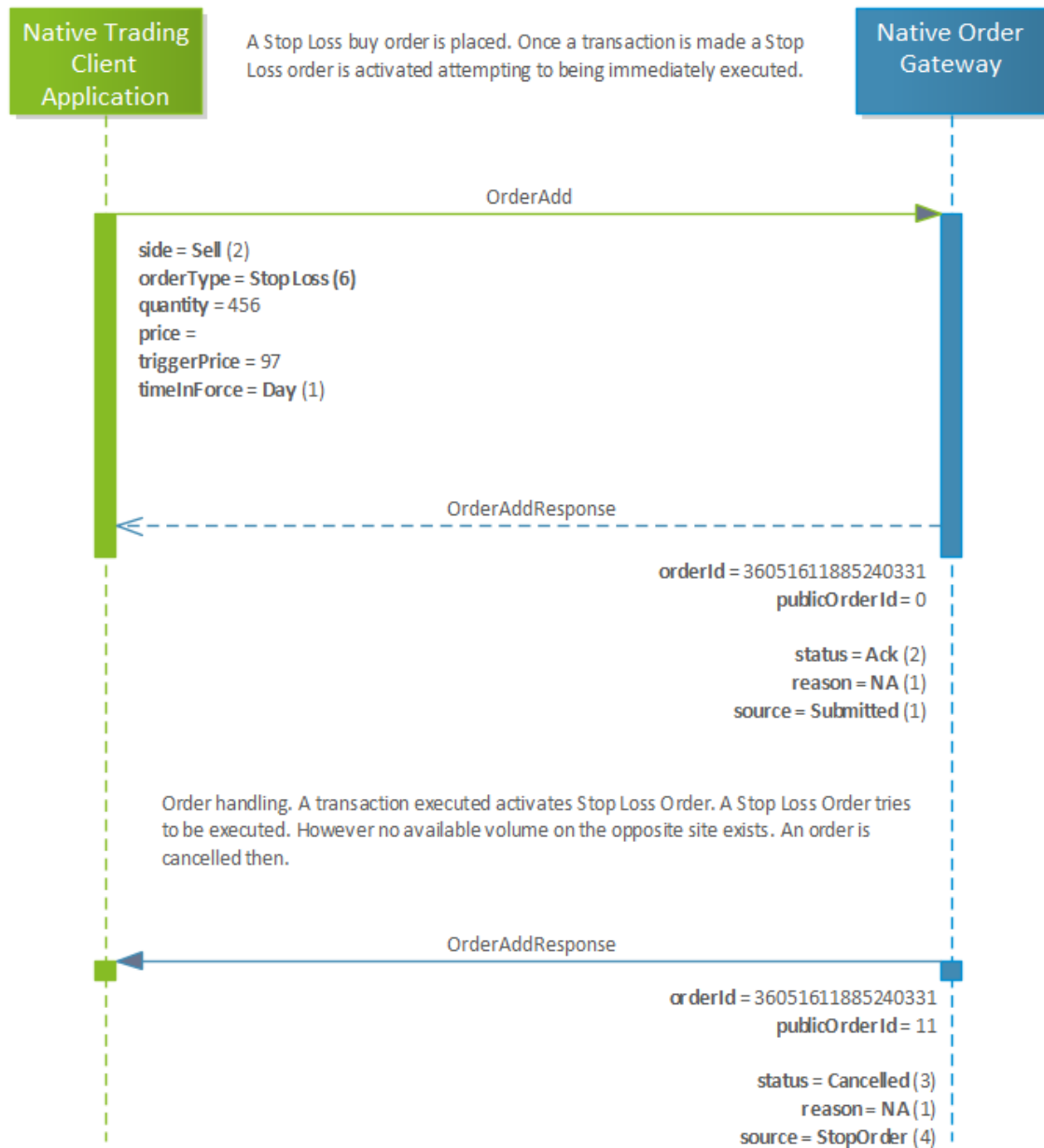


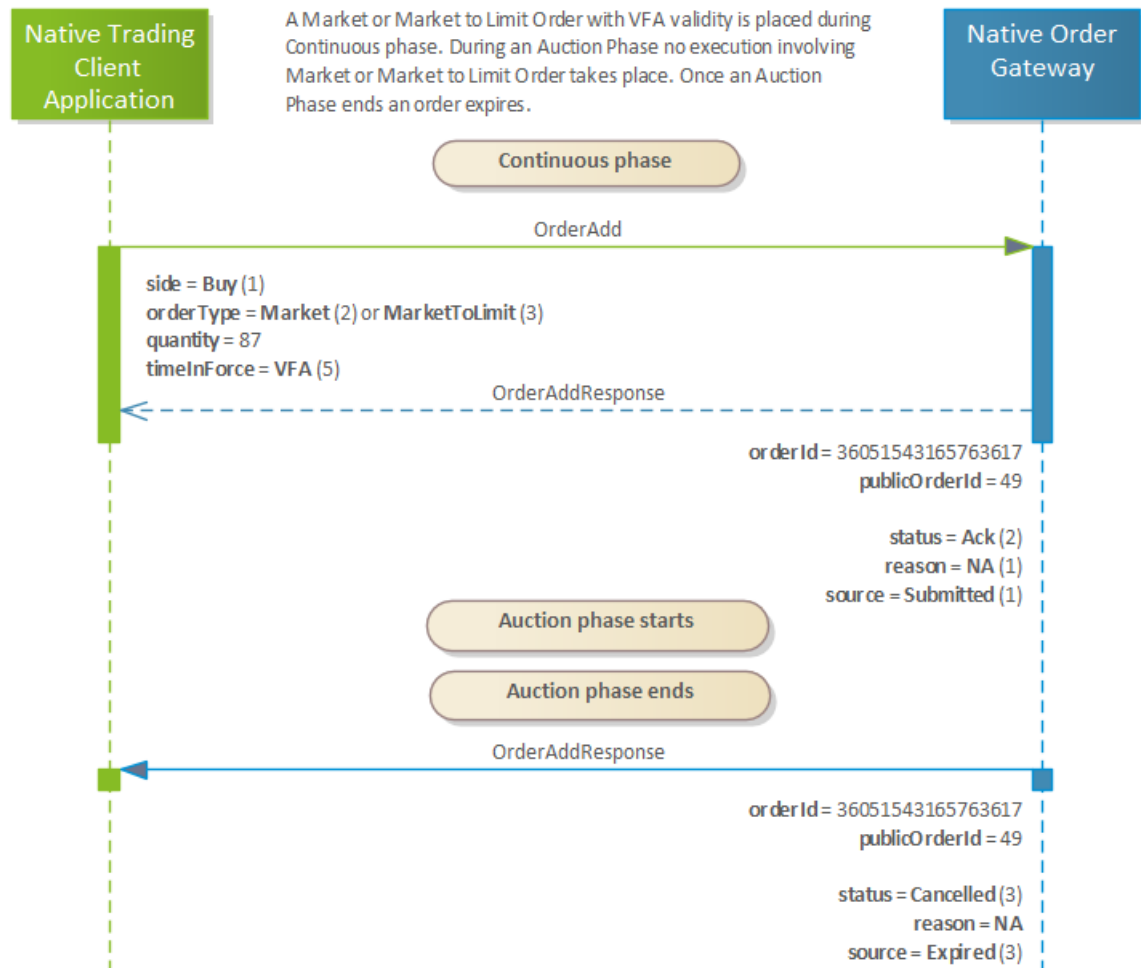
Order – Accepted and Filled – Full and Partially

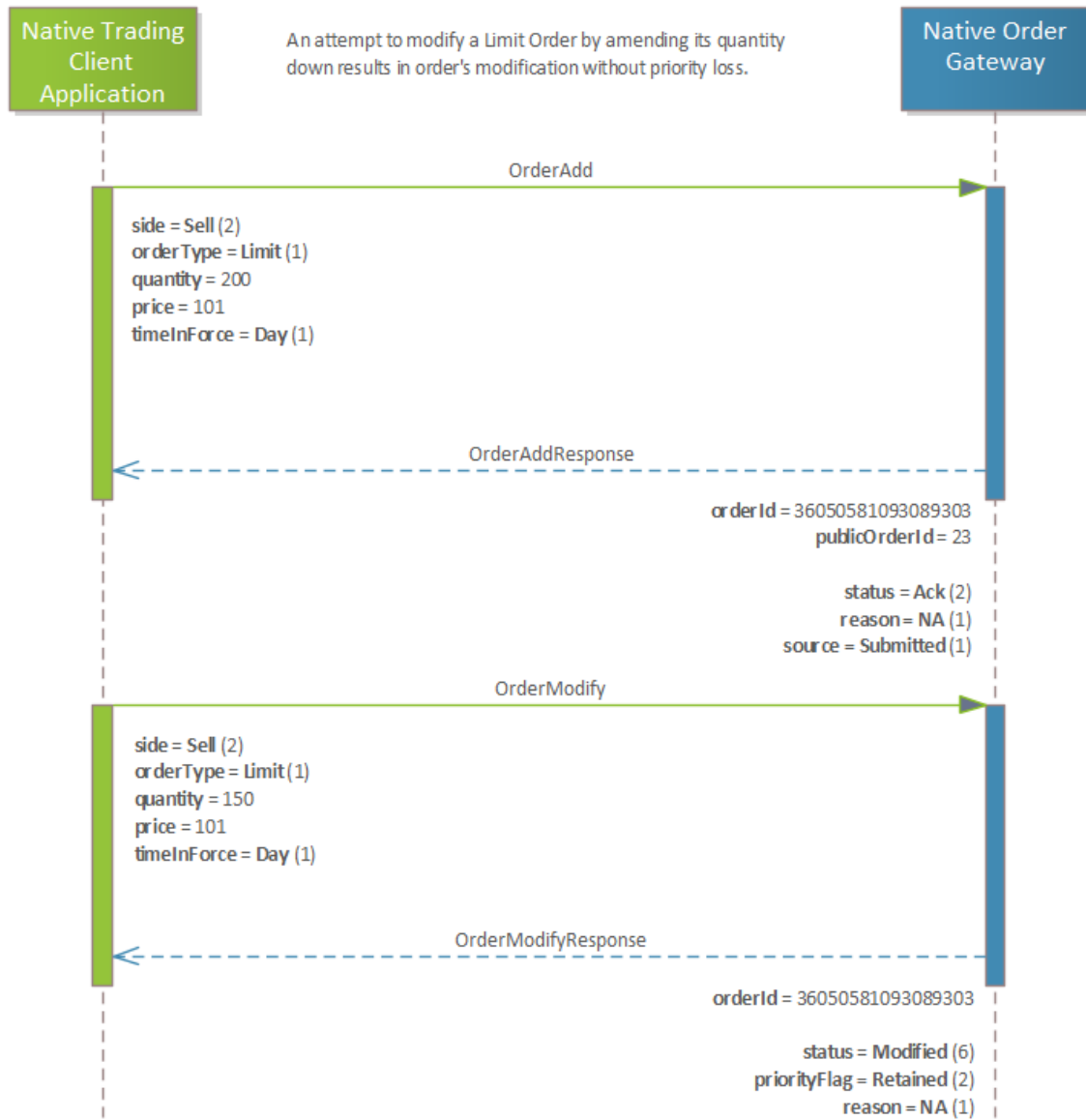
Iceberg Order – Accepted

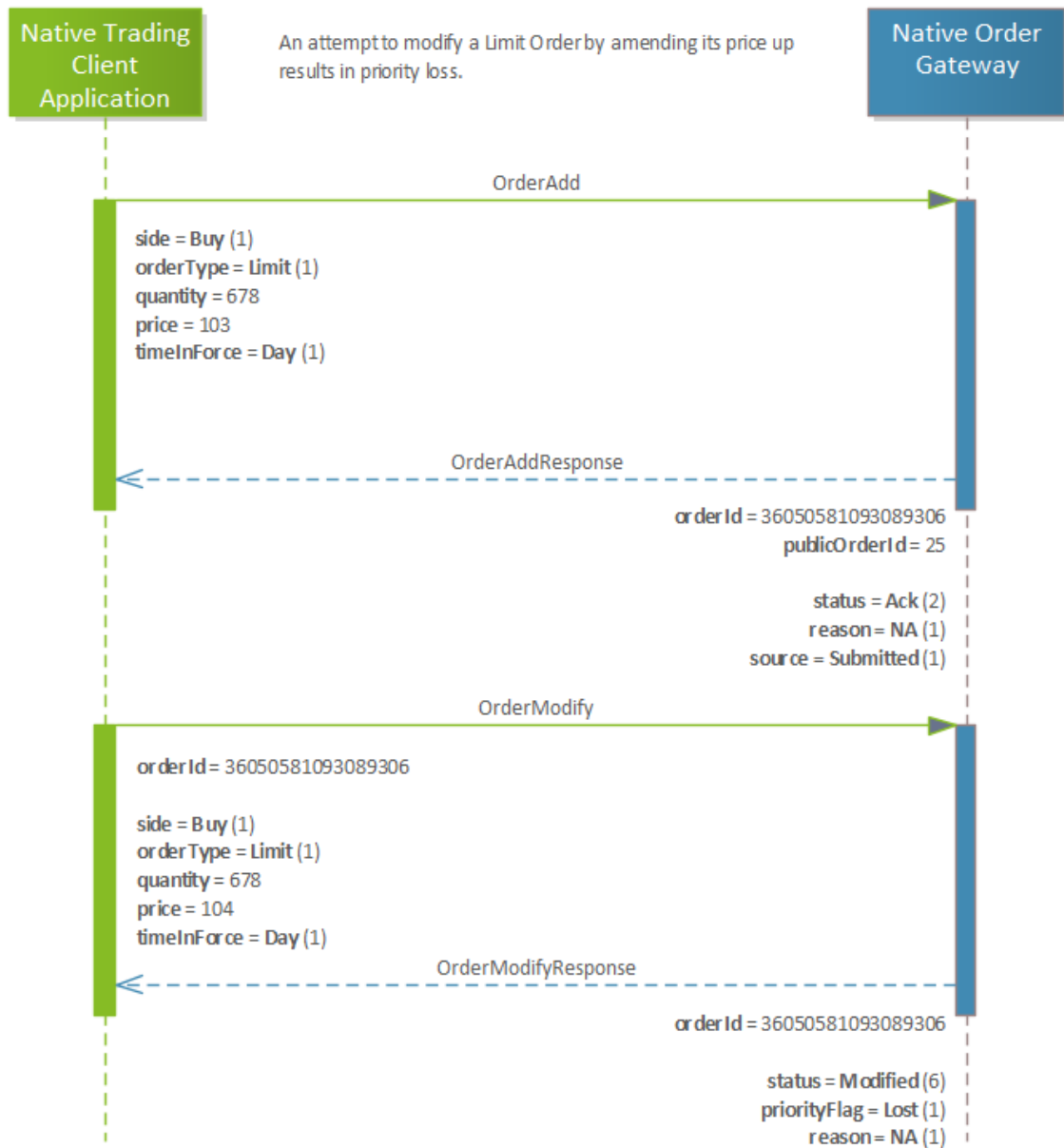
Iceberg Order – Accepted Partially Filled and Restated

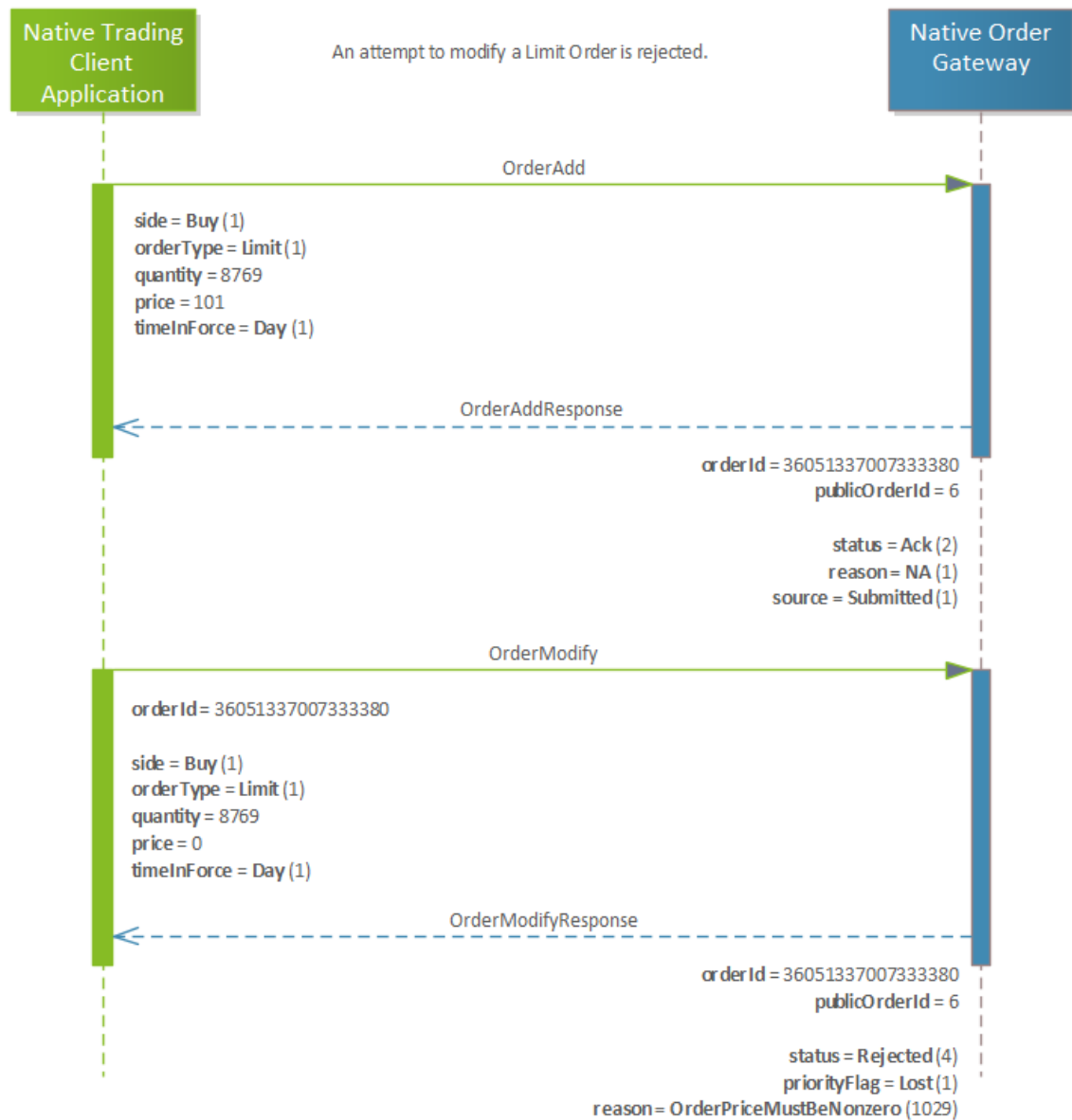
Stop Limit Order – Accepted and Triggered

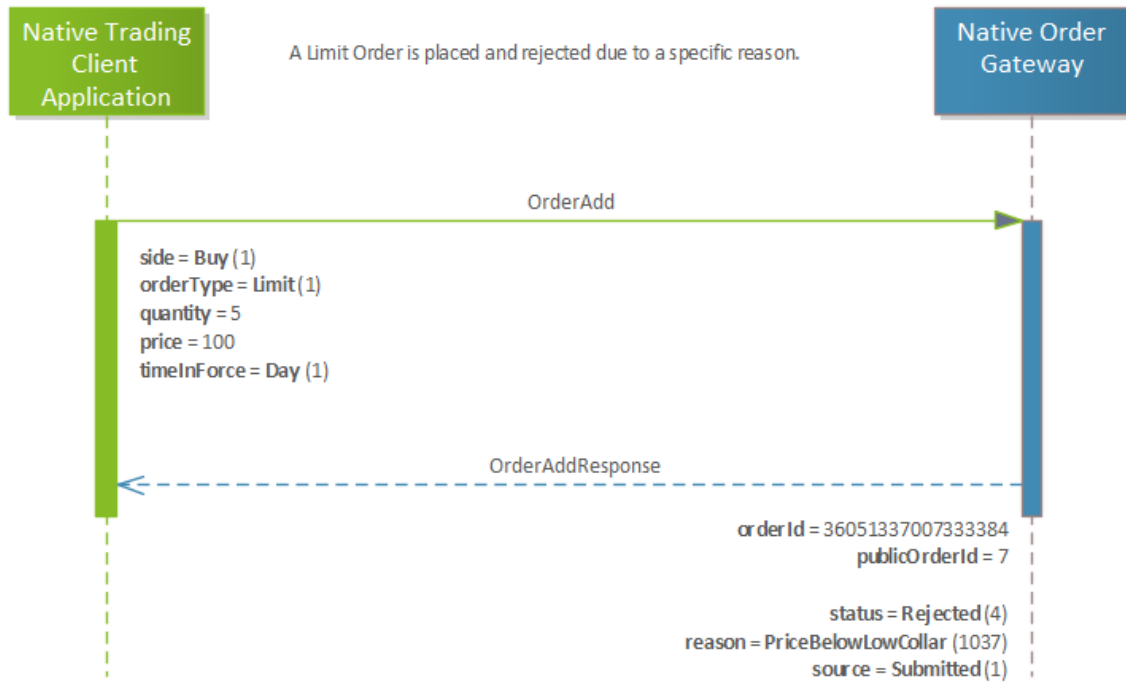
Stop Loss Order – Accepted, Triggered and Cancelled

Market Order and Market to Limit Order with TIF VFA – Triggered during Auction

Order Modification without priority loss

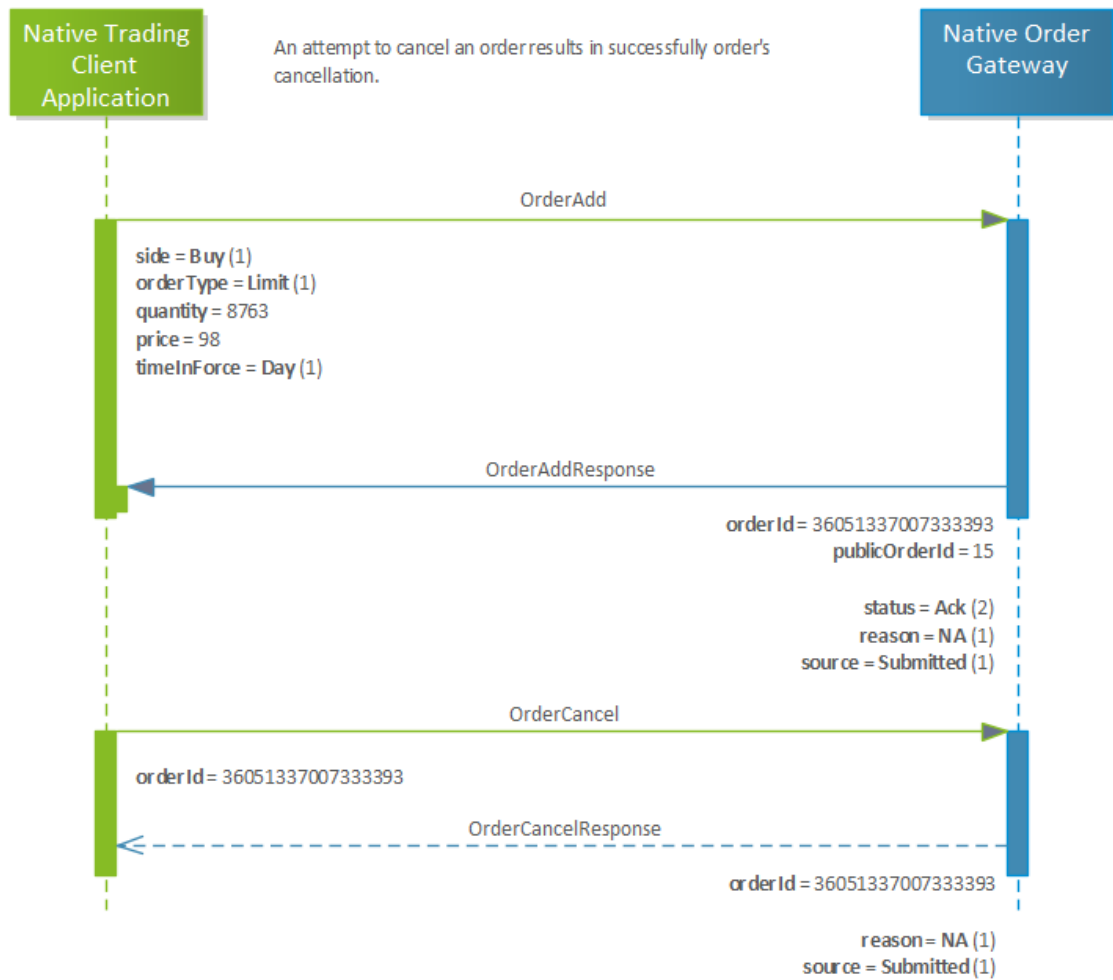
Order Modification with priority loss

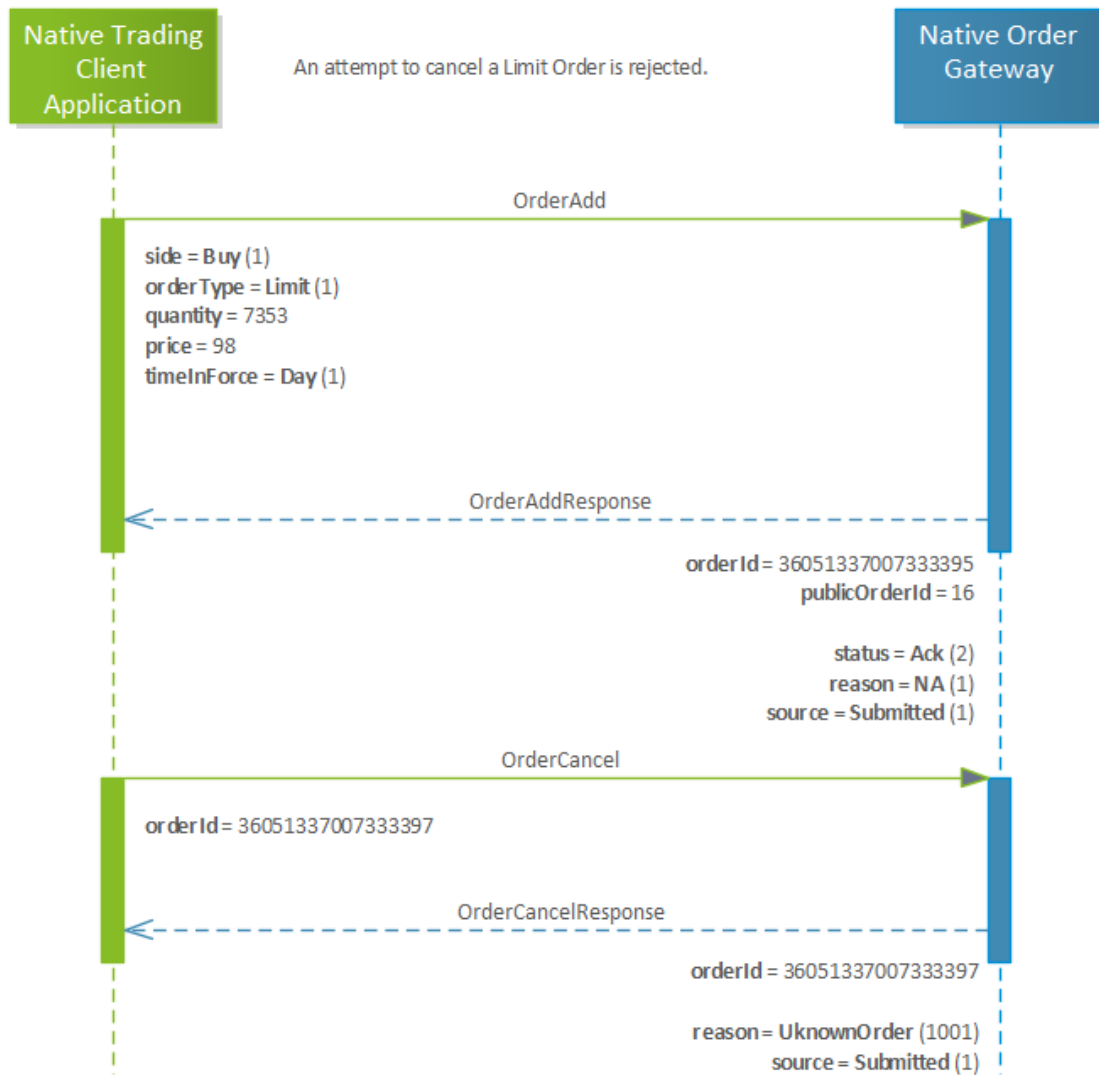
Order Modification – Rejected

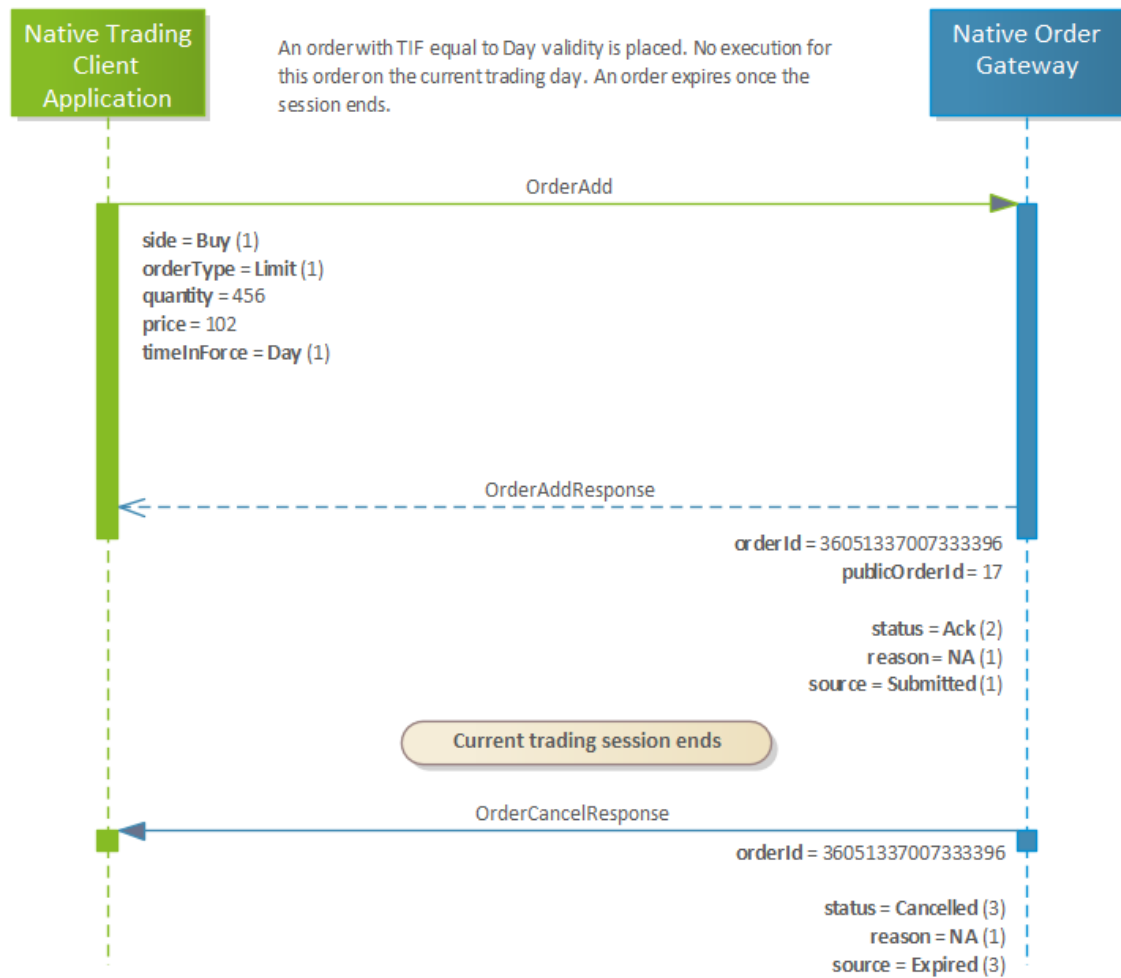
Order – Rejected

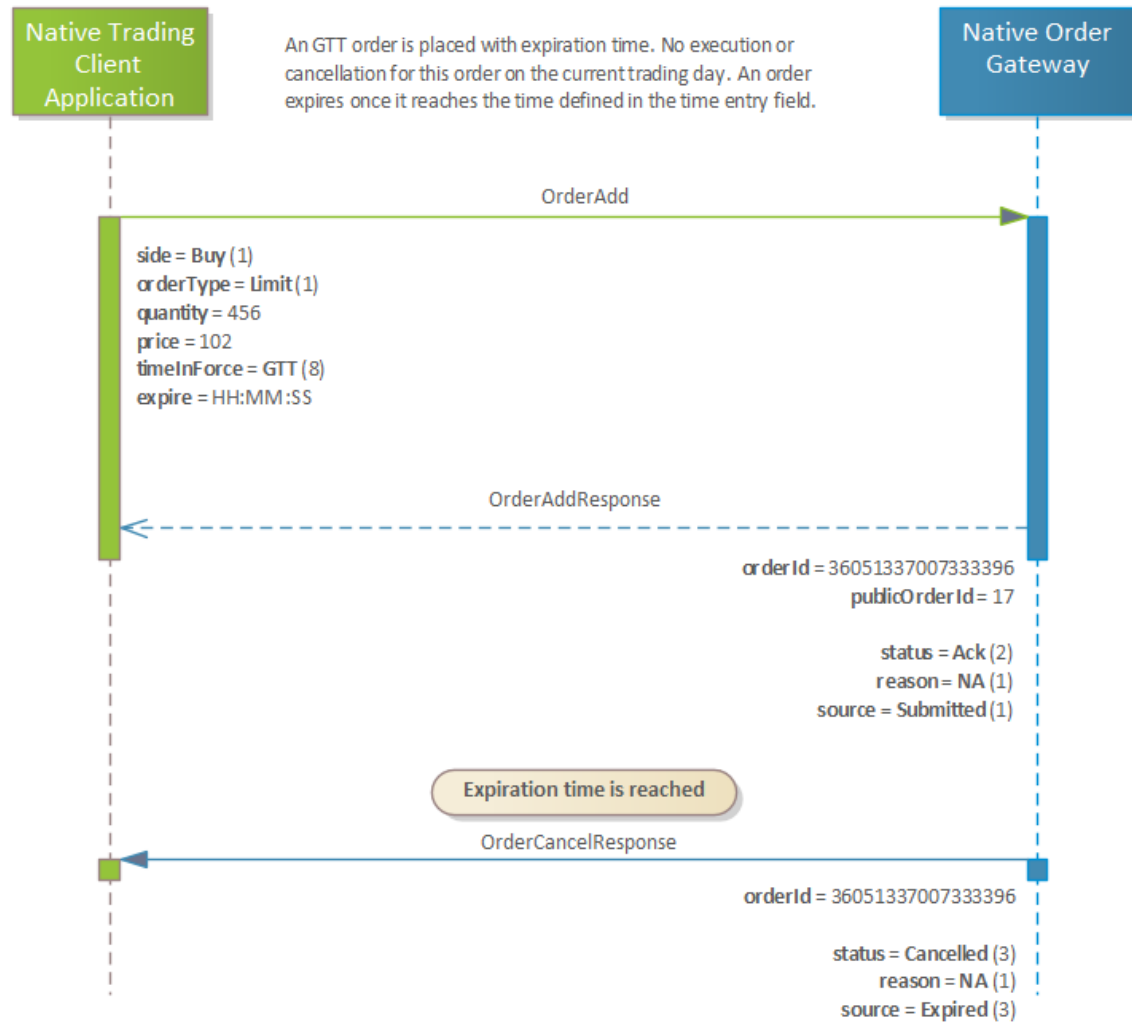
Stop Limit Order – Rejected

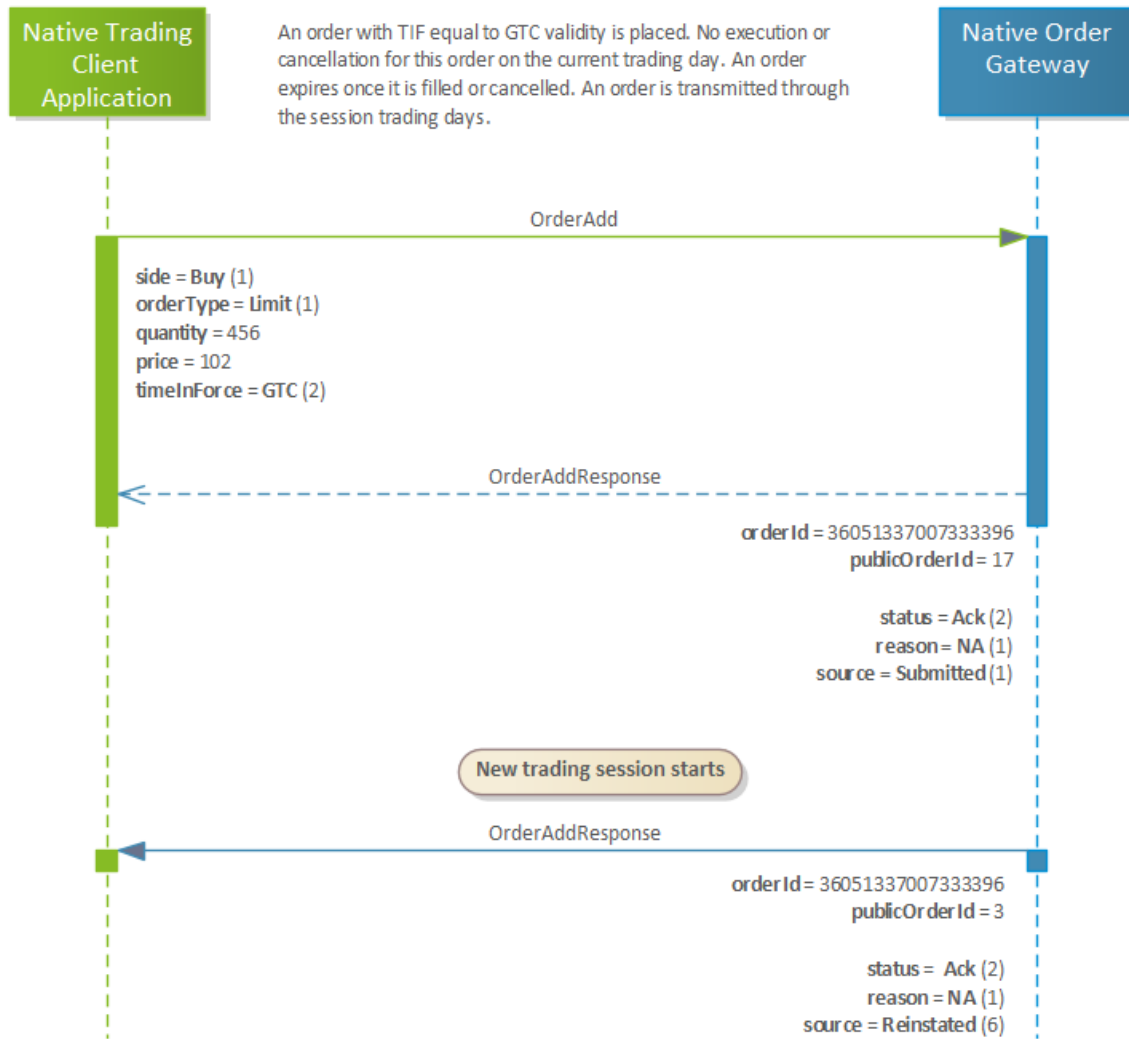
Market Order and Market to Limit Order ICO/FOK – Rejected during Auction

Order Cancellation – Accepted

Order Cancellation – Rejected

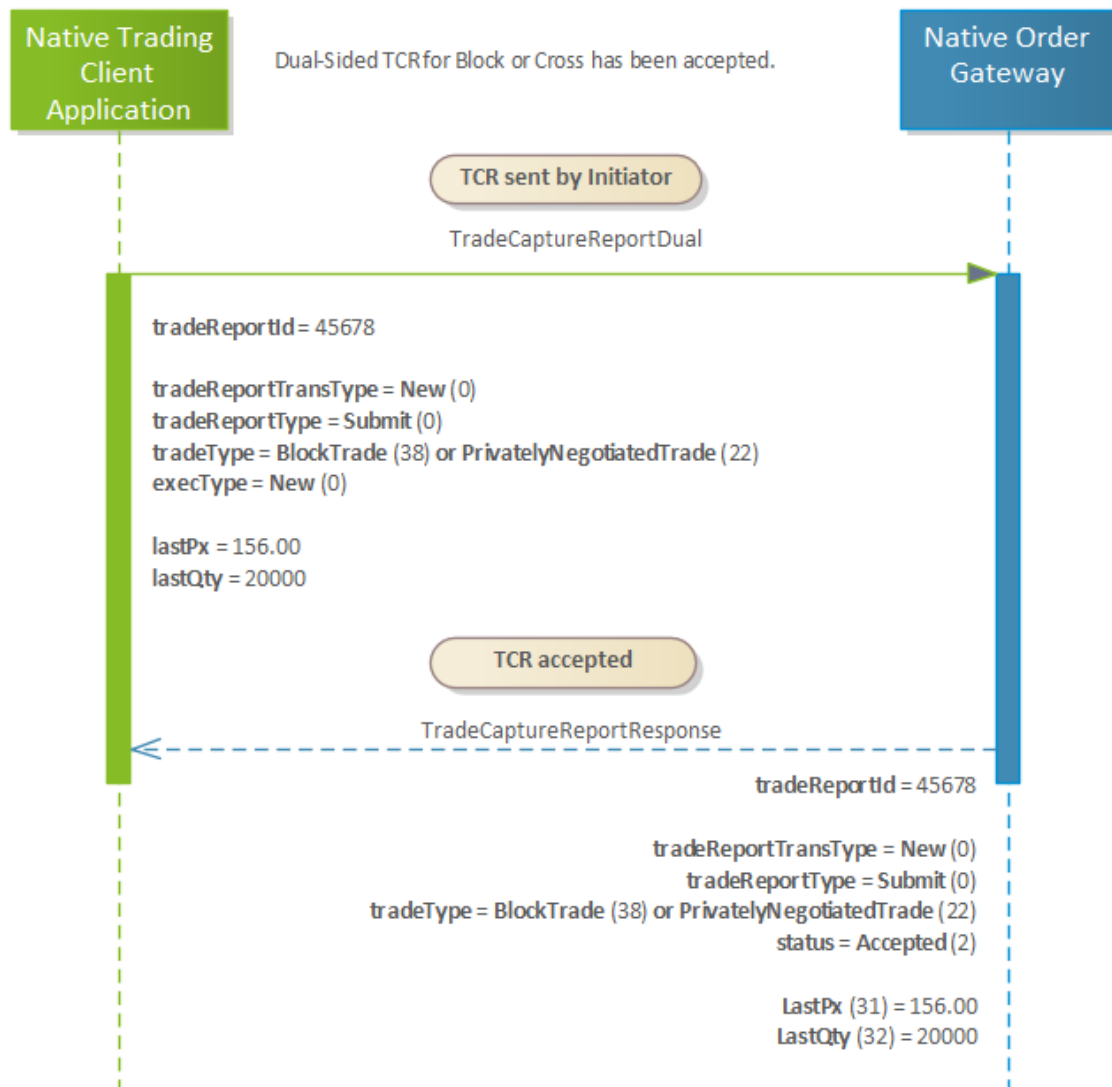
Day Order – Expired

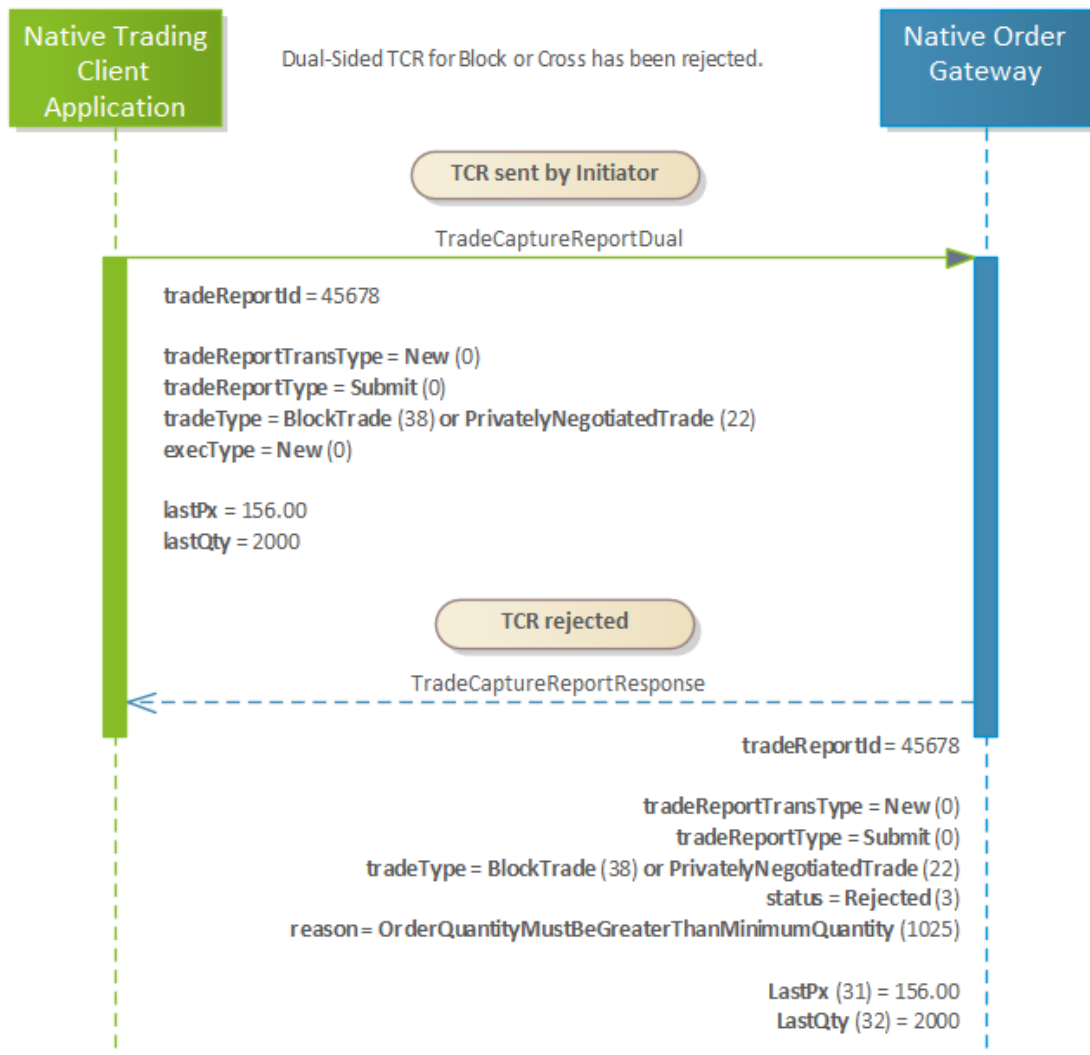
GTT Order – Expired

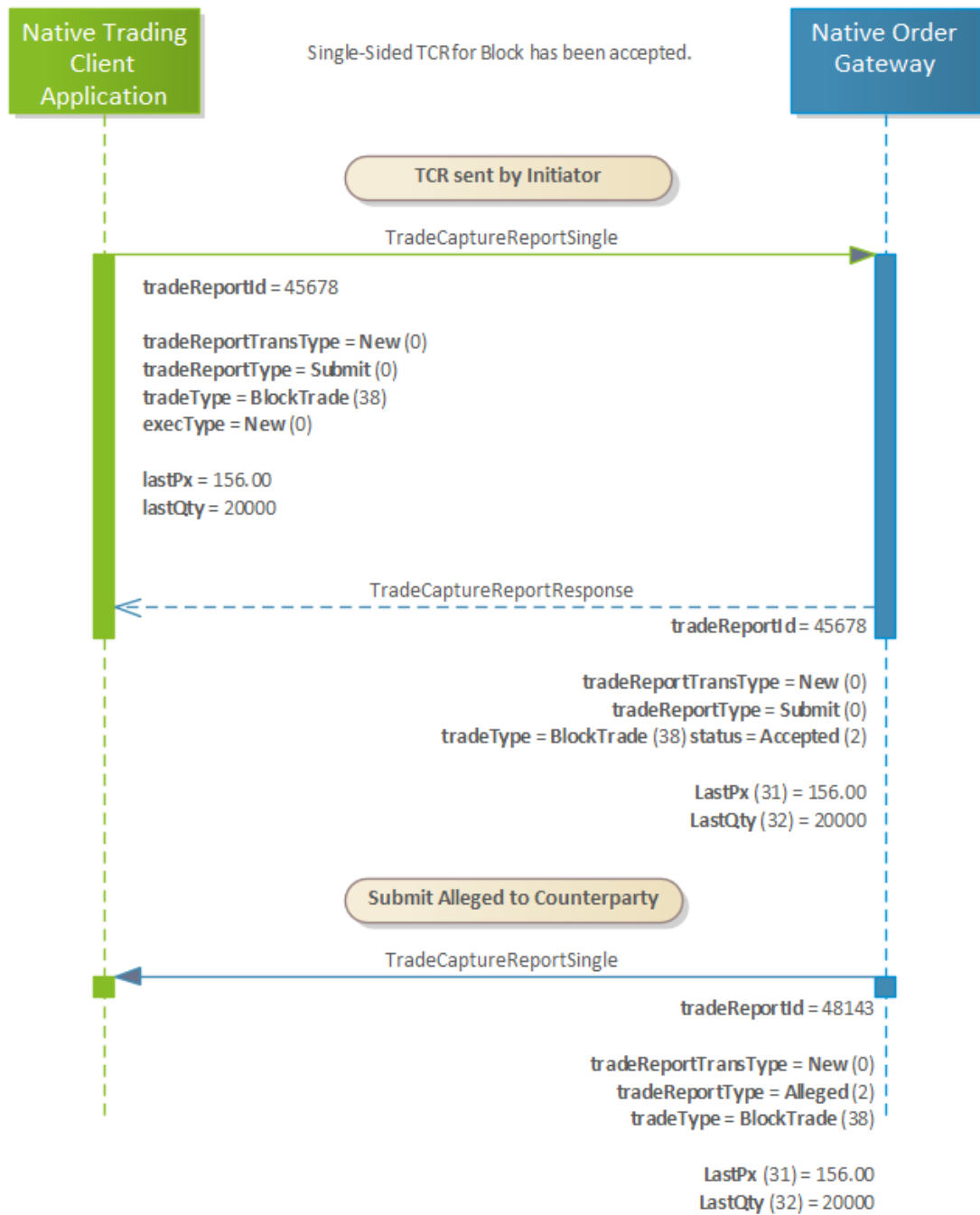
GTC Order – Reinstated

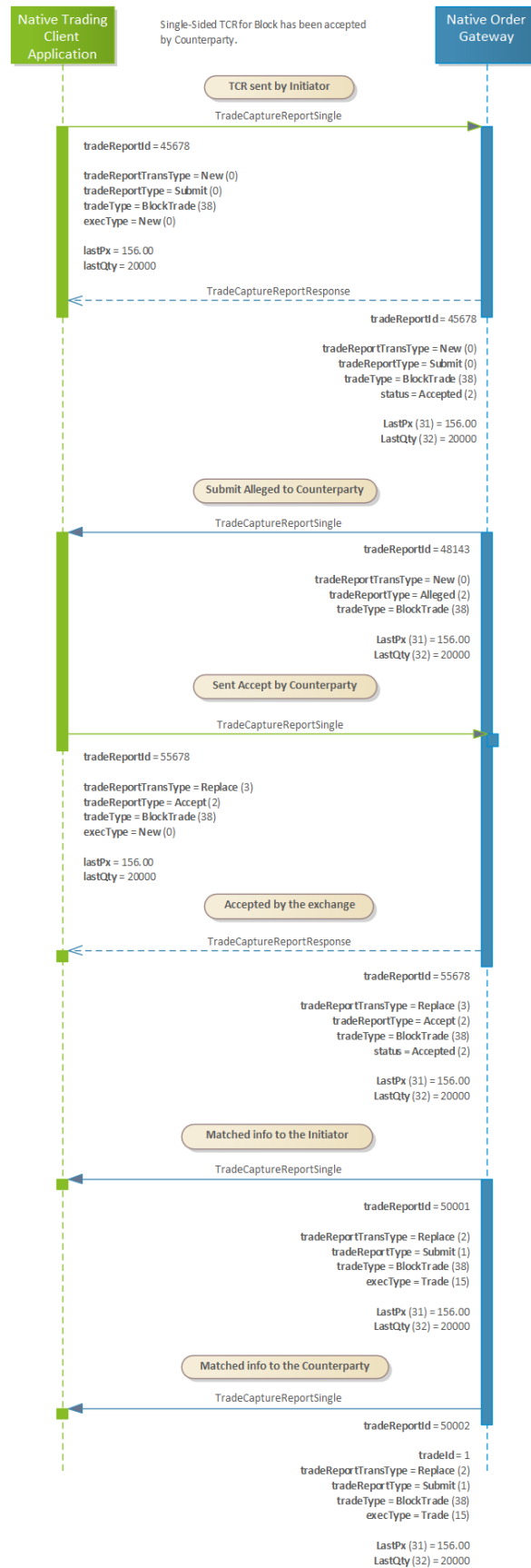
8.2. OFF-BOOK (BLOCK OR CROSS)

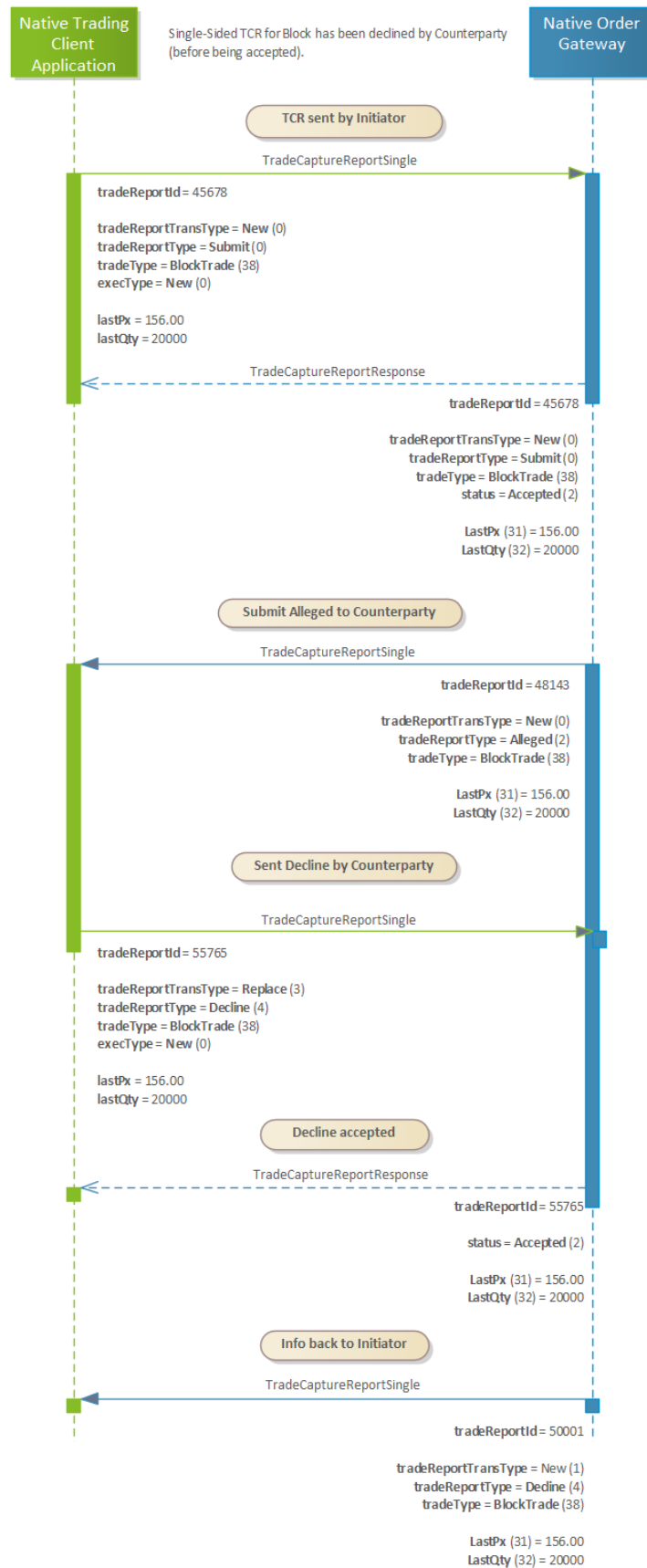
Dual Sided – Accepted

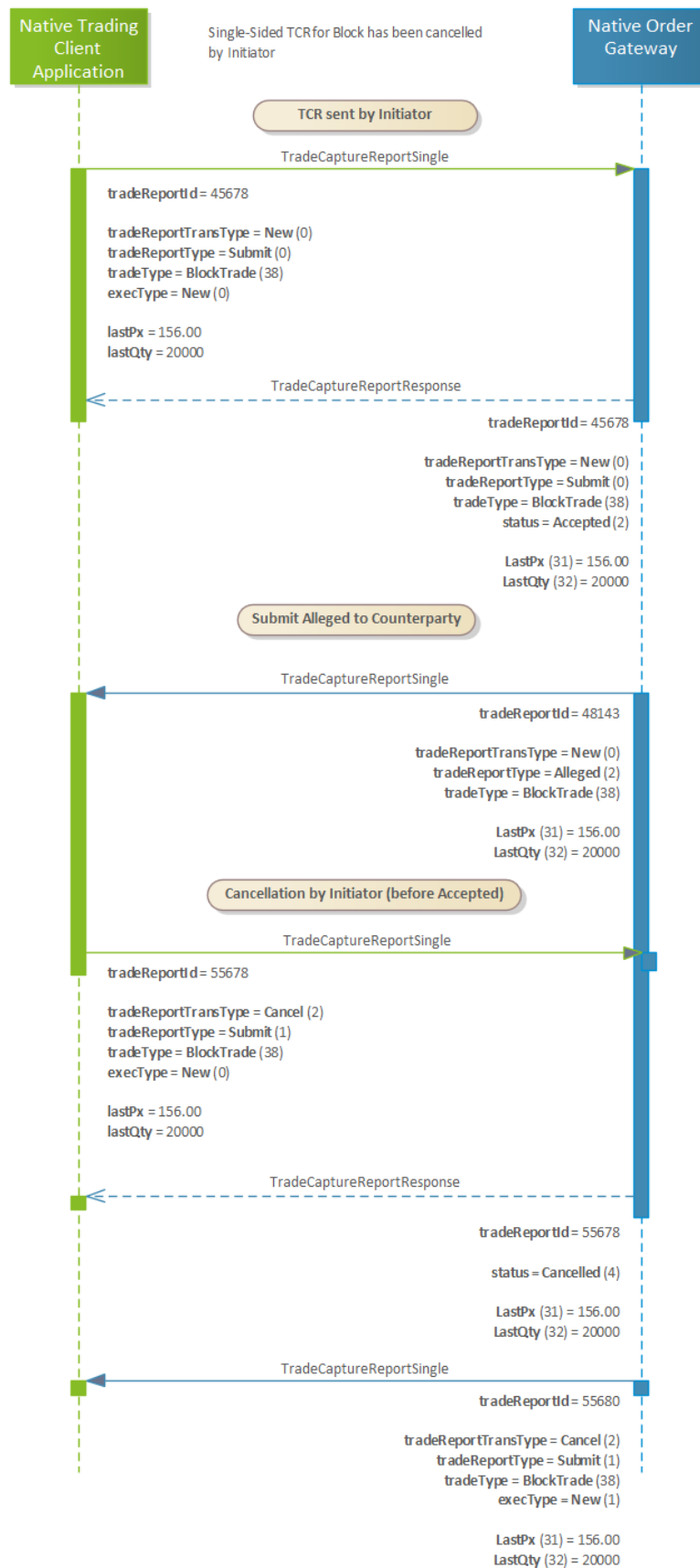


Dual Sided -Rejected

Single Sided – Accepted

Single Sided – Accepted by Counterparty

Single Sided – Denied by Counterparty

Single Sided – Cancelled by Initiator

Single Sided – Rejected